Paeds Final — one-page summaries

all 165 topics · in the words of your Anki cards · **tested answers** highlighted in their branch colour

Section I · Differential diagnostic titles

I-1 The unconscious child

- **Def**
  - **Unconscious child**: an airway problem before a neurology problem.
- **Cause**
  - **AEIOU-TIPS** is the mnemonic list of **causes of coma**.
  - Alcohol, epilepsy, insulin, opiates, uraemia, trauma, infection, psychiatric, shock, space-occupying lesion.
- **Patho**
  - **Structural** causes: **focal and asymmetric**; metabolic/toxic: **global and symmetric**.
  - **CPP** = **MAP** − **ICP**; raised ICP → **Cushing triad** (**hypertension**, **bradycardia**, irregular breathing).
  - **Uncal herniation**: CN III compression → **dilated pupil**; tonsillar → **cardiopulmonary arrest**.
- **Signs**
  - **Opioids**: respiratory and CNS depression, **pinpoint pupils**.
  - **Decorticate**: **flexed** arms; **decerebrate**: **extended** arms.
  - Expose: purpura, odd bruises (abuse), needle marks.
- **Dx**
  - Never skip a **bedside glucose**.
  - **Head CT** if **GCS ≤ 14**, skull fracture or focal signs.
  - Never do a lumbar puncture with **raised ICP or shock**.
- **Rx**
  - ABCDE; **GCS of 8 or less** → **intubation**; **naloxone** for pinpoint pupils.
  - Meningitis possible: **ceftriaxone immediately**, don't wait for lumbar puncture.
  - Raised ICP: **elevate** head, **hyperventilate**, **mannitol**.

I-2 Causes and treatment of convulsion in childhood

- **Def**
  - **Status epilepticus**: seizure lasting ≥5 min → treat on the 5-5-5 ladder.
- **Cause**
  - A neonatal seizure is always **pathological**: HIE, hypoglycaemia, low Ca/Na, sepsis, haemorrhage.
  - Toddler: **febrile seizure** commonest (**3–5%** of children), meningitis, NAI, West syndrome.
  - Older: epilepsy, trauma, tumour, AVM.
- **Patho**
  - **Excitatory** synapses mature before **inhibitory** ones → **low seizure threshold**.
  - Rapid temperature rise or metabolic insult → threshold crossed.
- **Signs**
  - **Absence**: **blank stare**, provoked by **hyperventilation**.
  - **Focal impaired awareness**: staring, head tilting, mouth chewing, confusion.
  - **West syndrome**: bilateral jerking, **hypsarrhythmia**.
- **Dx**
  - **Bedside glucose** at 0 min; check Ca, Na; screen for infection.
  - First seizure: EEG and **CT head without contrast** (emergency) or **MRI**.
- **Rx**
  - 5 min **first benzodiazepine**; 10 min second benzodiazepine; 15 min **levetiracetam, phenytoin or valproate**; 25 min RSI.
  - Neonates: **phenobarbital**, **pyridoxine** if resistant; West: **ACTH**.

I-3 Headache in childhood

- **Def**
  - **Primary** (migraine, tension-type — benign) vs **secondary** headache.
- **Cause**
  - Primary: migraine and tension-type = overwhelming majority.
  - Childhood CNS tumors are usually **infratentorial**: **pilocytic astrocytoma**, **medulloblastoma**, **ependymoma**.
  - Also **idiopathic intracranial hypertension**, **medication overuse headache**.
- **Patho**
  - Raised ICP → worse on waking, coughing, lying flat; vomiting **without nausea**.
  - Raised ICP → new **CN VI** palsy = **false localising sign**.
- **Signs**
  - **Migraine**: throbbing, **nausea**, **photophobia**; **4** to **72** hours; in children often **bilateral**, as short as **2 hours**.
  - **Tension**: **bilateral**, constant, no aura; chronic = **15** days/month for more than **3** months.
- **Dx**
  - Three questions: acute or chronic, **progressive or not**, **red flags**.
  - Red flags: **awakening from sleep**, coordination difficulties, focal signs, age under **5 years**.
  - Any red flag → urgent neuroimaging, preferably **MRI**.
- **Rx**
  - Acute migraine: **NSAIDs** or **acetaminophen** first-line; also **triptans**.
  - **Idiopathic intracranial hypertension**: **acetazolamide** (± furosemide).
  - Headache diary for triggers; prophylaxis if frequent.

I-4 Febrile infant

- **Def**
  - Infant with fever ≥38 °C; under 3 months they **cannot localise infection**.
- **Cause**
  - Neonatal sepsis: **group B streptococcus**, **E. coli** and **Listeria**; consider HSV.
  - Commonest serious bacterial infection: **urinary tract infection**.
- **Patho**
  - Immature immunity + no vaccine cover + no localising signs → appearance unreliable.
- **Signs**
  - Meningitis = poor feeding, 'not right'; no neck stiffness.
  - Non-blanching rash → meningococcal sepsis; fever >5 days → think Kawasaki.
- **Dx**
  - Under **28 days**: full septic screen incl. **lumbar puncture** for all.
  - 1–3 months: FBC, CRP, culture, urine; LP if unwell.
  - Over 3 months: **traffic-light** system; urine in everyone: **clean catch**, not a bag.
- **Rx**
  - Under 28 days: admit + IV **ampicillin (amoxicillin)** + gentamicin/cefotaxime; ± aciclovir.
  - Listeria is resistant to **cephalosporins**; non-blanching rash → antibiotics first.
  - Well over 3 months with viral source → home with safety-netting.

I-5 Rash and fever in children (incubation periods as well!)

Def
- Fever + rash: first ask **does it blanch?**, then pattern-match.

Cause
- First **measles**, second **scarlet fever**, third **rubella**, fifth parvovirus, sixth **roseola**.
- **Chickenpox**: **varicella-zoster virus (VZV)**, **airborne**.
- Emergencies: meningococcaemia, Kawasaki, SJS/TEN, toxic shock.

Patho
- Non-blanching + fever → **meningococcal sepsis** until proven otherwise.
- Petechiae **above the nipple line** after coughing or vomiting: raised venous pressure.
- **Chickenpox** lesions are in **different** stages.

Signs
- Measles rash starts **behind the ears**; **Koplik spots**; death from **pneumonia**.
- Scarlet fever: **sandpaper** rash, strawberry tongue; parvovirus: **slapped cheek**.
- Rubella: suboccipital nodes. Coxsackie A: **hand, foot and mouth disease**.

Dx
- **Glass test**; then morphology, site of onset, timing vs fever.
- Incubation: varicella **10–21 days**; rubella **14–21 days**; measles 10–14 d; scarlet 2–5 d.

Rx
- Suspected meningococcal sepsis → **benzylpenicillin or ceftriaxone immediately**.
- Severe chickenpox: **IV acyclovir**; newborn of infected mother: **VZIG**.
- Kawasaki → IVIG within 10 days.

I-6 Fever of Unknown Origin (FUO) and Kawasaki syndrome

Def
- **FUO**: fever >38 °C ≥8 days, no cause after initial evaluation.
- **Kawasaki**: fever ≥**5 days** plus **4 of 5** principal features.

Cause
- FUO is most often **infection** (~50%), then **inflammatory disease** (~20%), then malignancy (~10%).
- Kawasaki classically affects **Asian children**.

Patho
- Kawasaki hits the **coronary arteries** → **aneurysm** → rupture or myocardial infarction.
- sJIA with **falling ESR**, cytopenias, high ferritin → **macrophage activation syndrome**.

Signs
- Kawasaki conjunctivitis: bilateral, **non-purulent**, **limbal sparing**.
- **Thrombocytosis** in the **second week**; high CRP/ESR, sterile pyuria.
- sJIA: daily **spiking fevers**, **salmon-pink macular** rash.

Dx
- FUO: **repeated daily history and examination** is the highest-yield investigation.
- Suspected Kawasaki: **CRP or ESR**; **echocardiogram** at diagnosis, then 1–2 and 4–6 weeks.

Rx
- **IVIG 2 g/kg** plus aspirin within **10 days**; aspirin exception despite **Reye syndrome**.
- No defervescence → second IVIG; refractory → steroids/infliximab.
- Incomplete Kawasaki: **under 1 year**, **highest** aneurysm risk; delay vaccines 8 months.

I-7 Differential diagnosis and treatment of neonatal respiratory distress

Def
- Newborn with tachypnoea >60, grunting, flaring, recession, cyanosis.

Cause
- **RDS**: **prematurity**, maternal diabetes, caesarean; **TTN**: term caesarean; MAS: term and post-term.
- Sepsis; cyanotic CHD, diaphragmatic hernia, choanal atresia, TE fistula.

Patho
- **Prematurity** → inadequate **surfactant** → alveoli fail to open → poor oxygenation.
- **TTN**: delayed clearance of fetal lung fluid.
- PPHN: pulmonary vascular resistance fails to fall; meconium **inactivates surfactant**.

Signs
- TTN: term C-section baby with grunting, settles in **24–72 hours**.
- CDH: **scaphoid** abdomen; **choanal atresia**: **pink when crying**.
- TE fistula: **coughing, choking, cyanosis**.

Dx
- RDS CXR: **ground-glass** with air bronchograms; lecithin:sphingomyelin ratio **≥ 2**.
- No response to **100% oxygen** → duct-dependent CHD; PPHN: **echocardiography**.
- NG tube coiling in esophagus → esophageal atresia.

Rx
- **CPAP** for all; RDS: **maternal antenatal glucocorticoids**, surfactant when FiO_2 exceeds **0.30**.
- CDH: never bag-mask; intubate with **NG tube on suction**.
- Sepsis: **ampicillin + gentamicin** (cefotaxime if meningitis); PPHN: **nitric oxide**.

I-8 Neonatal jaundice

Def
- Classify by **onset** and **conjugated vs unconjugated**; **UGT** conjugates bilirubin.

Cause
- Jaundice in the first **24 hours** is always **pathological**: Rh/ABO, G6PD, spherocytosis, sepsis.
- **Breast milk** (**β-glucuronidase**); Gilbert and Crigler-Najjar: unconjugated.
- Conjugated: **biliary atresia**, hepatitis, choledochal cyst, TORCH, galactosaemia.

Patho
- **Phototherapy** makes bilirubin **water-soluble**, excreted without hepatic conjugation.
- High unconjugated bilirubin → **kernicterus**.
- **Rh** disease: Rh− mother, Rh+ fetus; not the first pregnancy.

Signs
- Prolonged jaundice beyond **2 weeks** with **pale stools and dark urine** = biliary atresia.
- **Cephalohematoma** is subperiosteal, does not cross sutures; encephalopathy: high-pitched cry, opisthotonus.

Dx
- Early work-up: blood group, **direct Coombs**, FBC, film.
- **Split the bilirubin**: conjugated is >**25 µmol/L** or >**20%** of total.

Rx
- **Phototherapy** above the line; **exchange transfusion** if it fails or encephalopathy.
- Breastfeeding jaundice: **increase feeds**; breast milk jaundice: continue breastfeeding.
- Anti-D at **28–32 weeks** and within **72** hours; **Kasai** before **60 days**.

I-9 Neonatal cyanosis

Def
- **Central** cyanosis (lips, mucous membranes) = low arterial oxygen saturation.

Cause
- Respiratory, cardiac, PPHN (**SSRIs** in pregnancy), methaemoglobinaemia, polycythemia (hematocrit >**65**%).
- Five Ts: truncus, **transposition** (day 1), **tricuspid atresia**, **tetralogy of Fallot** (commonest overall), total anomalous pulmonary venous return.

Patho
- PPHN: **pulmonary vascular resistance** fails to fall → right-to-left shunt at **PDA and foramen ovale**.
- Duct closes day 2–3 → collapse.

Signs
- **Comfortably blue** = cardiac, as respiratory drive responds mainly to **CO_2**; distressed and blue = pulmonary.
- Methaemoglobinaemia: **chocolate-brown** blood, **normal PaO_2**.

Dx
- **Hyperoxia test**: pulmonary cyanosis **improves** on 100% O_2; cardiac and PPHN **do not**.
- Pre-ductal saturation on the **right hand**; difference over **10%** suggests PPHN.
- **Echocardiography** is the key investigation.

Rx
- Duct-dependent lesions: **prostaglandin E1**, avoiding **excess oxygen**.
- Transposition: **balloon atrial septostomy** before **arterial switch**.
- PPHN: inhaled **nitric oxide** (**cGMP**); refractory → **ECMO**; methaemoglobinaemia: methylene blue.

I-10 Hypoglycaemia and hyperglycaemia

Def
- Neonatal hypoglycaemia: blood glucose below **2.6 mmol/L**.
- **HHS**: glucose >**600** mg/dL, osmolarity >320.

Cause
- **Ketotic** hypoglycaemia of childhood is commonest at 1–5 y; GSD, GH/cortisol deficiency.
- **No ketones** → **hyperinsulinism** (infant of diabetic mother) or fat-oxidation defect (**MCAD**).
- Hyperglycaemia → almost always **type 1 DM**.

Patho
- Maternal diabetes → fetal hyperinsulinism → neonatal hypoglycaemia, **macrosomia**.
- DKA insulin **bolus** → rapid osmotic shifts → **cerebral oedema**.

Signs
- Autonomic (sweating, tremor, tachycardia, hunger) → neuroglycopenic (confusion, seizures, coma).

Dx
- Take the **critical sample** (insulin, C-peptide, cortisol, GH, ketones) **before** giving glucose.
- Best discriminator: the **ketone level** at the time of the low sugar.
- DKA = glucose + ketosis + pH <7.3 or bicarbonate <18 mmol/L.

Rx
- **2 ml/kg of 10% dextrose**, then infusion; **glucagon** if no access.
- **Diazoxide** for hyperinsulinism; cornstarch (GSD); avoid fasting (MCAD).
- DKA: fluids first, then insulin **0.05–0.1 U/kg/h** an hour later, **no bolus**.

I-11 Differential diagnosis of lymphadenopathy

Def
- **Reactive** nodes (soft, tender, <2 cm, resolve 4–6 wk) vs malignant nodes.

Cause
- Localised: **Staphylococcus aureus** > **Streptococcus pyogenes**, **cat scratch** (Bartonella), atypical mycobacteria, TB.
- Generalised: **EBV**, CMV, toxoplasma, HIV, Kawasaki, JIA, SLE, **leukaemia, lymphoma**.

Patho
- EBV: T-cell hyperplasia in the node **paracortex**.
- Staph/Strep lymphadenitis → **suppuration**, **abscess**, **fluctuance**.
- Lymphoma diagnosis depends on **nodal architecture**; mediastinal mass threatens airway.

Signs
- EBV: fatigue, fever, **posterior cervical** nodes, pharyngitis, **hepatosplenomegaly**.
- Worrying: over **2 cm** and growing, beyond **6 weeks**, **supraclavicular**, painless.
- Cold, violaceous, non-tender node in a toddler: **atypical mycobacteria**.

Dx
- Highest-yield first test: **full blood count and film**; CRP, LDH, urate, EBV/CMV serology.
- **Chest X-ray** before biopsy to look for a **mediastinal mass**.

Rx
- Soft, mobile, non-tender nodes: **observation**; reactive nodes resolve in 4–6 weeks.
- Suspected lymphoma: **excisional** biopsy, not FNA.

I-12 Arthralgia

Def
- Joint pain: exclude **septic arthritis** first, then classify by cause and timing.

Cause
- Reactive: **rheumatic fever** (GAS M protein mimicry); inflammatory: JIA, SLE, HSP; malignant: **leukaemia**.
- Mechanical by age: DDH infancy, **Perthes** (**4–8** years), **SUFE** obese adolescent.
- Septic: **Staphylococcus aureus**; **Salmonella** osteomyelitis in sickle cell.

Patho
- Osteomyelitis starts in the **metaphysis** (slow-flowing **hairpin capillary loops**).

Signs
- Septic: febrile, toxic, refuses to weight-bear; **transient synovitis**: afebrile, well, limps.
- Morning = **inflammatory**, after activity = **mechanical**, night = **malignant**.
- **Jones**: carditis, polyarthritis, **Sydenham chorea**, **erythema marginatum**; **sJIA**: quotidian fever.

Dx
- **Kocher**: non-weight-bearing, fever >38.5, ESR >40, WBC >12.
- Urgent **aspiration** + blood cultures; MRI for osteomyelitis.
- JIA: onset before **16**, lasting at least **6 weeks**.

Rx
- **Aspirate the joint** before antibiotics → IV antibiotics + surgical washout.
- Transient synovitis: **rest and NSAIDs**; JIA: **methotrexate** weekly.
- Rheumatic fever: penicillin prophylaxis; **macrolides** if anaphylaxis.

I-13 Causes and differential diagnosis of bloody stool in children

Def
- Bloody stool: **digital rectal examination** and haemodynamics first, then narrow by age and story.

Cause
- Neonate: swallowed maternal blood, **NEC**, CMPA, volvulus, vitamin K deficiency.
- Toddler: **fissure** (commonest), intussusception, **Meckel's**, polyp.
- Older: infectious colitis, IBD, HSP, **HUS**.

Patho
- Meckel's bleeds because it contains ectopic **gastric mucosa**.
- **Rule of 2s**: 2% of people, 2 feet from **ileocaecal valve**, 2 inches, under age 2.
- E. coli O157 toxin → HUS; antibiotics increase toxin release.

Signs
- Painful, paroxysmal + sausage mass (3 mo–3 y) → intussusception.
- **Painless**, transfusion-requiring bleeding without diarrhoea → Meckel's.
- Bilious vomiting + blood → volvulus.

Dx
- **Apt test** for maternal blood.
- Meckel's → **technetium-99m pertechnetate scan**; intussusception → ultrasound target sign.
- HUS: haemolytic anaemia, **thrombocytopenia**, **acute kidney injury**.

Rx
- Shocked → resuscitate first: wide-bore access, fluids, cross-match.
- Intussusception: enema reduction. Volvulus: theatre. NEC: stop feeds, decompress, antibiotics.
- HUS: **no antibiotics** for E. coli O157.

I-14 Upper gastrointestinal bleeding in children (hematemesis)

Def
- Haematemesis: confirm the blood is the child's own and GI.

Cause
- Non-variceal: oesophagitis, gastritis, peptic ulcer (H. pylori, NSAIDs), Mallory-Weiss, **button battery**.
- Variceal: **portal hypertension** after **umbilical venous catheterisation** (portal vein thrombosis), biliary atresia.
- Neonate: swallowed maternal blood, **vitamin K deficiency bleeding**.

Patho
- Portal vein thrombosis → pre-hepatic portal hypertension → **varices**.
- Button battery → necrosis within hours → can erode into the aorta.

Signs
- **Splenomegaly**, ascites, caput medusae, thrombocytopenia → variceal.
- Neonate bleeding from several sites → vitamin K deficiency.

Dx
- Exclude impostors: epistaxis, cracked nipple (**Apt test**), red foods.
- Split **variceal vs non-variceal** — decides treatment.

Rx
- Two large cannulas, cross-match, blood; vitamin K + FFP for coagulopathy.
- Variceal: terlipressin or **octreotide** + prophylactic **antibiotics** + **band ligation**.
- Non-variceal: high-dose **proton pump inhibitor** + endoscopy; button battery → emergency removal.

I-15 Proteinuria and leukocyturia (pyuria)

Def
- Proteinuria: is it real, is it big? Leukocyturia = UTI until culture negative.

Cause
- Commonest: **transient proteinuria**; **orthostatic** (upright only, overweight adolescents).
- Nephrotic age 1–10 → **minimal change disease** (~90%).
- Sterile pyuria: partially treated UTI, **renal tuberculosis**, **Kawasaki disease**, appendicitis.

Patho
- Nephrotic: heavy proteinuria, oedema, albumin below **30 g/L**, **hyperlipidaemia**.
- PSGN: **low C3**, normal C4, normal by **8 weeks**; follows pharyngitis by **1–2 weeks**, skin by **3–6 weeks**.

Signs
- Nephrotic complications: **spontaneous bacterial peritonitis**, renal vein thrombosis, PE.
- Nephritic: haematuria with **red cell casts**, **hypertension**, milder proteinuria.

Dx
- Start with **urinalysis**; confirm on **first-morning** urine; quantify with **protein:creatinine ratio**.
- UTI: **nitrites** specific, **leucocyte esterase** sensitive; nephritic → complement, ASOT.
- After UTI: ultrasound → MCUG (VCUG), DMSA.

Rx
- Typical nephrotic 1–10 y → **steroids without biopsy**.
- Nephritic → urgent nephrology.

I-16 Macroscopic and microscopic haematuria

Def
- Haematuria: confirm red cells on **microscopy** via **urinalysis**, then glomerular vs non-glomerular.

Cause
- Glomerular: **IgA**, post-streptococcal GN, HSP, **Alport**, thin BM, MPGN/lupus, HUS.
- Non-glomerular: **UTI** (commonest overall), stones, trauma, Wilms, coagulopathy, sickle cell.

Patho
- **Red cell casts** prove haematuria is **glomerular** in origin.
- IgA **1–2 days** after URTI, normal complement; post-strep **1–3 weeks**, **low C3**.

Signs
- Glomerular: **brown or cola-coloured**, **without clots**, proteinuria, hypertension.
- **Initial** haematuria = **urethral**; **terminal** = bladder or prostate.
- Alport: **sensorineural deafness** and family history of renal failure.

Dx
- Dipstick positive with **no red cells** → myoglobinuria or haemoglobinuria; exclude beetroot, rifampicin.
- Glomerular: check **complement levels**; BP, protein dipstick, renal function, ultrasound.

Rx
- Red cell casts → glomerular work-up; stop looking at the bladder.
- Non-glomerular → target cause: UTI, stones, trauma, tumour (Wilms spreads to **lungs**).

I-17 Causes of enuresis in childhood

Def
- Involuntary voiding at age ≥**5**: at least **2** times/week for ≥**3** months; **primary** vs **secondary**.

Cause
- Greatest risk factor: **family history**; both parents → ~**70%**.
- Secondary: UTI, **diabetes mellitus**, DI, CKD, constipation, stress/abuse, neuro, structural.

Patho
- Failure to **wake**, nocturnal **polyuria** (low **ADH**), small bladder capacity.
- **Constipation** compresses the bladder, causing detrusor overactivity.

Signs
- Continuous dampness in a toilet-trained girl → **ectopic ureter**.
- Sacral dimple/hairy patch → spinal dysraphism; polyuria, polydipsia, weight loss → DM.

Dx
- First **rule out organic causes**: dipstick everyone for infection, **glucose**, specific gravity.
- Examine lumbosacral spine, gait, lower-limb reflexes.

Rx
- First-line: **lifestyle modification and behavioral training** (night **fluid restriction**, timed voiding).
- Next: **enuresis alarm** ± oral **desmopressin**; **imipramine** rarely used.
- Desmopressin → restrict evening fluids (**hyponatraemia**); the alarm is the best long-term cure.

I-18 Polydipsia, polyuria

Def
- True polyuria: >2 l/m²/day, with polydipsia.

Cause
- Three causes: **osmotic diuresis (diabetes mellitus)**, **diabetes insipidus**, primary polydipsia.
- Central DI: craniopharyngioma, **germinoma**, **Langerhans cell histiocytosis**, trauma.
- Nephrogenic: AVPR2, low K, high Ca, lithium.

Patho
- Central = no ADH; nephrogenic = kidney ignores ADH.
- Primary polydipsia → low-normal serum osmolality.

Signs
- Glycosuria, weight loss despite appetite, ketones → DM.
- DI: high serum osmolality with dilute urine.

Dx
- First bedside test: **urine dipstick** for **glucose**; then paired serum + urine osmolality.
- **Water deprivation test**: stop at **3–5%** body-weight loss.
- Concentrates after desmopressin → **central** DI → **MRI of the hypothalamic-pituitary axis**.

Rx
- Central DI: **desmopressin** + assess other pituitary hormones (cortisol).
- Nephrogenic: free water, low-solute diet, **thiazides**, sometimes indometacin (indomethacin).

I-19 Differential diagnosis and treatment of hypertension. Pheochromocytoma.

Def
- BP ≥**95th centile** for age, sex, height, confirmed on 3 occasions.
- **Pheochromocytoma**: tumor of **chromaffin cells**; rule of **10s**.

Cause
- Before adolescence most often **kidney disease** (~70–80%), then renovascular, coarctation, endocrine, drugs.
- **ARPKD** (infants, **PKHD1**); **ADPKD** (young adults, **renin**).
- Pheochromocytoma: **MEN 2A/2B**, VHL, NF1, SDH; familial more often **bilateral**.

Patho
- Catecholamine-secreting tumour → **episodic** surges.
- Beta-blocker first → **unopposed alpha** → hypertensive crisis.
- Neuroblastoma (adrenal medulla): **N-myc**, **Homer-Wright rosettes**.

Signs
- Episodic **headache**, **sweating**, palpitations; fasting hyperglycemia.
- Weak femorals → coarctation; hypertension + bradycardia = **Cushing triad**.
- **MEN 2B**: tall, kyphoscoliosis, lip neuromas.

Dx
- Cuff bladder covers **80%** of arm; too small **raises** reading; BP in all four limbs.
- Pheo: **metanephrines** → adrenal **CT/MRI** → **I-123 MIBG** scan.
- Ultrasound for PKD; **string of beads** = fibromuscular dysplasia.

Rx
- Pheo: **alpha**-blockade first (**phenoxybenzamine**), then **beta**-blockade, then **surgical excision**.
- Hyperkinetic hypertension: **beta-blocker**; Cushing triad: do **not** lower BP.

I-20 Differential diagnosis and treatment of cough in children

Def
- **Acute** cough under 3 weeks; **chronic** cough lasts more than **4 weeks**.

Cause
- Acute: **viral infection of the respiratory tract** commonest; **bronchiolitis** (RSV), croup, **foreign body**.
- Chronic: **pertussis**, asthma, GERD, **cystic fibrosis**, **protracted bacterial bronchitis**, **primary ciliary dyskinesia**.

Patho
- Bronchiolitis: small airway inflammation, usually **RSV**.
- Pertussis: **catarrhal** 1–2 weeks → **paroxysmal** → **convalescent**; coughing risks **pneumothorax**.
- Chronic wet cough + clubbing → **suppurative lung disease**, never asthma.

Signs
- Bronchiolitis: wheezing/crackles, not stridor. Pertussis: whoop, post-tussive vomiting; infants **apnoea and cyanosis**.
- Barking → croup; nocturnal dry + wheeze → asthma; meconium ileus → CF.
- Sudden cough in a well toddler → foreign body.

Dx
- Plot growth, check **clubbing**, immunisations, choking history, TB contacts.
- Suspected foreign body → **rigid bronchoscopy**; normal CXR doesn't exclude.

Rx
- Viral: **reassure + safety-net**; no over-the-counter cough medicines.
- Bronchiolitis: **supportive**; RSV prevention **nirsevimab or palivizumab**.
- Pertussis: **macrolide**, also prophylaxis for contacts.

I-21 Sore throat (viral versus bacterial pharyngitis) and upper respiratory tract infection

Def
- About **70–80%** of sore throats in children are viral; the task is finding **group A strep**.

Cause
- Viruses; **EBV** in teenagers; strep in age **3–14**.
- Emergencies: epiglottitis, quinsy, retropharyngeal abscess, diphtheria, Lemierre syndrome.

Patho
- Untreated strep → **acute rheumatic fever**; antibiotics do not prevent PSGN.
- **Cough, coryza or hoarseness** = viral.

Signs
- Strep: abrupt fever, exudate, tender nodes, **no cough**; EBV: posterior nodes, **splenomegaly**.
- Quinsy: trismus, uvula deviated **away**; epiglottitis: drooling, muffled voice, tripod, **no cough**.

Dx
- **Rapid strep** has low sensitivity: negative → throat culture.
- EBV: atypical lymphocytes, raised LFTs; diphtheria: gray-white pseudomembrane.

Rx
- Strep: **penicillin V for 10 days**, to prevent rheumatic fever.
- EBV given amoxicillin → **non-allergic maculopapular rash**; no contact sport **4–6 weeks** (splenic rupture).

I-22 Inspiratory stridor, croup syndrome

Def
- **Inspiratory stridor** localises narrowing above the cords (extrathoracic); acute cause is usually **croup**.

Cause
- Croup: **parainfluenza virus**, age **6 months to 6 years**.
- Mimics: epiglottitis, bacterial tracheitis (Staph aureus), foreign body, laryngomalacia, vascular ring.

Patho
- The subglottis swells inward because it is ringed by the **cricoid**, the only complete cartilage ring.

Signs
- Croup: barking cough, stridor, not toxic; **laryngomalacia**: improves prone.
- Epiglottitis: drooling, toxic, **no cough**; tracheitis: toxic, no response to adrenaline.

Dx
- Severity = **stridor at rest**, recession, conscious level, not **oxygen saturation**.
- Epiglottitis: do not examine the throat.

Rx
- Every child with croup gets oral **dexamethasone**; severe: nebulised **adrenaline**, wears off in about **2 hours**.
- Epiglottitis: secure the **airway** first; foreign body: **rigid bronchoscopy**.

I-23 Haemolytic anaemias in childhood

Def
- Haemolysis is proven by anaemia with high **reticulocytes**, high **unconjugated bilirubin**, high LDH, low **haptoglobin**.

Cause
- Positive direct Coombs = **immune**: warm AIHA **IgG**, cold AIHA **IgM**.
- Coombs negative: hereditary spherocytosis, **G6PD**, sickle cell, thalassaemia, HUS/TTP/DIC.

Patho
- Hereditary spherocytosis: membrane tethering defect, **autosomal dominant**.
- G6PD: triggered by **infection**, fava beans, drugs → bite cells, Heinz bodies.

Signs
- Jaundice, splenomegaly, dark urine; sickle cell: dactylitis, **acute chest syndrome**.
- **Low** reticulocytes in chronic haemolysis = **parvovirus B19** aplastic crisis.

Dx
- Spherocytes, Coombs **negative**, positive EMA binding, raised MCHC = hereditary spherocytosis.
- Schistocytes = **microangiopathic haemolysis**; Hb electrophoresis for sickle and thalassaemia.

Rx
- Warm AIHA: stop offending drug, **steroids**; splenectomy last resort.
- Sickle: daily **penicillin** to 5 years; fever → **ceftriaxone**; stroke or priapism → **exchange transfusion**.

I-24 Nonhemolytic anaemias

Def
- The first branch point in any anaemia is the **MCV**; the commonest in childhood is **iron deficiency**.

Cause
- Microcytic: iron deficiency, thalassaemia trait, **lead poisoning**.
- Macrocytic: **B12/folate** deficiency, Diamond-Blackfan; pancytopenia: Fanconi, aplastic anaemia.

Patho
- Excess cow's milk: **iron-poor**, **inhibits iron absorption**, occult GI blood loss.
- B12/folate are needed for **DNA synthesis**, so cells grow large.

Signs
- Lead: bluish gum-tooth line, LEAD mnemonic; B12: paresthesias, megaloblasts.
- Fanconi: short stature, café-au-lait spots, malformed thumbs.

Dx
- RDW **high** in iron deficiency, **normal** in thalassaemia trait; red cell count normal or high.
- Ferritin is an **acute phase protein**, so inflammation can mask deficiency.

Rx
- Iron deficiency: **oral elemental iron**, continued **3 months** after Hb normalises.
- Lead 45–69: **DMSA (succimer)**; ≥70 mcg/dL: **IM dimercaprol** + **EDTA**.

I-25 Causes of failure to thrive

Def
- Failure to thrive classically means crossing **two or more** weight centile lines downward.

Cause
- Commonest cause: **inadequate intake (non-organic)**; child neglect is the commonest maltreatment.
- Organic: cystic fibrosis, galactosaemia, congenital heart disease, enteropeptidase deficiency.

Patho
- Order of loss: **weight** → **height** → **head circumference**.
- Congenital heart disease: **increased energy expenditure** plus **reduced intake**.

Signs
- Short and plump with delayed bone age suggests an **endocrine** cause.
- Small head from early on: **intrauterine, genetic or congenital infection**.
- Kwashiorkor: oedema, swollen abdomen; marasmus: wasting.

Dx
- Highest-yield 'investigation': **observe a feed**.
- Meconium ileus → **cystic fibrosis**; neonatal jaundice and E. coli sepsis → **galactosaemia**.

Rx
- Treat the cause; feeding advice for intake problems.
- Suspected abuse: first **contact child protective services (CPS)**.

I-26 Hepatomegaly and splenomegaly

Def
- Enlarged liver, spleen or both; a palpable liver edge is normal in infants, so measure the liver **span**.

Cause
- EBV/infectious mononucleosis, CMV, hepatitis A, malaria, visceral leishmaniasis.
- Hereditary spherocytosis, systemic JIA, **glycogen storage disease type I**.

Patho
- A spleen must enlarge to about **2–3 times** normal size before it is palpable.
- **Hypersplenism**: enlarged spleen sequesters blood cells and lowers the platelet count.

Signs
- Infant hepatomegaly with **fasting hypoglycaemia** suggests von Gierke.
- CMV newborn: blueberry muffin rash, jaundice; systemic JIA: spiking fevers, salmon-pink rash.

Dx
- First test: **full blood count and film**.
- Synthetic function: **prothrombin time (INR)**, not the ALT.

Rx
- Treat the cause.
- EBV splenomegaly: avoid **contact sports** for 4–6 weeks.

I-27 Differential diagnosis of diarrhoea

Def
- In acute diarrhoea the priority is assessing **hydration**, not the stool.

Cause
- Acute: mostly rotavirus and norovirus; bacterial if bloody.
- Chronic: pivotal question is whether the child is **growing**; toddler's diarrhoea is the thriving form.

Patho
- ORS works because the **sodium-glucose cotransporter (SGLT1)** stays intact.
- Antibiotics in E. coli O157 raise the risk of **haemolytic uraemic syndrome**.
- **Osmotic** diarrhoea **stops** with fasting; **secretory** **continues**.

Signs
- Loose stools with recognisable food in a thriving 2-year-old = **toddler's diarrhoea**.
- Blood, high fever, severe pain suggest bacterial cause.

Dx
- Grade dehydration clinically.
- Faltering growth: coeliac screen with tTG and total IgA, sweat test, calprotectin.

Rx
- **ORS** and continue feeding; IV only if ORS fails.
- **Loperamide** is not used; no routine antibiotics.

I-28 Differential diagnosis of vomiting

Def
- Green (bilious) vomit means obstruction distal to the **ampulla of Vater**.

Cause
- Neonate: **malrotation with volvulus**, atresias, Hirschsprung, salt-wasting CAH.
- Infant: **pyloric stenosis**, intussusception; older: gastroenteritis, appendicitis, DKA, raised ICP.

Patho
- Pyloric stenosis causes a **hypochloraemic, hypokalaemic metabolic alkalosis**.
- Urine is acidic despite alkalosis: **paradoxical aciduria**.

Signs
- Pyloric stenosis: **2–8 weeks**, firstborn boy, projectile vomiting, hungry afterwards.
- Neonate with shock, hyponatraemia, hyperkalaemia = salt-wasting **congenital adrenal hyperplasia**.

Dx
- Bilious vomiting: urgent exclusion of malrotation with volvulus.
- Pyloric stenosis imaging of choice: **ultrasonography**.

Rx
- Correct fluids and electrolytes before surgery; alkalosis risks post-operative **apnoea**.
- Operation: **Ramstedt pyloromyotomy**; volvulus → emergency surgery.

I-29 Acute abdominal disorders in children

Def
- The commonest cause of an acute abdomen in children is **appendicitis**.

Cause
- Sort by age: neonate NEC and volvulus; toddler intussusception; school age appendicitis.
- Adolescent: testicular or ovarian torsion, gynaecological causes; trauma: splenic rupture.

Patho
- Pain starts periumbilically: appendix is a midgut organ, afferents enter the cord at **T10**.
- **Two-stage splenic rupture**: subcapsular haematoma, shock **hours to days** later.

Signs
- Neonate with bilious vomiting = **malrotation with midgut volvulus** until proven otherwise.
- Appendicitis: central pain moving to RLQ, anorexia, low fever.

Dx
- Four free checks: chest, urine dipstick, **glucose**, **scrotum** in boys.
- Polytraumatised child: **acute (whole-body) CT**.

Rx
- Peritonism → surgical referral; resuscitate if unstable.
- Salvage of a torted testis falls sharply after about **6 hours**.

I-30 Differential diagnosis of palpable abdominal mass in childhood

Def
- Exclude benign causes, then separate **Wilms tumour** from **neuroblastoma**.

Cause
- Neonate: most masses are **renal** (hydronephrosis, multicystic dysplastic kidney).
- Wilms: WAGR (**aniridia**), **Beckwith-Wiedemann syndrome**, hemihypertrophy; neuroblastoma: **neural crest**, N-myc.

Patho
- Wilms arises from **metanephric mesenchyme (blastema)**; it spreads mostly to the **lungs**.
- Do not palpate repeatedly: risk of rupture.

Signs
- Wilms: peaks **3–4 years**, well child, smooth mass, does **not cross the midline**, haematuria.
- Neuroblastoma: **under 2 years**, unwell, crosses midline, raccoon eyes, **opsoclonus–myoclonus–ataxia**.

Dx
- Abdominal **ultrasound** first.
- Neuroblastoma: raised urinary **VMA and HVA**; small blue round cells, Homer-Wright rosettes.

Rx
- A suspected Wilms tumour must not be **repeatedly palpated**.
- Europe (**SIOP**): chemotherapy before surgery; US (**COG**): usually operate first.

I-31 Chronic abdominal pain

Def
- Most chronic abdominal pain in children is **functional** and should be diagnosed positively.

Cause
- Functional: irritable bowel syndrome, functional abdominal pain.
- Organic: **constipation** is the commonest treatable cause; also coeliac, IBD, H. pylori, choledochal cyst, abuse.

Patho
- The further the pain from the **umbilicus**, the more likely an organic cause.
- IBS is not related to **structural abnormalities**.

Signs
- Functional abdominal pain never **wakes the child at night**.
- Key question in infancy: **how the infant is growing/developing**.

Dx
- Best non-invasive test to separate IBD from functional pain: faecal **calprotectin**.
- H. pylori is urease-positive: **urea breath** test.

Rx
- IBS first line: lifestyle modification, low FODMAP diet, **fiber**.
- Best evidence: **cognitive behavioural therapy**; treat constipation.

I-32 Differential diagnosis of oedema

Def
- Four Starling mechanisms: raised **hydrostatic** pressure, low **oncotic** pressure, increased capillary permeability, lymphatic obstruction.

Cause
- Commonest cause of generalised oedema in a young child: **nephrotic syndrome**.
- Also kwashiorkor, protein-losing enteropathy, liver disease, hereditary angio-oedema.

Patho
- Kwashiorkor: oedema from decreased plasma **oncotic** pressure.
- Membranous nephropathy: GBM thickening, granular immune-complex deposits; amyloid deposits in the **mesangium**.

Signs
- Oedema first appears **periorbitally**, in the morning.
- Recurrent non-pitting, non-itchy swelling without urticaria suggests **hereditary angio-oedema**.

Dx
- Nephrotic: UPCR >200 mg/mmol, low albumin.
- Low albumin with a negative urine dipstick points to the **gut (protein-losing enteropathy)** or **liver**.

Rx
- Treat the mechanism.
- Nephrotic syndrome in children: mostly steroid-responsive; HAE is C1-esterase inhibitor deficiency.

I-33 Differential diagnosis of dehydration in childhood

Def
- Fluid deficit (ml) = % dehydration × **weight (kg) × 10**.

Cause
- Commonest cause: **acute gastroenteritis**; **rotavirus** is the commonest cause of severe diarrhoea.
- Cholera and Shigella (high fever, mucoid-bloody diarrhoea, seizures) also cause it.

Patho
- Diarrhoea: ECF volume **decreased**, haematocrit and plasma protein increased.
- Rotavirus causes villous destruction and atrophy.

Signs
- Doughy skin in a dehydrated infant suggests **hypernatraemic** dehydration.
- In children **hypotension** is a late, pre-terminal sign of shock.

Dx
- Check serum sodium; correct slowly, no faster than **0.5** mEq/L/hr.
- Hypernatraemia corrected rapidly causes **cerebral oedema**.

Rx
- Without shock: ORS **50 ml/kg over 4 hours** plus maintenance.
- Shock bolus: **10 ml/kg 0.9% sodium chloride**; rotavirus vaccine at 2, 4, 6 months.

I-34 Principles of fluid therapy in childhood. Maintenance fluid therapy plan for a 3kg neonate and for a 20kg child.

Def
- Plan = resuscitation + deficit + maintenance + ongoing losses; standard maintenance is **DSNS**.

Cause
- Holliday–Segar maintenance: **100** ml/kg first 10 kg, **50** ml/kg next 10 kg, **20** ml/kg above.

Patho
- Hypotonic saline alone can cause RBC lysis; use **isotonic** fluid beyond the neonatal period.
- Sodium is withheld from neonatal fluids until the **postnatal diuresis** begins.

Signs
- Overload: oedema, hepatomegaly, crackles, weight gain.
- High ADH (meningitis, pneumonia, post-op, head injury): restrict to **50–80%**.

Dx
- Maintenance for a 20 kg child: **1500 ml/day (≈60 ml/h)**; hourly 4/2/1 rule.
- 3 kg neonate day 1: about **180 ml/day (7.5 ml/h)**.

Rx
- Term neonate day 1: **50–60 ml/kg/day** of **10% glucose**, rising to 120–150 ml/kg/day by day 5.
- Older child: **0.9% NaCl with 5% glucose**; shock bolus 10 ml/kg.

I-35 Disorders of sexual differentiation

Def
- Mismatch of chromosomal, gonadal or phenotypic sex; ambiguous genitalia is an emergency.

Cause
- Commonest cause of ambiguous genitalia in a newborn: **21-hydroxylase deficiency (CAH)**.
- 46,XY with female genitalia, testes and no uterus: **complete androgen insensitivity**.

Patho
- **SRY** makes the gonad a testis; **AMH** from Sertoli cells regresses the Müllerian ducts.
- Testosterone virilises; androgen insensitivity is an **androgen receptor** defect.

Signs
- Salt-wasting CAH presents at **1–3 weeks** with low sodium and **high potassium**.
- Androgen insensitivity: pubic hair **absent**; Müllerian agenesis: **present**.

Dx
- Do **karyotype**; screening test for CAH: raised **17-hydroxyprogesterone**.
- Non-palpable gonads in a 'male' infant = **CAH** until proven otherwise.

Rx
- Involve the MDT before assigning sex.
- Androgen insensitivity: **bilateral gonadectomy after puberty**; do not circumcise hypospadias.

I-36 Disorders of sexual development/maturation

Def
- Premature puberty: signs before 8 years in girls, 9 in boys.
- Delayed puberty: no breasts by 13, testes by 14.

Cause
- Central precocious puberty: girls mostly **idiopathic**; boys often **organic (CNS lesion)**.
- Commonest delay: **constitutional delay**; Kallmann syndrome (anosmia); Turner and primary ovarian insufficiency.

Patho
- Central: leuprolide stimulation **increases** LH; peripheral: LH/FSH **suppressed**.
- Oestrogen closes the epiphyseal growth plate.

Signs
- First sign: girls **thelarche**; boys **testicular volume ≥4 ml**.
- Boy with small testes suggests **CAH/adrenal** cause.

Dx
- Central precocious puberty needs **MRI of the brain** before treatment.
- Primary amenorrhea: **pelvic ultrasound**; Müllerian agenesis has normal secondary sex characteristics.

Rx
- Central precocious puberty: **GnRH agonist depot** (leuprolide).
- Treat the cause; premature adrenarche and newborn breast enlargement need reassurance.

I-37 Arrhythmias

Def
- Commonest symptomatic tachyarrhythmia in children: **SVT**; presents with abrupt onset tachycardia.

Cause
- WPW: ventricular pre-excitation; **long QT**: Romano-Ward (AD), Jervell and Lange-Nielsen (AR, deafness).
- Congenital complete heart block: maternal **anti-Ro/La** antibodies; Brugada: AD.

Patho
- Long QT predisposes to **torsades de pointes**.
- Adenosine inhibits the AV node, causing transient heart block.

Signs
- Infants with SVT (rate >220) present with **heart failure**, not palpitations.
- Sudden syncope without prodrome = **arrhythmia**.

Dx
- WPW resting ECG: short PR, widened QRS, **delta wave**.
- Brugada: pseudo-RBBB with ST elevation in V1–V3.

Rx
- Stable SVT: **vagal manoeuvres** first, then **rapid IV adenosine**; **verapamil** contraindicated in infants.
- Unstable: **synchronised cardioversion**; long QT: **beta-blockers**; complete heart block: pacemaker.

I-38 Obesity in childhood

Def
- Excess weight judged by BMI centiles; mostly **exogenous** obesity.

Cause
- Exogenous: child is **tall** with **advanced** bone age.
- Endocrine: child is **short** with slowing height velocity; commonest monogenic cause **MC4R**; Prader-Willi.

Patho
- Prader-Willi: gene **deletion**, raised ghrelin.
- Insulin resistance underlies **acanthosis nigricans**, PCOS and fatty liver.

Signs
- Velvety dark skin in neck folds = **acanthosis nigricans**.
- Knee pain and limp in an obese teenager: **SCFE**; also obstructive sleep apnea.

Dx
- Genetic red flags: onset before **5 years**, severe hyperphagia, developmental delay or dysmorphism.
- Metabolic syndrome: at least 3 of 5 criteria.

Rx
- Family-based lifestyle change first.
- **GLP-1 receptor agonists** (semaglutide) licensed for obesity from age 12.

I-39 Causes of short stature

Def
- Short stature = height below **−2 SD**.

Cause
- Normal variants: familial, constitutional delay.
- Most frequent endocrine cause: **hypothyroidism**; also growth hormone deficiency, Turner, Prader-Willi, Fanconi anaemia.

Patho
- Growth hormone and thyroid hormone increase bone growth; GH raises **IGF**.
- Pathological short stature = low height velocity, crossing centiles **downward**.

Signs
- Familial: bone age **equal to** chronological age.
- Constitutional delay: bone age **delayed**, final height **normal**.

Dx
- Every unexplained short girl needs a **karyotype** to exclude Turner syndrome.
- GH deficiency needs **two** GH stimulation tests.

Rx
- Treat the cause: levothyroxine for hypothyroidism.
- **Growth hormone** for deficiency, Turner and Prader-Willi; constitutional delay: reassurance.

I-40 Muscle weakness, paralysis and other myopathies

Def
- Localise the weakness: horn cell, nerve, neuromuscular junction or muscle.

Cause
- SMA: deletion of **SMN1**; Duchenne: X-linked absence of **dystrophin**.
- Myasthenia: antibodies to **postsynaptic ACh receptors**; infant botulism: honey spores; GBS.

Patho
- Myasthenia: weakness **worsens** with muscle use; Lambert-Eaton (presynaptic calcium channels) **improves**.
- SMA severity depends on **SMN2** copy number.

Signs
- Tongue fasciculations and areflexia in a floppy infant: **SMA**; infant botulism begins with **constipation**.
- Duchenne: **Gowers sign**; myasthenia: fatigable **ptosis** and diplopia.

Dx
- GBS: CSF **high protein** with **normal cell count**; serial **spirometry**.
- Duchenne: serum CK raised **10–100x**; myotonic dystrophy: EMG then genetic testing.

Rx
- GBS: **IVIG** or plasmapheresis; infant botulism: **human-derived botulism immune globulin**.
- Duchenne: **steroids**; myasthenic crisis: intubation, then prednisone plus IVIG.

I-41 Disorders of coagulation

Def
- Mucocutaneous bleeding suggests **platelets/VWF**; joint and muscle bleeds suggest **clotting factors**.

Cause
- Haemophilia A = factor **VIII**; haemophilia B = factor **IX**.
- Commonest inherited bleeding disorder: **von Willebrand disease**; ITP: IgG against GPIIb/IIIa.

Patho
- Desmopressin increases release of **vWF** from Weibel-Palade bodies.
- ITP antibodies are produced in the **spleen** and can cross the placenta.

Signs
- ITP: scattered petechiae, isolated **thrombocytopenia**, increased bleeding time.
- Severe haemophilia: spontaneous joint bleeds.

Dx
- Haemophilia screen: **APTT** prolonged, PT normal.
- Chronic ITP: low platelets for >1 year.

Rx
- Haemophilia with head trauma, even without fainting: **immediate factor concentrate** and nearest haemophilia centre.
- Inhibitors: **bypassing agent**; prophylaxis **emicizumab**; ITP with only petechiae: observation.

I-42 Behavioural disorders (ADHD, autism spectrum disorder)

Def
- **ADHD**: **hyperactivity**, **impulsivity**, and/or **inattention** in ≥ 2 settings.
- **Autism spectrum** disorder: **poor social interactions**, **repetitive / ritualized behaviors**.

Cause
- ADHD: ~70–80% heritable, prematurity, fetal alcohol/tobacco.
- Autism: genetic; fragile X, Rett, fetal valproate; **not MMR**; more common in **males**.

Patho
- ADHD: weak **dopamine** and noradrenaline signalling → impulsivity.
- Autism: altered connectivity → social cues not rewarding → routines calm.

Signs
- ADHD: onset before **12**, in **two or more** settings, for at least 6 months.
- Autism: **early childhood** onset; poor eye contact, insistence on sameness, stereotypies.

Dx
- ADHD: clinical; exclude hearing, vision, absences, OSA.
- Autism screening at **18** months and **2** years: **M-CHAT-R/F** (16–30 months).
- **Hearing test** before diagnosing autism in non-verbal child; ADOS.

Rx
- ADHD from age 5: **stimulants** ± CBT; **methylphenidate** first-line, then atomoxetine, guanfacine.
- Monitor heart rate, blood pressure, **height and weight** on stimulants.
- Autism: early intensive developmental therapy; risperidone for severe irritability.

I-43 Acid-base disorders

Def
- Disturbed pH; follows the **HCO3 : pCO2 ratio**.
- **SMORE**: Same direction (pH and pCO2) → Metabolic; Opposite → REspiratory.

Cause
- **High anion gap** = acid added: **DKA**, lactate, renal failure, toxins.
- **Normal** anion gap: **diarrhoea**, RTA (**HARDASS**); alkalosis: vomiting, diuretics.
- Pyloric stenosis causes **hypochloraemic, hypokalaemic metabolic alkalosis**.

Patho
- Metabolic acidosis: **decrease** in HCO3 → **hyperventilation** → decreased pCO2.
- Respiratory compensation is **immediate**; kidneys slow (**3–5 days**).
- Pyloric stenosis: HCl loss + aldosterone → **paradoxical aciduria**.

Signs
- Kussmaul breathing, vomiting, lethargy.
- A normal or rising pCO2 in acute severe asthma signals **exhaustion**.

Dx
- Anion gap = **Na − (Cl + HCO3)**, normal about 8–12.
- **Winter's**: expected pCO2 = **1.5 × HCO3 + 8** (±2).
- Metabolic alkalosis: next step **check urine chloride**.

Rx
- Treat the cause: fluids, **insulin for DKA**, dialysis; bicarbonate not routine.
- Pyloric: **chloride + volume** first.
- Operate once Cl^- and HCO3 normal.

I-44 Hypo- and hypernatremia

Def
- Abnormal plasma sodium → water shifts into (low) or out of (high) brain cells.

Cause
- Low: **SIADH + hypotonic fluids**, diarrhoea and vomiting, salt-wasting CAH, nephrotic, heart failure.
- High: gastroenteritis, **poor breastfeeding** in week 1, diabetes insipidus, bad formula.

Patho
- In hyponatraemia water moves **into** brain cells, causing cerebral oedema.
- High Na^+ → brain shrinks → bridging veins tear → haemorrhage.

Signs
- Low: headache, vomiting, **seizures**; neonate low Na^+ + high K^+ → CAH.
- High: high-pitched cry, doughy skin, seizures; shock late.

Dx
- Plasma and urine osmolality, urine Na, volume status.
- SIADH: low plasma osmolality, inappropriately **concentrated** urine, euvolaemia, urine Na >30.

Rx
- Hyponatraemic seizures: bolus of **3% saline**; SIADH → fluid restriction; **isotonic** maintenance fluids.
- Too-rapid correction of chronic hyponatraemia causes **osmotic demyelination**; raise ≤8–10 mmol/L/24 h.
- Hypernatraemia: corrected over **48** hours, no more than **0.5** mmol/L per hour; 0.9% saline for shock.

I-45 Hypo- and hyperkalaemia

Def
- Abnormal K^+; K^+ sets the **resting membrane potential**.

Cause
- High: AKI, CAH, DKA, tumour lysis, ACE inhibitors, heparin, trimethoprim.
- Commonest cause of high K in a well infant: **a haemolysed sample**.
- Low: diarrhoea and vomiting, pyloric stenosis, diuretics, insulin in DKA, refeeding.

Patho
- High K^+ → inactivates fast Na^+ channels → slow conduction, fast repolarisation.
- Salbutamol and insulin lower K by stimulating the **Na/K-ATPase**.
- Low K may not correct until **magnesium** is replaced.

Signs
- Earliest ECG sign of hyperkalaemia: **peaked T waves** → wide QRS → sine wave.
- Low: weakness, ileus, flat T waves, **U waves**.

Dx
- First exclude **haemolysed heel-prick**; repeat free-flowing venous sample.
- Low K: urine K (>20 = renal loss), acid-base.

Rx
- High with ECG changes: IV **calcium gluconate 10%** first (protect).
- Shift: **salbutamol**, insulin-glucose, sodium bicarbonate; then resin, loop diuretic, dialysis.
- Low: oral KCl; IV in fluids, never bolus; replace Mg.

I-46 Causes of polyhydramnios and oligohydramnios

Def
- Abnormal amniotic fluid: **poly** AFI >24–25 cm; **oligo** AFI under 5 cm.

Cause
- Poly: **oesophageal and duodenal atresia**, maternal diabetes (fetal **osmotic diuresis**), TTTS **recipient**.
- Oligo: no urine (renal agenesis, ARPKD), blocked (PUV), leak (**PPROM**, commonest), FGR.
- ACE inhibitors / ARBs in utero may cause **oligohydramnios**.

Patho
- After 20 weeks fluid comes mainly from **fetal urine**, removed by **fetal swallowing**.
- Early oligohydramnios → no lung distension → lethal **pulmonary hypoplasia**.

Signs
- Poly: preterm labour, cord prolapse, **placental abruption** after rapid decompression.
- Oligo: **Potter** sequence (pulmonary hypoplasia, twisted facies, extremity deformities, renal agenesis).
- Also growth restriction, cord compression.

Dx
- Ultrasound AFI; double bubble (duodenal atresia), keyhole bladder (PUV).
- Oligo before ~24 weeks → most severe lung hypoplasia.

Rx
- Polyhydramnios newborn: **NG tube** before first feed to exclude oesophageal atresia.
- Severe polyhydramnios: **amnioreduction** or **indomethacin**; term, mild, asymptomatic: expectant.
- Oligohydramnios: treat cause; pulmonary hypoplasia is the killer.

I-47 Neonatal and infant resuscitation

Def
- Newborn arrest is **respiratory** — aerate the lungs.
- Best indicator of success: the **heart rate**.

Cause
- Hypoxia → primary apnoea → gasping → terminal apnoea.
- Meconium aspiration: **term and post-term** infants.

Patho
- First breaths clear lung fluid, create FRC → pulmonary vascular resistance falls.
- Meconium **inactivates surfactant**; bradycardia = hypoxia until proven otherwise.

Signs
- Assess **tone, breathing, heart rate**; HR under 100 or not breathing → act.
- Apgar at 1 and 5 minutes: Appearance, Pulse, Grimace, Activity, Respiration.

Dx
- Stethoscope, then ECG/**pre-ductal SpO_2**.
- Targets: 70–75% at 3 min, 80–85% at 5 min, 85–95% at 10 min.
- PPV with brighter left chest transillumination → **left pneumothorax**.

Rx
- Dry, warm; **5** inflation breaths, each **2–3** seconds; term babies start in **air (21%)**.
- Compressions **3:1** if HR under **60**/min after **30 s** effective ventilation.
- Adrenaline **20 µg/kg** via UVC; HIE: **33–34 °C** for **72 h**, within 6 h.

I-48 Pediatric resuscitation

Def
- Paediatric arrest = end of progressive hypoxia or shock.
- Key to survival: **elapsed time to effective resuscitation**.

Cause
- Respiratory failure or shock → tiring → hypoxia → bradycardia → asystole.
- Non-shockable rhythms (asystole, PEA) are commonest.

Patho
- **Paradoxical breathing**: chest moves **inward**, abdomen **outward** during inhalation.
- PEA: organized rhythm without a palpable pulse.

Signs
- ABCDE: stridor, recession, grunting, head bobbing, CRT, AVPU, glucose.

Dx
- Pulse oximetry not informative in **carbon monoxide intoxication**, **methaemoglobinaemia**, **severe anaemia**, cardiac failure, cold extremities.
- SpO_2 ignores ventilation → capnography or blood gas.

Rx
- Conscious, breathing: face mask with **reservoir**, high flow **10–15 l/min**.
- BLS: **5 rescue breaths**, then **15:2**; asystole/PEA: CPR + adrenaline **10 µg/kg** IV/IO every 3–5 min.
- VF / pulseless VT: unsynchronized shock **4 J/kg**; **amiodarone** after 3rd and 5th shock.

I-49 Fever, management of fever, antipyretics, pain relief

Def
- Fever: cytokine-driven rise in hypothalamic **set point**.

Cause
- Microbial products → IL-1, IL-6, TNF-α release.
- Early-onset neonatal sepsis occurs within **72 hours** of birth.

Patho
- Cytokines induce **PGE_2** in the hypothalamus, raising the set point → shivering, vasoconstriction.
- Antipyretics block COX; fever itself not harmful.

Signs
- NICE red: child under **3 months** with temperature ≥38 °C, mottled, grunting, neck stiffness.
- Amber: 3–6 months with ≥39 °C, fever ≥5 days, CRT ≥3 s.

Dx
- **NICE traffic light** for under-5s; red → septic work-up.
- Neonatal sepsis: blood culture, serial CRP, **lumbar puncture**; antibiotics within **1 hour**.
- Pain scales: FLACC (preverbal), Wong–Baker (4+), numeric (8+).

Rx
- Only if distressed: paracetamol **15 mg/kg** every 4–6 h, ibuprofen **10 mg/kg** three times daily.
- Antipyretics **do not** prevent febrile seizures; aspirin avoided under 16 (**Reye syndrome**).
- Neonatal sepsis: **ampicillin + gentamicin**; codeine contraindicated under **12** (CYP2D6).

I-50 Causes of shock in childhood

Def
- Inadequate oxygen delivery; types hypovolaemic, distributive, cardiogenic, obstructive, dissociative.

Cause
- Gastroenteritis, haemorrhage, **sepsis**, anaphylaxis, myocarditis, SVT, tamponade.
- Neonate 1–2 weeks with weak femoral pulses: duct-dependent lesion.

Patho
- Children compensate with tachycardia + vasoconstriction until 30–40% volume lost.
- **Hypotension** is a late sign of shock; **tachycardia** is early.

Signs
- Tachycardia (earliest), CRT >2 s, cool mottled peripheries, oliguria, lactic acidosis.
- Bradycardia + hypotension = decompensation / pre-arrest.

Dx
- Cultures, gas, lactate, glucose.
- Hypotension in children 1–10 y: systolic below **70 + 2 × age** mmHg.
- Reassess HR, CRT, BP, liver edge, crackles after each bolus.

Rx
- Septic shock: O2, IV/IO, fluid bolus **20 ml/kg** body weight; antibiotics within 1 hour.
- Fluid-refractory: start **adrenaline or noradrenaline**; TSS: clindamycin **suppresses toxin synthesis**.
- Anaphylaxis: IM adrenaline first (**150 µg** under 6); duct-dependent: **prostaglandin E**.

I-51 Neonatal birth injuries

Def
- Injuries from delivery, determined by anatomical layer and nerve root.

Cause
- Risk factors: **difficult** deliveries, shoulder dystocia, macrosomia, vacuum, forceps.
- Maternal hyperglycemia → macrosomia.

Patho
- **Caput succedaneum** crosses suture lines; **cephalhaematoma** is subperiosteal and does not.
- **Subgaleal** haemorrhage: injury to **emissary veins**, between aponeurosis and periosteum.
- Erb: lateral traction on neck during delivery.

Signs
- Cephalhaematoma: swelling limited to one cranial bone; resorption can cause **jaundice**.
- Subgaleal haemorrhage: hypovolaemic shock, typically after vacuum delivery.
- **Erb** C5–C6 waiter's tip; **Klumpke** C8–T1 claw hand ± Horner; clavicle: crepitus, asymmetric Moro.

Dx
- Subgaleal: monitor head circumference, haemoglobin.
- Erb palsy with tachypnoea suggests **phrenic nerve** palsy (C3–C5).
- Clavicle fracture: check **subclavian artery**, **brachial plexus**.

Rx
- Shoulder dystocia: McRoberts (flex hips) + **suprapubic pressure**.
- Commonest fracture: **clavicle**, heals rapidly; plexus palsy resolves within 3 months (80%).
- No biceps recovery by **3-6 months** → **surgery**; subgaleal: volume and blood replacement.

I-52 Poisoning in childhood (toxicology)

Def
- Ingestion: accidental in toddlers (1–4 y), deliberate in adolescents.

Cause
- Medicines, cleaners, **button batteries**, paracetamol, iron, TCAs, opioids, caustics.
- Lead poisoning: homes built before **1978**.

Patho
- Paracetamol: toxic metabolite **NAPQI**, detoxified by **glutathione** until exhausted.
- Iron: corrosive, then mitochondrial poison; TCA: Na^+ channel block → wide QRS.

Signs
- Toxidromes: anticholinergic, cholinergic, opioid (pinpoint pupils), sympathomimetic.
- Paracetamol day 1 near asymptomatic; lead: **bluish** gum-tooth line.
- Aspirin: early pure respiratory alkalosis; late mixed with metabolic acidosis.

Dx
- **ABCDE + glucose**; dose in mg/kg, assume worst case.
- Paracetamol level on nomogram from **4 hours**; iron X-ray + serum iron; battery CXR halo.
- Lead: capillary level raised → measure **venous lead level**.

Rx
- **Never induce vomiting**; charcoal only within **1 hour**, does not bind iron, lithium, alcohols.
- Antidotes: **N-acetylcysteine** (paracetamol); **desferrioxamine** (iron); **bicarbonate** (TCA); **naloxone** (opioid).
- Oesophageal battery or coin: **urgent endoscopy**; lead moderate **DMSA**, severe dimercaprol + EDTA.

I-53 Child abuse and shaken baby syndrome

Def
- Physical, sexual, emotional abuse or **neglect**; neglect is the commonest form of maltreatment.

Cause
- Most significant parental risk factor: being abused themselves; abuser usually primary caregiver.
- Shaken baby: infants under 1 y, peak 2–4 months; sexual abuse peak 9–12 years.

Patho
- Shaking tears the **bridging veins**, causing subdural haemorrhage.
- Vitreous pulls retina → retinal haemorrhages; grip → posterior rib fractures.

Signs
- History: delay, changing story, injury not fitting development.
- Any bruise in a **non-mobile infant**; scalds with sharp lines of demarcation.
- Abusive head trauma: **subdural haemorrhage**, **retinal haemorrhages** and encephalopathy.

Dx
- CT then MRI, **skeletal survey** repeated at **11–14 days**, ophthalmology, coagulation.
- Most specific: **posterior rib** and **classic metaphyseal (corner)** fractures.

Rx
- Suspected abuse: contact **child protective services** first; abuse and neglect both reported.
- Document exactly, keep child safe — admit if needed; welfare overrides confidentiality.

I-54 Airway foreign bodies in children, aspiration

Def
- Inhaled object in larynx, trachea or bronchus.

Cause
- Toddlers 1–3 y: no molars, eat while running; nuts classic.
- **Right** lung commonest: its bronchus is wider and more vertical.

Patho
- Partial block = **check valve** → air trapped on **expiration**.
- Complete block → atelectasis; nuts radiolucent.

Signs
- Choking → symptom-free interval → cough, unilateral wheeze, recurrent pneumonia.
- Laryngotracheal: stridor, cyanosis.

Dx
- **Holzknecht sign** = **pendulum movement of the mediastinum** = airway foreign body.
- **Expiratory and inspiratory chest radiograph** or fluoroscopy; normal X-ray never excludes.

Rx
- Choking infant under 1 year: 5 back blows then 5 **chest thrusts** (not abdominal).
- Over 1 y: abdominal thrusts.
- Definitive: **rigid bronchoscopy**; symptomatic swallowed coin: emergency endoscopy.

I-55 Pediatric imaging techniques

Def
- Imaging chosen by **ALARA** (as low as reasonably achievable): justification, optimisation, non-ionising first.

Cause
- Children radiosensitive: dividing cells, longer remaining life, organs close to beam.

Patho
- Blood echogenic, CSF anechoic → cranial US via anterior fontanelle.
- Preterm IVH arises in the **germinal matrix**.

Signs
- Thymus shadow visible **till 3 years of age**; absent → **DiGeorge** or **SCID**.
- Mediastinal shadow narrowing trachea → mass, not thymus.

Dx
- Ultrasound first, MRI next; cranial US: **brain haemorrhage** and **hydrocephalus**.
- Urinary: US, MCUG, **DMSA** 4–6 months after infection shows scars, MAG3.

Rx
- Polytrauma → **acute CT**; head trauma: non-contrast CT if GCS ≤14 or skull fracture.
- Suspected stroke with normal CT → MRI / MR angiography.
- US for pyloric stenosis, intussusception; upper GI contrast for bilious vomiting.

Section II · Thesis line

II-1 Standard and recommended immunizations for children and adolescents

Def
- Vaccines prime **memory B and T cells** → fast high-affinity IgG on real infection.

Cause
- **Live**: MMR, varicella, rotavirus, BCG; avoid in SCID, pregnancy.
- Hungary: **BCG** to all newborns; varicella mandatory since **2019**.

Patho
- **Maternal antibodies** neutralise MMR earlier → given after 12 months.
- Conjugate vaccines work in infants because the protein carrier recruits **T-helper cells**.

Signs
- MMR fever and rash appear **7–12 days** after vaccination.
- Varicella vaccine rash with few lesions = vaccine-strain replication.

Dx
- Only universal contraindication: **anaphylaxis** to a previous dose or component.
- DTaP: anaphylaxis → no future doses; encephalopathy → hold pertussis.
- Not contraindications: mild illness; preterms use **chronological** age.

Rx
- At **2, 4, 6 months**: DTaP, IPV, PCV15, rotavirus; hepatitis B at birth, 2, 6 months.
- **MMR** and varicella at **1 year** and 4 years; annual influenza from **6 months**.
- Age 11: Tdap, meningococcal, HPV through **26**.

II-2 Normal psychomotor development in childhood (different age groups). Child growth percentiles. Anthropometry.

Def
- Development in 4 domains: gross motor, fine motor/vision, speech/hearing, social.

Cause
- Any **loss of skills** is an urgent red flag.
- Hand preference before **12 months** is a red flag.

Patho
- Myelination → cephalocaudal and proximal→distal progression.
- Primitive reflexes normal till **6-12 months**; persistence suggests **upper motor neuron lesion** (e.g. CP).

Signs
- Social smile 6 weeks; lifts head **2 months**; rolls over **6 months**.
- Sits alone, stranger anxiety **9 months**; pulls to stand 9–12 months.
- Walks **12–18 months** (limit **18 months**); kicks ball 2 years.

Dx
- Weight, length, head circumference on centile charts; judge trajectory.
- Growth failure: weight below **5th** percentile or crossing two major lines.
- Failure to thrive: **weight** → **height** → **head circumference**.

Rx
- Birth weight regained by **2** weeks, doubles by **4–5 months**, triples by **12 months**.
- Length rises **50%** by 1 year; growth velocity is the most sensitive marker.

II-3 Anthropometry of a newborn. Intrauterin growth restriction (IUGR). Characteristics of a premature baby.

Def
- Term birth weight **2500–4000 g**, head circumference ~**35 cm**; preterm under 37 weeks.
- **SGA** = birth weight below the **10th centile** for gestation.

Cause
- Leading cause of low birth weight in developed countries: smoking.
- **Asymmetric** FGR usually **placental insufficiency**; fetal: trisomies, TORCH.

Patho
- Asymmetric (late, placental): head spared; symmetric (1st trimester infection): global growth lag.
- Chronic hypoxia → EPO → polycythaemia (haematocrit over **65**%) → hyperviscosity, jaundice.

Signs
- IUGR: **hypoglycaemia**, **hypothermia**, **polycythaemia**.
- Preterm: thin red skin, lanugo, soft ears, hypotonia.
- RDS (**surfactant deficiency**), IVH, **PDA**, NEC, ROP.

Dx
- Low birth weight under **2500** g; VLBW under 1500; ELBW under 1000 g.
- Gestation by dating ultrasound, New Ballard score.
- FGR: Doppler of umbilical artery; absent/reversed end-diastolic flow → urgent delivery.

Rx
- Before birth: maternal **glucocorticoids** (24–34 wk) + MgSO4; after birth: artificial **surfactant**.
- NEC: NPO, NG tube, IV fluids, antibiotics; surgery for perforation.
- ROP: **laser photocoagulation**.

II-4 Nutrition of the healthy infant: breastfeeding and weaning. Breastmilk composition. Energy requirements.

Def
- Breastfeed **on demand**, **exclusively for 6 months**, continue to 2 years.

Cause
- Contraindications: infant **galactosaemia**; maternal HIV, untreated tuberculosis, varicella.
- Hepatitis B and mastitis are not.

Patho
- Suckling → **prolactin** (synthesis) + **oxytocin** (let-down); supply follows demand.
- Breast milk protein is **one third** of cow's; colostrum contains **IgA**.

Signs
- Adequacy: ≥5 wet diapers per day at 5 days; loses up to **10%** of birth weight.
- Birth weight regained by **2** weeks.
- Breast milk jaundice: **β-glucuronidase**.

Dx
- Cow's milk first given after **12 months** → else iron deficiency, renal solute load.
- Energy ~100–110 kcal/kg/day early; fluid ~150 ml/kg/day.
- Kwashiorkor: oedema, swollen abdomen; marasmus: wasting.

Rx
- Solids not before **17 weeks**, ~6 months; no honey under 12 months (infant botulism).
- Breastfeeding jaundice: feed more often (q2h); milk jaundice: continue.
- All: **vitamin D 400 IU** daily + vitamin K 1 mg IM at birth.

II-5 Psychosomatic diseases. Anorexia nervosa. Bulimia nervosa.

Def
- Psychosomatic = real symptoms driven by stress; commonest: **functional abdominal pain**.
- **Anorexia**: restriction, low weight; **bulimia**: binge + compensation weekly for **3 months**.

Cause
- Stress: school, bullying, family conflict, abuse; separation anxiety → school avoidance.
- Anorexia peaks in adolescent girls.

Patho
- Stress → HPA + brain–gut axis → visceral hypersensitivity.
- Anorexia: loss of pulsatile GnRH → low LH/FSH/estrogen → amenorrhea.
- Vomiting → HCl loss + aldosterone.

Signs
- AN: **bradycardia**, hypotension, **lanugo**, amenorrhea, mitral valve prolapse.
- BN: parotid hypertrophy, enamel erosion, **Russell sign** (knuckle calluses).
- Conversion: **la belle indifférence**.

Dx
- Exclude red flags; avoid endless tests.
- BN: **hypokalaemic hypochloraemic metabolic alkalosis**.
- Admit AN: HR under **40**, SBP under 80, temp under 35.5 °C, %mBMI under 70%, syncope.

Rx
- **Family-based therapy** first-line for adolescent AN; bulimia: psychotherapy ± SSRIs (**fluoxetine**).
- Refeeding: key electrolyte disturbance **hypophosphataemia**.
- Somatic symptoms: monthly visits with a single physician; screen self-harm.

II-6 Newborn screening tests. Preventive care.

Def
- Screening for diseases silent at birth, harmful, treatable (Wilson–Jungner).

Cause
- Hungary screens **26** congenital metabolic diseases: PKU, hypothyroidism, galactosaemia, biotinidase + tandem-MS.
- Aminoacidopathies, fatty-acid oxidation defects (MCAD), organic acidurias.

Patho
- Placenta clears toxic metabolites in utero → feeding unmasks the defect.
- PKU: deficiency of **phenylalanine hydroxylase**.
- Galactosaemia: jaundice, cataract, E. coli sepsis after milk feeds.

Signs
- MCAD: fasting → **hypoketotic** hypoglycaemia, sudden death.
- PKU: fair skin, musty odour.

Dx
- Heel prick at **48–72 hours** → **tandem mass spectrometry** on dried blood spot, TSH, enzyme assays.
- Positive screen ≠ diagnosis → urgent recall + confirmation.
- Also hearing, pulse oximetry, hips, red reflex; vision every well-child visit (amblyopia).

Rx
- PKU low-phe diet; galactosaemia stop lactose; MCAD avoid fasting.
- All newborns receive **vitamin K** to prevent haemorrhagic disease; vitamin D 400 IU.
- Hepatitis B mother: immune globulin + vaccine within 12 hours.

II-7 Sudden infant death syndrome. Infantile and paediatric mortality data.

Def
- **SIDS** is sudden death under 1 year, a diagnosis of exclusion after **autopsy**, scene investigation, history.

Cause
- **Smoking**, **bed-sharing** (sofa, alcohol), prone/side sleep, soft bedding, overheating, prematurity.
- **Low birth weight**: **less than 2500 grams (5.5 pounds)**.

Patho
- Triple-risk model: vulnerable infant + **critical developmental period** + **external stressor**.
- Prone → rebreathes CO2 → hypoxia → vulnerable infant **fails to arouse**.

Signs
- SIDS peaks at **2–4 months** of age, during sleep.
- More often boys, winter, preterm, low birth weight.

Dx
- Unexpected death → full **autopsy** + skeletal survey, metabolic screen; exclude infection, MCAD, abuse.
- Infant deaths: **perinatal conditions (prematurity)** and **congenital malformations**; over 1 year, accidents **36–40%**.

Rx
- "**Back** to sleep" (**supine**), avoid bed **sharing**, **smoking** cessation, pacifier during sleep.
- Room-share in own cot, firm mattress; breastfeed, immunise; bereavement support.
- Accident prevention: car seats, pool fencing, locked medicines.

II-8 Clinical symptoms of infections, laboratory diagnostics, normal parameters.

Def
- Infection signs are mostly the host response: fever, tachycardia, tachypnoea, poor perfusion.

Cause
- Any infant under **3 months** with temperature **≥38 °C** is high risk (NICE red); mottled skin, non-blanching rash.

Patho
- IL-1, IL-6, TNF induce **PGE2** in hypothalamus → raised set-point → fever.
- Vasoconstriction → mottling, prolonged CRT → early shock.

Signs
- Grunting, RR >60, bulging fontanelle, neck stiffness, petechiae, lethargy.
- Tachycardia out of proportion to fever → **shock**.

Dx
- **CRP** peaks at **36–48 hours** (early CRP may reassure falsely); **procalcitonin** rises within **2–4 hours**, more bacterial-specific.
- Normal CSF WBC <5/µL; bacterial meningitis → **low** glucose.

Rx
- Blood cultures **before antibiotics**; **number of cultures and volume of blood** decide sensitivity.
- Infant <3 months ≥38 °C → **full septic screen**.
- Shocked child: cultures → antibiotics within the hour; **never delay for LP**.

II-9 Antimicrobial therapy

Def
- Choose by **bug (age, site), penetration, severity, local resistance**; culture first.
- Review antibiotic therapy at **48–72 hours** to narrow or stop.

Cause
- Neonate: **GBS, E. coli, Listeria**. Older: pneumococcus, meningococcus, Hib.
- **Listeria** is resistant to cephalosporins, so neonatal cover needs amoxicillin.

Patho
- β-lactams inhibit **penicillin-binding protein transpeptidases**; **β-lactamase** hydrolyzes the β-lactam ring.
- Aminoglycosides show **concentration**-dependent killing; β-lactams show **time**-dependent killing.

Signs
- **Chloramphenicol** → **gray baby syndrome**; **tetracyclines** → permanent tooth staining.
- **Ceftriaxone** in neonates displaces **bilirubin** and precipitates with **calcium**.
- **Aminoglycosides**: acute tubular necrosis, ototoxicity; **fluoroquinolones**: tendon rupture, cartilage damage.

Dx
- **Blood culture before first dose** → review at **48–72 hours**.
- Narrow to culture, IV → oral when afebrile, shortest effective duration.

Rx
- Neonatal sepsis: **ampicillin + gentamicin**; suspected meningitis: **ampicillin + cefotaxime**. Beyond neonate: **ceftriaxone**.
- CAP: oral **amoxicillin**; **MSSA**: nafcillin, oxacillin, dicloxacillin; newborn gonorrhea: ceftriaxone.
- No antibiotics for colds, bronchiolitis, viral wheeze.

II-10 Borreliosis (Lyme-disease) and parotitis epidemica (mumps)

Def
- **Lyme**: spirochete **Borrelia burgdorferi**, **Ixodes** tick (Europe: **Ixodes ricinus**). Mumps = paramyxovirus parotitis.

Cause
- Lyme: Northeast and upper Midwest US; needs >24–36 h attachment. Mumps: incubation **16–18 days**.

Patho
- Spirochetes spread centrifugally → **expanding erythema** with central clearing.
- Mumps: viraemia seeds glands + CNS → explains every complication.

Signs
- Lyme **stage 2**: **facial palsy**, meningitis, carditis (**AV block**); **stage 3**: encephalopathy, arthritis.
- Mumps: painful **parotid swelling**; **orchitis** in **post-pubertal** males; meningitis, pancreatitis, deafness.

Dx
- **Erythema migrans** is clinical: **tick bite** + **growing erythema more than 5 cm**; no serology.
- Early disseminated/late: **ELISA then Western** if confirmation needed. Mumps: raised **amylase**, RT-PCR/IgM.

Rx
- Erythema migrans: young child **amoxicillin**; older **doxycycline** (US any age, 10 days).
- Neuro/high-grade block: IV **ceftriaxone**. Prophylaxis if tick attached **≥36 hours**, within **72 hours**.
- Mumps: supportive; **MMR** ×2.

II-11 Clinical manifestations of Epstein-Barr virus infection.

- **Def**
  - **Infectious mononucleosis** ("kissing disease") is caused by **EBV (HHV-4)**: fever, pharyngitis, nodes, splenomegaly.
- **Cause**
  - Most common in **teenagers**; spread by **saliva** or respiratory secretions; young children mild or silent.
  - EBV: **Burkitt** lymphoma, nasopharyngeal carcinoma.
- **Patho**
  - EBV first infects **oropharyngeal** epithelium, then B cells via **CD21**.
  - Reactive **CD8 T cells** = Downey type II atypical lymphocytes.
  - Polyclonal B activation → **heterophile antibodies**.
- **Signs**
  - Fatigue, fever, **posterior cervical** lymphadenopathy, **pharyngitis**, **hepatosplenomegaly**; grey-white tonsillar exudate.
  - Raised **LFTs** (hepatitis) and lymphocytosis.
  - **Amoxicillin** in EBV causes a maculopapular rash — not true penicillin allergy.
- **Dx**
  - Atypical lymphocytes >10% on smear; **Monospot** often negative under **4–5 years** → **VCA IgM**.
  - Cytopenias or blasts → exclude **leukaemia**.
- **Rx**
  - **Supportive**: rest, fluids, paracetamol (acetaminophen); aciclovir useless.
  - Steroids only for **airway obstruction, haemolysis or thrombocytopenia**.
  - No contact sport for 3–4 weeks because of **splenic rupture**.

II-12 Measles, rubella, scarlet fever.

- **Def**
  - **Measles** (**Paramyxoviridae**), **rubella** (togavirus), **scarlet fever** (group A strep exotoxin).
- **Cause**
  - Measles most contagious (R0 12–18), aerosol; scarlet fever 2–10 years; rubella in **first** trimester → congenital rubella.
- **Patho**
  - Measles kills memory cells → **immune amnesia**; rubella crosses placenta → organogenesis damage.
  - GAS **pyrogenic exotoxins** = superantigens → capillary dilation.
- **Signs**
  - Measles: 3 Cs + **Koplik spots** → rash day 3–4, ears downward.
  - Rubella: **postauricular and suboccipital** nodes; **CRS**: **cataract**, **sensorineural deafness**, PDA (**80%** foetal injury).
  - Scarlet: **sandpaper rash**, strawberry tongue, circumoral pallor, desquamation.
- **Dx**
  - Measles **pneumonia** is the commonest death; SSPE years later.
  - Scarlet: throat swab; **Rapid Strep** has low sensitivity, culture confirms; ASO later.
- **Rx**
  - Measles: supportive + **vitamin A**; contacts MMR ≤72 h or Ig ≤6 days.
  - Scarlet fever: **phenoxymethylpenicillin 10 days** (US also amoxicillin) → prevents rheumatic fever.
  - Prevent: **MMR** ×2; avoid if **pregnant** or **immunocompromised**.

II-13 Helminthic and protozoan infections.

- **Def**
  - Worms (helminths) and single-cell parasites (protozoa) in children.
- **Cause**
  - **Enterobius** (commonest Europe), Ascaris, **Toxocara** (puppies), Giardia, Entamoeba, Toxoplasma, malaria.
  - **Giardia** is **fecal-oral** (daycare, camping streams); **Cryptosporidium** in HIV/AIDS.
- **Patho**
  - Tissue-migrating worms → Th2 / **IL-5** → **eosinophilia**.
  - Giardia flattens villi → fat/lactose **malabsorption**; **Entamoeba** trophozoites invade colon.
- **Signs**
  - Threadworm: **nocturnal perianal itch**. Toxocara: **leukocoria** (ocular larva migrans), mimics retinoblastoma.
  - Giardia: bloating, foul-smelling **floating, fatty** stools. Entamoeba: **bloody** diarrhea, **right lobe** liver abscess.
  - Congenital toxo: **chorioretinitis**, **hydrocephalus**, **diffuse intracranial calcifications**. Malaria: fever, anemia, splenomegaly.
- **Dx**
  - **Sticky tape** (scotch tape) test; Giardia: **trophozoites or cysts in stool**.
  - Entamoeba: trophozoites with endocytosed **RBCs**; biopsy with **flask-shaped lesions**.
  - Fever after endemic travel = **malaria** → urgent thick/thin films.
- **Rx**
  - Threadworm: **mebendazole**, repeat after 2 weeks, treat the **whole household**.
  - Giardia/Entamoeba: **metronidazole** (or tinidazole); Toxocara: albendazole.
  - Congenital toxo: **pyrimethamine + sulfadiazine + folate** (leucovorin) for 12 months.

II-14 Clinical manifestations of Herpes simplex virus, Parvovirus, Coxsackie virus infection.

- **Def**
  - **HSV** (vesicles, latency), **parvovirus B19** (**single-stranded DNA**, fifth disease), **Coxsackie** (enterovirus).
- **Cause**
  - HSV: direct contact; neonatal from **genital tract lesions**. Enteroviruses faecal-oral, summer/autumn.
- **Patho**
  - HSV stays latent in sensory ganglia (HSV-1: **trigeminal ganglion**) → recurrence.
  - B19 infects erythroid progenitors via **P antigen** → **aplastic crisis** in sickle cell or spherocytosis.
- **Signs**
  - HSV: **herpes labialis**, **dendritic corneal ulcers**; **eczema herpeticum** = punched-out erosions + fever.
  - B19: **slapped cheek**, lacy rash; fetal **hydrops fetalis**.
  - Coxsackie A: **hand, foot and mouth disease**; B: **myocarditis**, **Bornholm disease**.
- **Dx**
  - HSV encephalitis: **temporal** lobe, CSF **HSV PCR**, MRI; **Tzanck smear** also detects HSV.
  - Aplastic crisis: falling Hb + **reticulocytopenia**.
  - Pregnant, B19 antibody positive: reassurance; >20 weeks **weekly ultrasounds**.
- **Rx**
  - Suspected HSV encephalitis → **IV aciclovir immediately**; eczema herpeticum: **aciclovir**.
  - Neonatal HSV: high-dose IV aciclovir **14–21 days**; genital HSV in labor: **cesarean delivery**.
  - Aplastic crisis: transfusion; fetal anemia → **intrauterine RBC transfusion**.

II-15 Varicella infection, herpes zoster

Def
- **Varicella** (chickenpox) = primary **VZV**; **herpes zoster** = its reactivation, unilateral dermatomal rash.

Cause
- **Airborne**; incubation **10–21** days; infectious 1–2 days before rash until all lesions **crusted**.

Patho
- Repeated viraemia waves → **crops**: lesions in different stages of healing together.
- VZV becomes latent in **dorsal root (sensory) ganglia** → falling T-cell immunity → zoster.

Signs
- Itchy **centripetal** rash (trunk first), headache, fever. Zoster: painful, does not cross midline.
- Fever returning after **day 3–4** → **invasive group A streptococcus**. **Hutchinson sign** → eye.

Dx
- Clinical (Tzanck smear). Maternal varicella before 20 weeks → **congenital varicella syndrome**.
- Maternal rash −5 to +2 days of delivery → severe neonatal varicella.

Rx
- **Paracetamol**, emollients; avoid **ibuprofen**, never **aspirin** (Reye).
- **Aciclovir** for neonates, immunocompromised, severe disease (IV if disseminated); **VZIG** for high-risk contacts.
- **Live attenuated** vaccine: **2 doses** (1 and 4 y); post-zoster neuralgia: TCAs, gabapentin.

II-16 Skin infections

Def
- Bacterial, viral, fungal skin infections; **depth decides the picture**.

Cause
- **S. aureus** and **GAS** via breaks (eczema, bites, varicella); tinea capitis in young **African American children**.

Patho
- S. aureus **exfoliative toxin** cleaves **desmoglein-1** in the granular layer → bullous impetigo, **SSSS**.
- Necrotising fasciitis thromboses perforators → deep death, mild skin.

Signs
- Impetigo: **honey-coloured (golden)** crusts, well. **Erysipelas**: upper dermis, sharp raised edge; **cellulitis** ill-defined.
- SSSS: Nikolsky +, **spares mucous membranes** (desmoglein-3) — unlike TEN.
- Necrotising fasciitis: **pain out of proportion**, bullae, crepitus, shock.

Dx
- **Proptosis**/painful eye movement → **orbital cellulitis** → CT.
- SSSS blister fluid sterile → culture the focus. Tinea: **Wood lamp** for Microsporum; scabies: papules and burrows.

Rx
- Impetigo: topical (**mupirocin**), or oral flucloxacillin (US cephalexin) if widespread/bullous.
- SSSS/orbital: admit, IV antibiotics. Abscess: **incision and drainage**. Tinea capitis: **oral** antifungal.
- Necrotising fasciitis: **urgent surgery + clindamycin**, consider IVIG.

II-17 Staphylococcus aureus infections. Toxic shock syndrome

Def
- **S. aureus**: pyogenic invasive disease; **TSS** = **superantigen** cytokine storm.

Cause
- Nasal carriage ~1/3. TSS: tampons, retained gauze, burns, wounds, nasal packing, varicella lesions.
- Most common cause of **osteomyelitis**.

Patho
- **Coagulase** builds a fibrin wall → abscesses, furuncles, carbuncles.
- **TSST-1** superantigen binds MHC II and TCR Vβ → T-cell activation (up to **20%**).
- **MRSA** alters **penicillin-binding proteins**; S. aureus makes β-lactamase.

Signs
- TSS: high fever, **erythroderma**, **hypotension**, multi-organ failure.
- **Desquamation** of palms/soles **1–2 weeks** later.
- Food poisoning: vomiting **1–6 hours** after eating (preformed enterotoxin).

Dx
- Bacteraemia → search **bone, joint, heart** focus + repeat cultures.
- Staph TSS blood culture usually negative; strep TSS positive (toxic shock-like syndrome: **GAS**).

Rx
- TSS: **fluids**, vasopressors, **source removal**; β-lactam + **clindamycin** (suppresses toxin synthesis); IVIG if severe.
- **MSSA**: anti-staph penicillins (nafcillin, oxacillin, dicloxacillin), cefazolin; drain pus.
- **MRSA** → **vancomycin**, linezolid; **daptomycin** for bacteremia; ceftaroline binds altered PBPs.

II-18 Neonatal and childhood tetanus. Botulism.

Def
- Toxin diseases of **C. tetani** (**spastic** paralysis) and **C. botulinum** (**flaccid** paralysis).

Cause
- Tetanus: spores in **soil**/rusty nails, deep wounds, unclean cord with poorly immunized mother.
- Infant botulism: ingested **spores** from **honey** or juice (or inhaled dust) in immature gut.

Patho
- Tetanospasmin → **retrograde axonal transport** → blocks **GABA/glycine** release → spasms.
- Botulinum cleaves **SNARE** → no **acetylcholine** release → flaccid + autonomic block.

Signs
- Tetanus: jaw first — **trismus** (lock jaw), **risus sardonicus**, spasms, opisthotonus; conscious. Neonate **stops sucking**.
- Botulism: constipation, early **diplopia, ptosis**, **descending** symmetric flaccid paralysis, floppy head; afebrile, alert.

Dx
- Tetanus: **clinical assessment only**; toxin effects, consciousness and sensation preserved.

Rx
- Tetanus: human tetanus immune globulin (**HTIG**), debridement, **metronidazole**, benzodiazepines, ICU.
- Prevent: **maternal toxoid vaccination**; vaccinate after recovery (no immunity after tetanus).
- Infant botulism: supportive + **human-derived botulism immune globulin**; adults: horse-derived antitoxin; no aminoglycosides.

II-19 Inflammatory joint and bone diseases

Def
- **Osteomyelitis** (bone) and **septic arthritis** (joint) — bacterial, septic arthritis a surgical emergency.

Cause
- **S. aureus** most common; **Kingella kingae** 6 months–4 years; GBS/E. coli neonates.
- **Salmonella** in sickle cell; Pseudomonas after nail puncture.

Patho
- Children: **hematogenous** spread; bacteria settle **in the metaphysis** (slow flow in sinusoids) → pus → **sequestrum**, involucrum.
- Under **18 months**, transphyseal vessels spread infection to the joint.

Signs
- Fever, hot swollen joint, **won't bear weight**; pseudoparalysis in infants.
- Septic hip held **flexed, abducted and externally rotated**.

Dx
- **Blood culture**, CRP, ESR; ultrasound effusion; X-ray normal for 10–14 days.
- **Joint aspiration**: Gram stain, culture, PCR.
- **MRI** best for osteomyelitis (high ESR, normal X-ray → MRI).

Rx
- MSSA: **nafcillin, oxacillin, dicloxacillin**, IV cefazolin; clindamycin or vancomycin if MRSA.
- Short IV (2–4 days) → **oral**; about **3 weeks** total.
- Septic arthritis: **joint washout**; drain subperiosteal abscess, remove sequestrum.

II-20 Congenital bone and joint disorders

Def
- Congenital disorders: **DDH**, **clubfoot**, **osteogenesis imperfecta**, achondroplasia.

Cause
- DDH: **breech**, family history, female sex, first-born, oligohydramnios, tight swaddling.
- OI: **COL1A1/COL1A2** (usually AD); achondroplasia: **FGFR3 gain-of-function**, older fathers.

Patho
- Femoral head displaced → acetabulum fails to mould → **DDH**.
- OI: defective **type I collagen** → fragile bone, blue sclerae.
- FGFR3 overactive → poor endochondral growth → **rhizomelic** limbs.

Signs
- DDH: Barlow/Ortolani positive; later **limited abduction**, leg-length discrepancy.
- Clubfoot: rigid cavus, adductus, varus, equinus; OI: low-trauma fractures.
- Achondroplasia: macrocephaly, trident hands, **foramen magnum stenosis**.

Dx
- **Ortolani** relocates a dislocated hip; **Barlow** dislocates a located one.
- Hip **ultrasound** before about **4.5 months**; X-ray afterwards.
- Consider OI in non-accidental injury. Scoliosis: **forward bend test** → X-ray.

Rx
- DDH: **Pavlik harness** under 6 months → closed → open reduction.
- Clubfoot: **Ponseti** casting → Achilles tenotomy → bracing. OI: IV **bisphosphonates**.
- Scoliosis Cobb 10–30°: monitor; 30–40°: brace; >40–50°: surgery.

II-21 Juvenile idiopathic arthritis

Def
- **JIA**: arthritis of unknown cause, onset before **16**, lasting at least **6 weeks**.

Cause
- Subtypes: **oligoarticular** (~50%, young ANA+ girls), polyarticular RF−/RF+, systemic, enthesitis-related (HLA-B27), psoriatic.

Patho
- T cells + TNF-α, IL-1, IL-6 → synovial hyperplasia (**pannus**) → erosion.
- Systemic JIA: **IL-1** and **IL-6** driven; ANA and RF negative.
- **MAS**: failed cytotoxic killing → cytokine storm; ESR **falls** as fibrinogen is consumed.

Signs
- Swollen joint, **morning stiffness**; limb overgrowth.
- Oligoarticular ANA-positive: **silent anterior uveitis** (iritis), painless white eye.
- Systemic: **quotidian fever**, **salmon-pink macular rash**, hepatosplenomegaly, serositis.

Dx
- Clinical; ANA → uveitis risk; RF/anti-CCP → erosive course.
- Exclude **leukaemia** (night pain, cytopenia, LDH) and septic arthritis.
- ANA+ oligoarticular: **slit lamp every 3 months**. MAS: **huge ferritin**, falling ESR.

Rx
- NSAIDs → **intra-articular triamcinolone** → **methotrexate** once weekly + folic acid.
- Biologics: anti-TNF, abatacept, **tocilizumab**; sJIA: **anakinra**/tocilizumab first-line.
- MAS (most serious complication): IV **methylprednisolone**, anakinra, ciclosporin.

II-22 Causes, diagnostic and treatment of acute viral hepatitis

Def
- Acute inflammation of the liver from hepatotropic viruses **A–E**.

Cause
- **A, E**: faecal-oral, self-limiting; **B, C, D**: blood/vertical, can become chronic.
- Others: EBV, CMV, HSV, paracetamol, autoimmune, Wilson.

Patho
- **Cytotoxic T cells** kill hepatocytes → ALT/AST in thousands; **zone I** first.
- Impaired bilirubin handling → jaundice, dark urine, pale stools, itch.
- Perinatal HBV becomes chronic in about **90%** (vs ~5% adults).

Signs
- Prodrome: fever, malaise, anorexia, nausea, vomiting → then **jaundice**; HAV in young children mostly **anicteric**.

Dx
- **IgM anti-HAV**; **HBsAg + IgM anti-HBc** = acute B; anti-HCV + **HCV RNA**.
- **HBeAg** predicts infectivity; severity by **INR** (not corrected by vitamin K), glucose, mental state.
- PALF: INR ≥1.5 with encephalopathy, or ≥2.0 without.

Rx
- **Supportive**: fluids, avoid hepatotoxins; admit if INR raised, hypoglycaemic, encephalopathic.
- Chronic only: HBV entecavir/tenofovir; HCV **DAAs**.
- Newborn of HBsAg+ mother: **HBV vaccine + HBIG** within 12 hours.

II-23 Tuberculosis

Def
- **Mycobacterium tuberculosis** infection; childhood TB = **recent infection from an adult contact**.

Cause
- Droplet nuclei from adult with cavitary pulmonary TB.
- Risk: age **under 2**, HIV, malnutrition, measles.

Patho
- Hilar nodes → **Ghon complex** = peripheral lung focus + hilar nodes.
- **Cord factor** → **TNF-α** → granulomas.
- Under 2 years: blood spread → **miliary TB and meningitis**.

Signs
- Primary TB mostly asymptomatic; cough **>2 weeks**, fever, night sweats, weight loss.
- Scrofula, Pott disease, erythema nodosum.

Dx
- Contact + **TST/IGRA** + CXR; IGRA has fewer BCG false positives.
- PPD positive: ≥**5 mm** high risk, ≥**10** intermediate, ≥**15** low risk.
- Early-morning **gastric aspirates** or induced sputum → Xpert MTB/RIF, culture. Meningitis: lymphocytes, high protein, very low glucose.

Rx
- RIPE: 2 months **HR2(E)** → 4 months **HR** (rifampin, isoniazid); pyridoxine with isoniazid.
- TB meningitis: **12 months** + corticosteroids.
- Contacts under 5 → **preventive therapy** even if tests negative.

II-24 Disorders of the humoral immune system

Def
- Primary antibody (B-cell) deficiencies: **XLA**, **CVID**, **selective IgA deficiency**, **hyper-IgM**.

Cause
- XLA: **BTK** mutation, boys; CVID: no B-cell differentiation into plasma cells.
- Hyper-IgM: mutated **CD40L** on Th cells. Secondary: nephrotic syndrome, rituximab, malignancy.

Patho
- XLA: **no B-cell maturation** → no opsonising IgG → **encapsulated bacteria** escape phagocytosis.
- Maternal IgG wanes → onset after **~6 months**; hyper-IgM: **defective class switching**.

Signs
- Recurrent **sinopulmonary infections** → bronchiectasis; enteroviral meningoencephalitis.
- XLA: **absent tonsils and nodes**; CVID: autoimmunity, lymphoma.
- IgA deficiency (most common): mostly asymptomatic; **transfusion anaphylaxis**.

Dx
- Quantitative **IgG, IgA, IgM**; IgA deficiency: normal IgG and IgM.
- **Vaccine responses** (tetanus, Hib, pneumococcal) before and 4–6 weeks after booster.
- Flow cytometry: **B cells (CD19)** absent in XLA; genetics (BTK).

Rx
- **Immunoglobulin replacement** (IVIG) every 3–4 weeks or subcutaneous weekly.
- Prompt antibiotics; **no live vaccines** in XLA.
- IgA deficiency: usually none; washed blood products.

II-25 Cellular immunodeficiencies

Def
- T-cell defects — almost always **combined** (antibody fails too): SCID, DiGeorge, Wiskott–Aldrich.

Cause
- SCID: X-linked **common γ-chain (IL2RG)** commonest, ADA deficiency, **RAG** mutations.
- DiGeorge: **22q11.2 deletion**; WAS: X-linked WAS protein; secondary: HIV, steroids.

Patho
- No T cells → no control of **viruses, fungi, intracellular opportunists**; no B-cell help.
- ADA deficiency: **dATP** toxic to lymphocytes; RAG → **VDJ recombination defects**.
- DiGeorge: abnormal 3rd/4th pharyngeal pouches → thymic + parathyroid hypoplasia.

Signs
- First months: **thrush**, **Pneumocystis**, chronic viral diarrhoea, failure to thrive.
- DiGeorge: **hypocalcaemia**, **conotruncal heart defects**, cleft palate.
- WAS triad: **eczema**, thrombocytopenia with **small platelets**, infections.

Dx
- **Lymphopenia**, absent **thymic shadow**; absent T cells on flow cytometry.
- Newborn screening: **TRECs**. DiGeorge: calcium, PTH, echo, FISH for 22q11.2.

Rx
- Isolation; **no live vaccines**; only **irradiated**, CMV-negative blood.
- TMP-SMX (co-trimoxazole) PJP prophylaxis, antifungals, immunoglobulin replacement.
- Cure: **stem cell transplant** (>90% if <3.5 months); ADA: **pegademase**, gene therapy.

II-26 Role of the complement system and diseases of the complement system

Def
- Complement: three pathways (classical begins **C1**, alternative **C3**) converging on **C3** → opsonise, recruit, lyse.

Cause
- Inherited deficiency of early classical (C1q, C2, C4), C3, terminal (C5–C9), properdin, factor H/I.
- **C1-inhibitor** deficiency (AD) → hereditary angioedema.

Patho
- Early classical deficiency → **SLE-like**; C3 → **encapsulated bacteria**.
- **C5–C9**/properdin deficiency → no MAC → recurrent **Neisseria** infection.
- No C1-inhibitor → **bradykinin** excess → vascular leak.

Signs
- Lupus-like disease; severe pyogenic infections; **recurrent meningococcal** disease.
- Factor H defect → **atypical HUS**.
- HAE: non-itchy swelling of face, gut, **larynx**; no urticaria.

Dx
- **CH50** (classical), **AH50** (alternative), C3, C4 levels.
- Second or unusual-serogroup meningococcal episode → complement screen.
- HAE screening test: **low C4** → C1-inhibitor level and function.

Rx
- Vaccinate: **meningococcus ACWY + B**, pneumococcus, Hib; early antibiotics for fever.
- aHUS: **eculizumab** (anti-C5) — requires meningococcal vaccination.
- HAE: **C1-inhibitor concentrate** or icatibant — adrenaline/antihistamines/steroids don't work.

II-27 HIV infections

Def
- **HIV** infects CD4 T cells; in children mainly **vertical**, most **intrapartum**.

Cause
- Transmission: in utero, **intrapartum**, breastfeeding; adolescents sexual or drug use.

Patho
- **gp120** binds CD4 + **CCR5** (macrophages, early) or **CXCR4** (T cells, late); **gp41** mediates fusion.
- → **CD4 depletion** and immune activation.
- Developing thymus, higher viral set points → faster progression than adults.

Signs
- Failure to thrive, lymphadenopathy, hepatosplenomegaly, parotid swelling, chronic diarrhoea.
- Persistent candidiasis, LIP, encephalopathy; infant 3–6 months with severe hypoxia → **PJP**.

Dx
- Under **18 months**: **HIV DNA/RNA PCR** because **maternal IgG persists**.
- Over 18 months: antibody/antigen tests as in adults.

Rx
- Prevent: maternal ART, neonatal **zidovudine**, avoid breastfeeding → transmission **<1%**.
- **ART at diagnosis, regardless of CD4**: 2 NRTIs + **dolutegravir**.
- **Co-trimoxazole** from 4–6 weeks against PJP; avoid BCG; rifabutin replaces rifampin with protease inhibitors.

II-28 Opportunistic infections

Def
- Infections by organisms harmless to healthy hosts; the **immune defect predicts the organism**.

Cause
- **HIV/AIDS**, **high-dose immunosuppression** (transplant), steroids; chemotherapy neutropenia, CGD.
- Antibody, complement, **asplenia** (XLA, CVID, sickle cell).

Patho
- No neutrophils → Pseudomonas, S. aureus, **Candida, Aspergillus**.
- No T cells → **Pneumocystis**, CMV, **Cryptococcus**, **MAC**.
- No antibody/spleen → **encapsulated bacteria**.

Signs
- Febrile neutropenia: **fever may be the only sign** of sepsis.
- PJP: **fever, dry cough, hypoxia**, high **LDH**; MAC: high fever, watery diarrhea.
- CMV: retinitis, **colitis**; Cryptococcus: fungal meningitis.

Dx
- Febrile neutropenia = **≥38°C** + neutrophils **<0.5 × 10^9/L**; cultures line + periphery.
- PJP: **bronchoalveolar lavage**; CMV retinitis: **CD4 ≤ 50**, cotton wool spots.

Rx
- **Piperacillin–tazobactam** within **1 hour**; add vancomycin for line/instability.
- PJP: **TMP-SMX** + steroids if **PaO2 ≤ 70 mmHg**; MAC: **clarithromycin** + **ethambutol**.
- Toxoplasmosis: **pyrimethamine & sulfadiazine**; varicella: **IV aciclovir**.

II-29 Poststreptococcal diseases

Def
- Non-suppurative sequelae of GAS: **acute rheumatic fever** and **PSGN**.

Cause
- ARF: GAS **pharyngitis 2–4 weeks** before.
- PSGN: pharyngitis **1–2 weeks** or skin **3–6 weeks** before.

Patho
- ARF: M protein cross-reacts with self-antigens (**molecular mimicry**); **type II HSR**.
- PSGN: immune complexes → complement → **low C3** → ↓GFR → salt retention.

Signs
- ARF (**J♥NES**): migratory polyarthritis, **carditis**, nodules, **erythema marginatum**, chorea.
- Mitral valve most common: acute **regurgitation**; chronic **fish-mouth** stenosis.
- PSGN: **cola urine**, periorbital oedema, **hypertension**, oliguria.

Dx
- **Jones**: 2 major, or 1 major + 2 minor, + evidence of GAS (ASO/anti-DNase B).
- Histology: **Aschoff bodies**, **Anitschkow cells**; echo for subclinical carditis.
- PSGN: **low C3, normal C4**, normalising within **8 weeks**.

Rx
- ARF: penicillin V 10 days (anaphylaxis: **macrolides**); **aspirin**/naproxen for arthritis.
- Secondary prophylaxis: **benzathine penicillin IM every 3–4 weeks**.
- PSGN: **supportive**; strep-throat antibiotics prevent ARF, not PSGN.

II-30 Pertussis infection

Def
- **Pertussis**: toxin-mediated Bordetella pertussis respiratory infection (whooping cough).

Cause
- Aerobic **gram-negative coccobacillus**, spread by **respiratory droplets**; attack rate **>80%**.
- Adolescents/adults often the source.

Patho
- Filamentous hemagglutinin attaches; **tracheal toxin** kills ciliated cells → violent coughing.
- **Pertussis toxin** disables Gi → ↑cAMP → **lymphocytosis**, hypoglycaemia.
- Lymphocytes **>30 × 10^9/L** → leukostasis → pulmonary hypertension.

Signs
- **Catarrhal** (most infectious) → **paroxysmal**: whoop, post-tussive vomiting → **convalescent**.
- Infants <6 months: **apnoea and cyanosis**, often no whoop; **pneumothorax** risk.

Dx
- **PCR of nasopharyngeal swab** early; culture on **Regan-Lowe** or Bordet-Gengou agar.
- FBC: marked **lymphocytosis**.

Rx
- Early **macrolides (azithromycin)**, also in neonates (erythromycin: **pyloric stenosis**).
- Admit infants <6 months with apnoea/cyanosis; **prophylactic macrolides** for all close contacts.
- Prevent: **Tdap in pregnancy** + DTaP at 2, 4, 6 months.

II-31 Salmonella infection

Def
- **Non-typhoidal salmonella** enterocolitis and **typhoid** (enteric) fever.

Cause
- NTS: raw chicken/eggs, unpasteurised milk, **reptiles** (turtles); large infective dose.
- Typhoid: **S. Typhi/Paratyphi**, humans the only reservoir; **travel** to endemic areas.

Patho
- Gram-negative, non-lactose fermenting, H_2S-positive rod with **type III** secretion system.
- Typhoid: **Peyer patches** → macrophages → bacteraemia; necrosis → **perforation** in week 3.

Signs
- NTS: **inflammatory diarrhea** with leukocytes; sickle cell → **osteomyelitis**.
- Typhoid: abdominal pain, **stepwise fever**, **relative bradycardia**, **rose spots**.

Dx
- NTS: stool culture.
- Typhoid: **blood culture** in week 1 (bone marrow most sensitive); leucopenia typical.

Rx
- NTS low-risk: **supportive care**; antibiotics **prolong faecal carriage**.
- Antibiotics only if **<3 months**, immunocompromised, sickle cell, invasive (sickle cell sepsis: **ceftriaxone**).
- Typhoid: **ceftriaxone** or azithromycin; **carbapenem** for XDR; vaccine.

II-32 Influenza in children

Def
- **Influenza A/B**: orthomyxovirus; **drift** → epidemics, **shift** (reassortment) → pandemics.

Cause
- **Respiratory droplets**; children main spreaders.
- High risk: **under 5**, chronic lung/heart/neuro disease, immunosuppression, aspirin therapy.

Patho
- Drift = point mutations in **hemagglutinin** or **neuraminidase**; HA binds **sialic acid**, NA cleaves it.
- Ciliated cells destroyed → **cytokines**; denuded airway → **secondary bacterial pneumonia**.

Signs
- Abrupt fever, myalgia, dry cough; infants mimic **sepsis**, febrile seizures.
- Influenza B: calf **myositis**; associated with **Guillain-Barré**; better then worse → pneumonia.

Dx
- Clinical in season; **PCR on nasopharyngeal swab** for hospitalised/high-risk.

Rx
- Supportive: paracetamol/ibuprofen; **no aspirin** (Reye syndrome).
- **Oseltamivir** (neuraminidase inhibitor) for high-risk/severe, ideally within **48 hours**.
- Annual vaccine from **6 months**; first time under **9** years: **two doses 4 weeks apart**.

II-33 Routine management of the healthy newborn infant. Disorders of the newborn adaptation.

Def
- Care of the healthy newborn and disorders of the **transition** to extrauterine life.

Cause
- Adaptation disorders: **TTN**, **PPHN**, hypothermia, hypoglycaemia, jaundice.
- Hypoglycaemia risk: infants of diabetic mothers, small, preterm, cold, asphyxiated.

Patho
- Labour catecholamines → **ENaC** clears lung fluid → aeration; TTN = **delayed clearance**.
- **Brown fat** (uncoupling protein 1) gives non-shivering heat.

Signs
- TTN: **term** baby after **C-section** with grunting; hypoglycaemia mnemonic **JOLTS**.
- Jaundice **within 24 h** is always pathological; cephalohematoma does not cross sutures.

Dx
- **Apgar 1, 5, 10** minutes; examination within **72 h**.
- Screening: **pulse oximetry**, hearing, heel-prick blood spot.
- Pre-/post-ductal saturation gap → **PPHN**.

Rx
- Dry and stimulate; delay **cord clamping ≥60 s**; skin-to-skin; breastfeed.
- **Vitamin K 1 mg IM**; HBV vaccine + **HBIG** within 12 hours if mother HBsAg-positive.
- Hypoglycaemia: feed, **dextrose gel**, IV dextrose; jaundice: phototherapy; TTN: supportive.

II-34 Respiratory Distress Syndrome (RDS)

Def
- **RDS**: **surfactant** deficiency in the preterm lung (mature levels around week **35**).

Cause
- **Prematurity** (most important), **maternal diabetes** (fetal insulin), caesarean delivery.

Patho
- **Type II pneumocytes** make surfactant; none → alveoli collapse (**Laplace**, $P = 2T/r$) → hypoxaemia.
- Prematurity → inadequate surfactant → alveoli fail to open → poor oxygenation.

Signs
- Within **4 h** of birth: tachypnoea, **grunting**, nasal flaring, recession, cyanosis.

Dx
- CXR: **ground-glass** with **air bronchograms**, atelectasis, low lung volumes (**BAD**).
- Mature lung: **L:S ratio ≥ 2**.

Rx
- Prevent: maternal **antenatal glucocorticoids**.
- **CPAP** from birth → artificial **surfactant** if FiO_2 **>0.30**.
- Caffeine for apnoea; antibiotics until GBS excluded; complications: **NEC**, ROP, BPD.

II-35 Meconium aspiration syndrome

Def
- **MAS**: respiratory distress from meconium aspiration in **term and post-term** infants.

Cause
- **Post-term** (postmaturity), growth restriction, pre-eclampsia, **oligohydramnios**, abnormal CTG.
- Meconium in a **preterm** → think **Listeria**.

Patho
- Fetal hypoxia → increased peristalsis and **anal sphincter relaxation** → meconium passage → aspiration.
- **Ball-valve** air trapping → pneumothorax; meconium **inactivates surfactant**; **PPHN**.

Signs
- Meconium-stained skin/nails/cord; distress from birth; **barrel chest**, crackles.
- Air leak (pneumothorax) in 10–20%.

Dx
- CXR: coarse patchy infiltrates + **hyperinflation**, flattened diaphragms.
- Pre-/post-ductal saturation gap → PPHN.

Rx
- Non-vigorous baby: **no routine tracheal suctioning**; start ventilation within the first minute.
- NICU: oxygen, CPAP/ventilation, **surfactant**, inhaled **nitric oxide** for PPHN.
- **ECMO** if refractory; antibiotics until sepsis excluded.

II-36 Primary Pulmonary Hypertension of the Newborn (PPHN)

Def
- **PPHN**: **pulmonary vascular resistance** fails to fall after birth.
- Right-to-left shunt at the **PDA and foramen ovale**.

Cause
- Commonest: **meconium aspiration**; also sepsis, asphyxia, RDS.
- Maternal **SSRIs** or NSAIDs in pregnancy; **diaphragmatic hernia**.

Patho
- Shunted blood bypasses alveoli → FiO_2 barely helps.
- Hypoxia + acidosis → more vasoconstriction → **vicious circle**.

Signs
- Term baby, cyanosis **out of proportion** to lung disease, labile saturations.
- Loud single S2, tricuspid regurgitation murmur.

Dx
- **Pre-post-ductal gap** (>3%); pre-ductal = **right hand**.
- **Echocardiography** confirms the shunt and excludes cyanotic heart disease (TAPVD).

Rx
- Minimal handling, sedation, normothermia; **oxygen**, avoid hyperoxia; surfactant.
- **Inhaled nitric oxide** 20 ppm if OI >20–25 (relaxes vessels via **cGMP**).
- Support BP (milrinone, adrenaline, vasopressin); **ECMO** if refractory.

II-37 Perinatal infections

Def
- Infections passed mother → baby **transplacentally**, **intrapartum** or **postnatally**.

Cause
- Transplacental: **TORCH**, parvovirus B19, VZV, Zika.
- Intrapartum: **GBS**, E. coli, HSV, HBV, HIV, gonococcus, chlamydia.
- Postnatal: NICU lines/hands (CoNS, Staph aureus, Candida), breast milk, contacts.

Patho
- Early pregnancy → organogenesis hit → **malformation**; later → active disease at birth.
- Parvovirus B19 → fetal **anaemia and hydrops**; HBV in infants → up to 90% carriers.

Signs
- Neonatal HSV: **seizures, vesicles, hepatitis**.
- Gonococcal conjunctivitis **2–5 days**; chlamydial **5–14 days**; silver nitrate: chemical conjunctivitis.

Dx
- Antenatal screening: **HIV, hepatitis B, syphilis**; maternal GBS status.

Rx
- HBV: vaccine + **HBIG** within **12 hours** (at latest 24 h); breastfeeding allowed.
- HSV → **IV aciclovir**; gonococcus → ceftriaxone; chlamydia → azithromycin; GBS → **penicillin G** prophylaxis.
- HIV → ART, no breastfeeding; peripartum chickenpox → **VZIG**; no fetal scalp electrodes in HIV/HCV.

II-38 TORCH infections

Def
- Congenital **T**oxoplasma, **O**ther (syphilis, VZV, parvovirus, Zika), **R**ubella, **C**MV, **H**SV.

Cause
- Maternal infection crossing the placenta (**transplacental**); **CMV** is the commonest; highest risk **1st trimester**.

Patho
- Cross placenta → attack **dividing fetal cells** → IUGR, CNS and eye damage.
- Parvovirus B19 → **hydrops fetalis**; VZV → **congenital varicella syndrome** (limb hypoplasia).

Signs
- CMV: **blueberry muffin rash**, hepatosplenomegaly, jaundice, **periventricular** calcification, **hearing loss**.
- Toxo: **chorioretinitis**, hydrocephalus, **diffuse** calcification; rubella: cataract, deafness, PDA.
- Syphilis: **snuffles**, desquamating rash, **saber shins**.

Dx
- CMV: urine/saliva **PCR within 21 days**; 80–90% asymptomatic at birth → hearing and eye follow-up.
- Toxoplasmosis: **antibody testing**.

Rx
- Symptomatic CMV: **valganciclovir 6 months**; toxo: **pyrimethamine** + sulfadiazine + folate.
- Syphilis: **penicillin G**; VZV: **VZIG**; rubella prevented by **MMR**.
- HSV lesions in labour: **C-section**; **acyclovir** from 36 weeks.

II-39 Neonatal and late onset neonatal sepsis

Def
- **Early-onset** (< **72 hours**) = vertical; **late-onset** (>72 h) = nosocomial/community.

Cause
- Early: **GBS**, E. coli, Listeria; risks: prolonged ROM, **chorioamnionitis**, prematurity.
- Late: **coagulase-negative staphylococci** (lines), Staph aureus, Gram-negatives, Candida.

Patho
- Small neutrophil reserve, low complement/Ig + lines breaching skin → rapid spread.

Signs
- Subtle: **apnoea**, temperature instability, grunting, poor feeding, mottling.

Dx
- **Blood culture**, serial CRP, **lumbar puncture**.
- GBS identified by **maternal vaginal screening**.

Rx
- Prevention: intrapartum **penicillin G** at least **4 hours** before delivery; not late-onset GBS.
- Without meningitis: **ampicillin + gentamicin** within **1 hour**; with meningitis: **ampicillin + cefotaxime**.
- Prior invasive GBS baby → prophylaxis in every subsequent pregnancy.

II-40 Central nervous system disorders of the neonates

Def
- Preterm: **IVH** and **PVL**; term: **HIE**, stroke, infection, kernicterus, birth trauma.

Cause
- IVH: preterm <32 weeks, within **72 hours**; HIE = commonest seizure cause.
- Subgaleal haemorrhage after **vacuum** delivery; Erb palsy: lateral traction at delivery.

Patho
- IVH: fragile **germinal matrix** bleeds → hydrocephalus.
- Watershed white matter ischaemia → **PVL** → spastic diplegia.

Signs
- Neonatal seizures are **subtle**; IVH: low tone, seizures, rapidly increasing head circumference.
- Scalp swelling crossing sutures + shock → **subgaleal haemorrhage**; Erb palsy: **waiter's tip**.

Dx
- **Cranial ultrasound**; Papile I–IV; **grade IV** = periventricular haemorrhagic infarction.
- Seizures: check **glucose and calcium** first, then sodium, magnesium, infection; EEG.

Rx
- Seizures: **phenobarbital** first line.
- IVH prevention: antenatal steroids, magnesium, delayed cord clamping; avoid pressure/CO2 swings.
- Erb/Klumpke palsy: **spontaneous resolution within 3 months** (80%).

II-41 Skin and umbilical disorders of the neonates

Def
- Benign newborn rashes, vascular lesions, skin infections and umbilical problems.

Cause
- Benign: **erythema toxicum**, milia, miliaria, nevus simplex, **dermal melanocytosis**.
- Infective: bullous impetigo/**SSSS**, candida, HSV, omphalitis.

Patho
- SSSS: exfoliative toxin cleaves **desmoglein-1** → mucosa spared.
- Omphalitis → umbilical vein → sepsis, **portal vein thrombosis**.

Signs
- Erythema toxicum: sterile pustules full of **eosinophils**; miliaria: sweat-duct blockage with heat.
- **Strawberry hemangioma** grows rapidly then regresses; umbilical hernia soft, reducible.
- Urine at navel = **patent urachus**; faeces = **vitelline duct**; granuloma: soft, moist, pink.

Dx
- Cord separates at **5–15 days**; delayed + neutrophilia → **leukocyte adhesion deficiency** (CD18).
- Single umbilical artery → congenital and chromosomal anomalies.

Rx
- Benign rashes → **reassurance**; hemangioma → **propranolol** if eye, airway, ulcer.
- Granuloma → **silver nitrate**; umbilical hernia: monitor, resolves by age 5.
- SSSS → IV flucloxacillin; omphalitis → IV antibiotics; HSV → aciclovir.

II-42 Necrotising enterocolitis (NEC). Meconium ileus

Def
- **NEC**: ischaemic-inflammatory bowel necrosis in preterms after feeds start.
- **Meconium ileus**: terminal ileum blocked by sticky meconium = **cystic fibrosis** (~15%).

Cause
- NEC: **prematurity** (biggest risk factor), **formula feeding**, abnormal flora, poor mucosal perfusion; breast milk protective.

Patho
- Immature barrier + abnormal flora + poor perfusion (e.g. hypoxaemia) → bacteria invade wall → **pneumatosis**.

Signs
- NEC: **abdominal distension, nutritional intolerance, bloody stool**; instability, low platelets, acidosis.
- Meconium ileus: distension, bilious vomiting, **no meconium**.

Dx
- NEC X-ray: **pneumatosis intestinalis**, portal venous gas; **Rigler sign** = perforation.
- Baby cannot be moved: **horizontal X-ray from side position**, lying on back.
- Meconium ileus: **soap-bubble**, **microcolon**; later **sweat test**.

Rx
- NEC: **NPO** (bowel rest), NG decompression, IV fluids, TPN, IV antibiotics (gram-negative, anaerobes).
- **Surgery** for perforation, peritonitis, or worsening despite medical therapy.
- Meconium ileus: **Gastrografin enema** → surgery if it fails.

II-43 Congenital diaphragmatic hernia. Oesophageal atresia

- **Def**
  - **CDH**: usually **left**-sided posterolateral **Bochdalek** hernia.
  - **OA**: proximal blind pouch, usually with **distal TEF** (85%).
- **Cause**
  - CDH: abdominal viscera herniate into chest; OA/TEF: associated **VACTERL** anomalies (mesodermal defects).
- **Patho**
  - CDH: **pulmonary hypoplasia + pulmonary hypertension** determine survival.
  - OA: fetus can't swallow → **polyhydramnios**; saliva pools → choking, aspiration.
- **Signs**
  - CDH: respiratory distress, **scaphoid abdomen**, heart sounds on right.
  - OA: **excess drooling**, choking at first feeding; distal fistula: abdominal distension.
  - Cough since birth, always during feeding → **H-type TEF**.
- **Dx**
  - OA: firm NG tube **coils** in pouch on X-ray; gastric gas = distal fistula.
  - Pure OA: **gasless abdomen**; VACTERL screen, echo.
- **Rx**
  - CDH: **intubate, no bag-mask**, NG tube, gentle ventilation, treat pulmonary hypertension.
  - CDH surgery after stabilisation: **physiological, not surgical, emergency**.
  - OA/TEF: **surgery** after Replogle tube; look for VACTERL.

II-44 Duodenal stenosis, bowel atresia, omphalocele, gastroschisis

- **Def**
  - Congenital gut obstructions (duodenal, jejunoileal, anal atresia) and abdominal wall defects.
- **Cause**
  - Duodenal: failed **recanalisation**, **trisomy 21**; jejunoileal: in-utero **vascular accident**.
  - Omphalocele: Beckwith–Wiedemann, trisomies 13, 18; gastroschisis: young mothers.
- **Patho**
  - Obstruction below ampulla → **bilious** vomiting from first feed, polyhydramnios.
- **Signs**
  - Duodenal: bilious vomiting, **flat abdomen**; jejunoileal: distension, many loops.
  - **Omphalocele**: midline, covered by **sac**; **gastroschisis**: **right** of cord, no membrane.
  - Omphalocele has other major malformations (cardiac, genitourinary, neural tube).
- **Dx**
  - Duodenal atresia: **double bubble** on abdominal X-ray; exclude malrotation.
- **Rx**
  - Duodenal: decompression, fluids → **surgical repair** (duodenoduodenostomy).
  - Wrap bowel in **sterile saline dressings**, NG tube; primary closure, else **staged silo repair**.
  - High anal atresia: **colostomy** → PSARP; low → anoplasty.

II-45 Follow-up of the premature baby. Bronchopulmonary dysplasia (BPD). Retinopathy of prematurity (ROP).

- **Def**
  - **BPD** = oxygen or respiratory support at **36 weeks PMA**.
  - **ROP** = abnormal retinal vessel growth in preterm babies.
- **Cause**
  - Prematurity plus **oxygen toxicity** (RIB), ventilator trauma, infection.
- **Patho**
  - BPD: arrested alveolar/vascular growth + injury → simplified alveoli, pulmonary hypertension.
  - ROP: high oxygen → hypoxic retina → **VEGF surge** → **neovascularisation**.
- **Signs**
  - BPD: month of oxygen need, ground-glass opacities on X-ray (scarring).
  - Follow growth, neurodevelopment (PVL → **spastic diplegia**), hearing, vision.
- **Dx**
  - **Corrected age** for growth/development (~2 years); **chronological** for vaccines.
  - ROP screen: **<31 weeks or <1501 g**; BPD with rising O2 → echo.
- **Rx**
  - Supplement **iron, vitamin D**; RSV prevention: **nirsevimab or palivizumab**.
  - BPD prevention: maternal glucocorticoids, surfactant, caffeine; oxygen, nutrition.
  - ROP: **laser photocoagulation** or **anti-VEGF**.

II-46 Lymphomas

- **Def**
  - **Hodgkin**: adolescent lymphoma with **Reed–Sternberg** cells; childhood **NHL** is high grade.
- **Cause**
  - NHL: **Burkitt** t(8;14) MYC, lymphoblastic, DLBCL, ALCL.
  - Hodgkin: **nodular sclerosis** commonest (70%).
- **Patho**
  - Hodgkin: reactive inflammatory cells, **contiguous spread**; cytokines → B symptoms.
  - Burkitt lysis → **TLS**: high K, phosphate, urate, **low calcium**.
- **Signs**
  - Painless rubbery cervical/supraclavicular nodes; B symptoms: fever, **drenching night sweats**, weight loss >**10%**.
  - Burkitt: abdominal mass; lymphoblastic: **anterior mediastinal mass**.
  - Suspicious node: >2 cm, hard, supraclavicular, persistent.
- **Dx**
  - **Excision biopsy**: RS cells **CD15**, **CD30**; CXR, **PET-CT**; **staging** predicts prognosis.
  - **No steroids** before biopsy.
- **Rx**
  - Hodgkin: response-adapted chemotherapy, cure >90%.
  - TLS: hyperhydrate, **rasburicase** (contraindicated in **G6PD deficiency**).
  - Mediastinal mass → **avoid general anaesthesia** and lying flat.

II-47 Leukaemia

Def
- Commonest childhood cancer; **ALL** ~80%, peak **2–5 years**.

Cause
- Risk factor: **Down syndrome**.

Patho
- Blasts fill marrow → **anaemia, bleeding, infection**; infiltration → bone pain, organomegaly; CNS a sanctuary.

Signs
- Pallor, bleeding, infection, bone pain, HSM, lymphadenopathy; mediastinal mass in T-ALL.

Dx
- **Bone marrow examination** + flow cytometry, genetics; LP.
- Good: age 1–10, WBC <50, hyperdiploidy/ETV6-RUNX1; **minimal residual disease** strongest.

Rx
- **Induction**, consolidation + **intrathecal methotrexate**, intensification, **maintenance** (~2 y); ~90% cure.
- **Febrile neutropenia** → IV antibiotics within 1 hour; **co-trimoxazole** vs Pneumocystis.
- High-risk/relapse: **allogeneic SCT**; **graft versus leukaemia** = donor cells attack leukaemic cells.

II-48 Disorders of platelets, thrombocytopenia, thrombocytosis

Def
- Low, dysfunctional or high platelets; bleed into **skin and mucosa**, not joints.

Cause
- **Destruction** (ITP, NAIT, TTP/HUS, DIC, hypersplenism); **underproduction** (leukaemia, aplasia).
- **Dysfunction**, normal count: **Glanzmann**, **Bernard–Soulier**, vWD, aspirin.

Patho
- ITP: IgG against **GPIIb/IIIa** → platelets consumed by **splenic macrophages**.
- TTP: **ADAMTS13** deficiency → uncleaved vWF → microthrombi; Glanzmann: **aggregation** defect; Bernard–Soulier: **adhesion**.

Signs
- Petechiae, purpura, epistaxis; feared complication **intracranial bleeding**.
- ITP: well child, purpura **3 weeks after viral illness**, no organomegaly.

Dx
- ITP: isolated low platelets, large platelets on film, increased megakaryocytes.
- Other cell line low, HSM or bone pain → **bone marrow**; TTP: raised **LDH**.

Rx
- ITP: cutaneous only → **observation** (75–80% resolve in 6 months); mucosal bleeding → **corticosteroids**.
- Significant bleeding → **IVIG**; chronic ITP (>12 months): **romiplostim**/eltrombopag.
- TTP: **plasma exchange**; thrombocytosis is **reactive** → no treatment, no aspirin.

II-49 Disseminated intravascular coagulation, haemolytic uremic syndrome (HUS)

Def
- **DIC**: consumptive coagulation. **HUS** triad: **microangiopathic haemolytic anaemia**, **thrombocytopenia**, **acute kidney injury**.
- **TTP**: deficiency or inhibition of **ADAMTS13** (vWF metalloprotease).

Cause
- DIC: sepsis, shock, trauma, malignancy.
- Typical HUS follows bloody diarrhoea from **Shiga toxin-producing E. coli (O157:H7)**.
- Atypical HUS: **alternative complement pathway** dysregulation.

Patho
- DIC: tissue factor → microthrombi + consumption + fibrinolysis → clots and bleeds.
- Shiga toxin binds **Gb3** receptors on glomerular endothelium → microthrombi → RBC fragmentation, AKI.

Signs
- DIC: bleeding plus thrombosis.
- HUS: anaemia, low platelets, AKI after bloody diarrhoea.

Dx
- DIC labs: low platelets, **prolonged PT and APTT**, **low fibrinogen**, high D-dimer.
- Clotting times are **normal** in HUS but prolonged in DIC; TTP has raised **LDH**.

Rx
- DIC: **treat the cause**; platelets, FFP, cryo only if bleeding.
- Typical HUS: supportive care, dialysis; avoid **antibiotics** and antimotility drugs.
- Atypical HUS → **eculizumab**; TTP → **plasma exchange**, glucocorticoids, rituximab.

II-50 Porphyria

Def
- Inherited **haem-synthesis** enzyme defects; precursors before the block accumulate.

Cause
- **Acute intermittent porphyria** (AIP): **autosomal dominant**, deficiency of **porphobilinogen deaminase (HMBS)**.
- **Porphyria cutanea tarda**: **UROD** deficiency, most common porphyria.
- Commonest in children: **erythropoietic protoporphyria**, **ferrochelatase** deficiency. Triggers: P-450 inducers, **barbiturates** contraindicated.

Patho
- Early haem precursors (ALA, PBG) are **neurotoxic** → acute attacks.
- Late porphyrins cause **photosensitivity**.

Signs
- AIP mnemonic **Wine In PAIN**: wine-coloured urine, inducers, psychological disturbance, abdominal pain, neuropathy; **no photosensitivity**.
- Hyponatraemia in an attack is due to **SIADH**.
- PCT: **blistering cutaneous photosensitivity** and **hyperpigmentation**.

Dx
- Screening test in an acute attack: urinary **porphobilinogen**; also raised **ALA**.

Rx
- Acute attack: IV **haem arginate (haemin)** and **glucose**, which inhibit ALA synthesis.
- PCT: low-dose **hydroxychloroquine** or **phlebotomy**.

II-S1 Histiocytosis

Def
- **LCH**: proliferation of Langerhans cells; **HLH**: systemic overactivation of **macrophages** and **cytotoxic T cells**.

Cause
- LCH: **BRAF V600E**, peak 1–3 years; Hand-Schüller-Christian often in children over 3.
- HLH: primary genetic or secondary to **EBV**, systemic JIA.

Patho
- LCH cells are positive for **CD1a** and **CD207 (langerin)**; infiltrate bone, skin, pituitary, organs.
- HLH: failed perforin cytotoxicity → cytokine storm.

Signs
- LCH: **lytic (punched-out) skull lesion**, painful lytic long-bone lesion with normocalcaemia, skin rash.
- Pituitary involvement causes **central diabetes insipidus**.
- HLH: **fever**, **pancytopenia**, **hepatosplenomegaly**.

Dx
- LCH: biopsy + staging; risk organs are **liver, spleen and bone marrow**.
- HLH shows very high **ferritin**, high triglycerides, low **fibrinogen**.

Rx
- Multisystem LCH needs **systemic chemotherapy**; single bone lesion may need only curettage or observation.
- HLH: **immunosuppression (steroids ± chemotherapy)**; primary (familial) HLH needs **HSCT**.

II-S2 Central nervous system tumours in children

Def
- CNS tumours are the commonest **solid** tumours of childhood.
- In children, primary CNS tumours are usually **infratentorial**; about half are posterior fossa.

Cause
- **Pilocytic astrocytoma**: commonest primary CNS tumour; good prognosis.
- **Medulloblastoma** (primitive neuroectodermal, cerebellar granule cells); **ependymoma** in the 4th ventricle.
- **Turcot** syndrome: FAP + malignant CNS tumours. Pineal tumours → **Parinaud syndrome**.

Patho
- Posterior fossa mass → obstructive **hydrocephalus** → raised ICP.
- Medulloblastoma grows rapidly, spreads via **CSF** (**drop metastasis**); pilocytic = cyst + **mural nodule**.

Signs
- Raised ICP headache worst **on waking** (nocturnal CO_2 retention → vasodilatation); vomiting.
- Ependymoma: relief in the **knee-chest** position. Parinaud: upward gaze palsy, **convergence-retraction nystagmus**.

Dx
- Urgent **MRI** of brain and spine, and no **lumbar puncture** with raised ICP.
- Ependymoma: perivascular pseudorosettes; medulloblastoma: **MYC** amplification.

Rx
- Maximal safe resection; dexamethasone, CSF diversion for hydrocephalus.
- Cranial radiotherapy is avoided under about **3 years**; chemotherapy; lifelong late-effect follow-up.

II-S3 Abdominal malignancies – Wilms tumour, neuroblastoma

Def
- **Neuroblastoma**: arises from **sympathetic neural crest** cells, most often the **adrenal medulla**.
- **Wilms tumour**: malignant kidney tumour of metanephric mesenchyme (blastema).

Cause
- Neuroblastoma: **N-myc (MYCN) amplification** is the strongest adverse marker.
- Wilms: 3–4 years; **WAGR** = Wilms, **aniridia**, genitourinary anomalies, developmental delay.

Patho
- Neuroblastoma: small blue round cells, **Homer-Wright rosettes**; markers bombesin and NSE.
- Wilms spreads most often to the **lungs**.

Signs
- Neuroblastoma: abdominal mass, periorbital ecchymoses (orbital bone metastases), **opsoclonus–myoclonus–ataxia**.
- Wilms: **smooth flank mass that does not cross the midline** in a well child aged **3–4 years**.

Dx
- Any abdominal mass → **same-day review** + ultrasound (Wilms: abdominal ultrasound).
- Neuroblastoma: urinary **VMA and HVA** raised; MIBG, marrow, biopsy.

Rx
- Neuroblastoma high-risk: chemo, surgery, stem cell rescue, radiotherapy, anti-GD2.
- Wilms: Europe (**SIOP**) gives chemotherapy before surgery; the US (**COG**) usually operates first.

II-S4 Malignant bone and soft tissue tumours

Def
- **Osteosarcoma**: malignant proliferation of osteoblasts; peak **10–20 years**.
- **Ewing sarcoma**: neuroectoderm-derived; **rhabdomyosarcoma**: commonest soft tissue sarcoma of childhood.

Cause
- Ewing carries **t(11;22)** (EWSR1–FLI1); boys.
- Osteosarcoma: hereditary retinoblastoma, Li–Fraumeni, prior radiation.
- Embryonal rhabdomyosarcoma (sarcoma botryoides): girls.

Patho
- Osteosarcoma arises in the **metaphysis** of long bones, often **knee**; cells make **osteoid**.
- Ewing: small round blue cells in the **diaphysis** (femur). Commonest metastasis site: **lungs**.

Signs
- Osteosarcoma: painful enlarging mass, pathological **fracture**, bone pain with swelling.
- Ewing may mimic **osteomyelitis**. Sarcoma botryoides: **grape**-like vaginal mass with bleeding.
- Soft tissue lump **>2 cm, deep, enlarging or painful** → urgent ultrasound.

Dx
- Osteosarcoma X-ray: **sunburst** and **Codman triangle**; Ewing: **onion-skin** periosteal reaction.
- Rhabdomyoblasts stain for **desmin** and **myogenin**; soft tissue sarcoma → core needle biopsy.

Rx
- Osteosarcoma: **en bloc resection** and **chemotherapy**; **radio-resistant**.
- Ewing is **radiosensitive**: chemo + surgery and/or radiotherapy.
- Soft tissue sarcoma: wide excision, chemotherapy, radiation.

II-55 Early and late onset adverse effects of chemotherapy

Def
- **Chemotherapeutic drugs** are effective against **rapidly** dividing cells → early and late toxicities.

Cause
- High-turnover tissues (marrow, mucosa, hair) plus drug-specific organ toxicity.
- Etoposide (topoisomerase II inhibitor) is linked to secondary **AML**.

Patho
- Marrow nadir **7–14 d** → neutropenia → infection risk.
- Tumour lysis syndrome: high **potassium**, high phosphate, high urate, low **calcium**.

Signs
- Anthracyclines → **dilated cardiomyopathy**; cisplatin → **high-frequency sensorineural hearing loss**, renal magnesium wasting.
- Cyclophosphamide/ifosfamide → **haemorrhagic cystitis** (acrolein); vincristine → **peripheral neuropathy**, constipation, paralytic ileus.
- Bleomycin → lung fibrosis, mucositis; vinblastine → myelosuppression.

Dx
- **Febrile neutropenia** = emergency → cultures.
- Late effects: cardiac, gonadal, endocrine, neurocognitive and **second cancers**.

Rx
- Febrile neutropenia: IV broad-spectrum antibiotics within **1 hour**.
- Protect with **dexrazoxane** (anthracyclines) and **mesna** (cyclophosphamide); rasburicase contraindicated in **G6PD deficiency**.
- Vincristine is fatal if given **intrathecally**; lifelong follow-up.

Section III · Thesis line

III-1 Diabetes mellitus. Management of diabetic ketoacidosis

Def
- **Type 1 DM**: autoimmune beta-cell destruction → absolute insulin deficiency; peaks 4–6 y, puberty.
- Four symptoms: **polyuria**, **polydipsia**, weight loss despite good appetite, fatigue.

Cause
- Autoimmune: antibodies to **GAD, IA-2, ZnT8, insulin**.
- Commonest associated autoimmune diseases: **coeliac disease** and **Hashimoto thyroiditis**.

Patho
- No insulin → hyperglycaemia → osmotic diuresis → dehydration.
- Ketone production → acidosis → **Kussmaul breathing**; total body K^+ depleted.

Signs
- Polyuria, polydipsia, weight loss despite good appetite, fatigue; secondary enuresis.
- DKA: vomiting, abdominal pain, Kussmaul breathing, ketotic breath, drowsiness.

Dx
- Typical symptoms + high glucose (18.2 mmol/L) → **paediatric diabetes centre immediately**.
- DKA: glucose >11 mmol/L, venous **pH , plus ketones.**

Rx
- DKA: fluids, then insulin infusion **0.05–0.1 U/kg/h** with **no bolus**; potassium, dextrose.
- Leading cause of death: **cerebral oedema** (headache, bradycardia) → mannitol or 3% saline.
- Most modern: **insulin pump with an integrated glucose sensor**.

III-2 Metabolic syndrome. Type 2 diabetes mellitus

Def
- **Metabolic syndrome**: central obesity + ≥2 of high TG, low HDL, HTN, high fasting glucose.
- **T2DM**: obese pubertal teen; islet **autoantibodies** negative, high **C-peptide**.

Cause
- Central obesity, puberty, family history of T2DM, high-risk ethnic group.
- MODY: autosomal dominant, **3 generations**, not obese.

Patho
- **Visceral fat → insulin resistance** → hyperinsulinaemia + free fatty acids → the cluster.
- Insulin resistance → **acanthosis nigricans**, fatty liver, PCOS.

Signs
- Thick, dark, velvety skin in neck and axillae: **acanthosis nigricans**.
- Often asymptomatic on screening; polyuria, polydipsia, occasionally DKA; raised ALT, sleep apnoea, SCFE.

Dx
- Type 1 vs type 2: islet **autoantibodies** and **C-peptide** in every new case.
- **GCK**-MODY: mild stable fasting hyperglycaemia. Diabetes <6 months → genetic testing.

Rx
- First-line drug without ketosis: **metformin** plus lifestyle; **insulin** if ketosis or HbA1c ≥8.5%.
- **HNF1A**-MODY very sensitive to **sulfonylureas**; GCK-MODY needs **no treatment**.
- Screen complications from diagnosis.

III-3 Diabetes insipidus

Def
- Central DI = deficient **vasopressin (ADH)**; nephrogenic DI = **renal resistance** to it.
- Dilute polyuria.

Cause
- Central: **craniopharyngioma**, germinoma, **LCH**, trauma, neurosurgery, meningitis.
- Nephrogenic: **X-linked AVPR2**, lithium, hypercalcaemia, hypokalaemia.

Patho
- No ADH or V2 resistance → no **aquaporin-2** in collecting duct → dilute urine.
- Free water loss → ↑ plasma osm → thirst; infants → **hypernatraemic dehydration**.

Signs
- Polyuria >2 L/m²/day, nocturia, secondary enuresis.
- Intense thirst, prefers cold water; infant FTT, vomiting, fever.

Dx
- DI confirmed by high plasma osmolality with urine osmolality below **300 mOsm/kg**.
- After water deprivation, desmopressin concentrates urine by >50% in **central** DI, not **nephrogenic**.
- Central DI needs **MRI**: craniopharyngioma, germinoma, **Langerhans cell histiocytosis**.

Rx
- Central → **desmopressin**; main risk is **hyponatraemia**.
- Nephrogenic → low-solute diet and **thiazide** diuretics; stop lithium.
- Correct hypernatraemia ≤0.5 mmol/L per hour.

III-4 Disorders of neuro- and adenohypophysis

Def
- Hypo- or hyperfunction of anterior (GH, ACTH, TSH, LH, FSH, prolactin) or posterior (vasopressin) pituitary.

Cause
- **Congenital**: midline defects — septo-optic dysplasia, cleft palate, single central incisor.
- **Acquired**: **craniopharyngioma**, cranial irradiation, trauma, histiocytosis, infection.

Patho
- GH deficiency → low growth velocity, **delayed bone age**, normal proportions.
- Craniopharyngioma compresses the **optic chiasm** → **bitemporal hemianopia**; DI.

Signs
- Newborn hypopituitarism: **hypoglycaemia**, **micropenis**, prolonged jaundice.
- Prolactinoma: galactorrhoea, amenorrhoea. Weight gain with **falling height velocity** suggests Cushing syndrome.
- Posterior: **DI**, or **SIADH**.

Dx
- Low IGF-1 → failure of **two** GH stimulation tests → pituitary MRI.
- Craniopharyngioma: suprasellar, cystic and **calcified**.

Rx
- GHD → **recombinant GH** until final height.
- Always replace **cortisol** before **thyroxine**.
- Prolactinoma: **cabergoline**; craniopharyngioma → surgery ± radiotherapy; SIADH → fluid restriction.

III-5 Congenital adrenal hyperplasia (CAH)

Def
- **CAH**: AR defect of cortisol synthesis; ~95% **21-hydroxylase** deficiency, commonest adrenocortical enzyme defect.

Cause
- Autosomal recessive **CYP21A2** mutation.
- Rare: **11β-hydroxylase** deficiency (virilisation and hypertension), 17α-hydroxylase.

Patho
- Low cortisol → no feedback → **ACTH up** → adrenal hyperplasia.
- Salt-wasting: insufficient **glucocorticoid** and **mineralocorticoid**, excessive **androgen**.

Signs
- Girls: **virilised genitalia at birth**.
- Boys: salt-wasting crisis at **1–3 weeks** — hyponatraemia, **hyperkalaemia**, low glucose, shock.
- Simple virilising: precocious pseudopuberty.

Dx
- Diagnostic marker: raised **17-hydroxyprogesterone** (newborn screening); high ACTH and renin.
- Karyotype, pelvic ultrasound, CYP21A2 genetics.

Rx
- Crisis: **IV hydrocortisone**, 0.9% saline 10–20 mL/kg, IV dextrose.
- Long-term: **hydrocortisone**, **fludrocortisone** and salt in infancy.
- **Stress dosing** in illness; emergency card.

III-6 Thyroid diseases

Def
- **Congenital** or **acquired** hypothyroidism; hyperthyroidism mostly **Graves** (Basedow's disease, TRAb-driven).

Cause
- Commonest congenital cause: thyroid **dysgenesis** (~85%), most often an **ectopic** gland.
- **Maternal iodine deficiency** → **temporary primary hypothyroidism** in the newborn.
- Acquired: **Hashimoto thyroiditis**, commonest in children.

Patho
- No hormone → slowed metabolism, growth and brain development → delayed bone age.
- TSH-receptor stimulating antibodies → gland overdrive → thyrotoxicosis.

Signs
- Neonate: **prolonged jaundice**, constipation, hypotonia, macroglossia, umbilical hernia.
- Hashimoto: painless goitre, **growth failure**.
- Basedow's: **weight loss**, **tachycardia**, **Graefe sign**.

Dx
- **Heel-prick TSH** day 3–5.
- Hashimoto: high TSH, low free T4, **anti-TPO**. Graves: suppressed TSH, high T4/T3, **TRAb**.

Rx
- Congenital: **levothyroxine 10–15 µg/kg/day** within the first **2 weeks**.
- Graves: **methimazole** + propranolol; **propylthiouracil** is avoided because of severe hepatotoxicity.
- Sore throat/fever → urgent blood count.

III-7 Disorders of calcium and phosphorus homeostasis. Rickets.

Def
- **Rickets** = failed growth-plate mineralisation; mostly **vitamin D deficiency** in unsupplemented breastfed infants.

Cause
- Commonest: **vitamin D deficiency** — unsupplemented breastfeeding, dark skin, little sunlight.
- Phosphopenic: X-linked hypophosphataemic rickets (PHEX, high FGF23), Fanconi syndrome.

Patho
- Low vitamin D → less Ca/P absorption → **PTH rises** → phosphate wasting.
- Low Ca × P → growth plate cannot mineralise → deformity.

Signs
- Soft, ping-pong-ball skull bones = **craniotabes**; rachitic rosary, Harrison sulcus, wide wrists, bowed legs.
- Infants <6 mo: hypocalcaemic seizures, tetany, laryngospasm.

Dx
- Nutritional rickets bloods: phosphate **low**, ALP **very high**, PTH **high**.
- XLH: low phosphate with normal calcium and normal **PTH**.
- X-ray: **cupping**, fraying and splaying of the metaphyses.

Rx
- Prevent: all infants receive **400 IU** of vitamin D daily.
- Nutritional: high-dose vitamin D ≥12 weeks + calcium; XLH: phosphate + calcitriol or burosumab.
- Hypocalcaemia that won't correct with IV calcium — check **magnesium**.

III-8 Systemic lupus erythematosus (SLE)

Def
- Multisystem autoimmune disease: immune complexes against nuclear antigens.
- cSLE: **adolescent girls**, more severe than adult disease.

Cause
- Female predominance after puberty; more common in non-white populations.
- Anti-**histone** antibodies suggest drug-induced lupus (hydralazine, procainamide, isoniazid).

Patho
- Immune complexes deposit → organ damage; in active SLE complement **C3 and C4** fall.
- Maternal **anti-Ro/La** antibodies cause **congenital heart block**.

Signs
- Fever, **malar rash** sparing nasolabial folds, photosensitivity, painless oral ulcers.
- Non-erosive arthritis, serositis, **nephritis 50–80%**, seizures/psychosis.
- Coombs-positive haemolysis, leucopenia, thrombocytopenia.

Dx
- Entry criterion: ANA at a titre of at least **1:80**.
- **Anti-dsDNA** tracks activity and nephritis; **anti-Sm** most specific.
- High **CRP** in a febrile lupus patient suggests infection, not flare.

Rx
- Every SLE patient takes **hydroxychloroquine** + sun protection.
- Moderate → azathioprine, methotrexate or mycophenolate.
- Severe → IV methylprednisolone → mycophenolate or IV cyclophosphamide; rituximab if refractory.

III-9 Dermatomyositis, scleroderma

Def
- **JDM**: autoimmune small-vessel vasculopathy of skin and muscle; commonest childhood myopathy.
- Commonest scleroderma in children: **localised (linear)**; systemic sclerosis rare.

Cause
- JDM peaks 5–10 y, girls more often.

Patho
- Complement deposition → capillary loss in muscle and skin → weakness, rash, ulcers.
- Rarely linked to malignancy (unlike adults) → no routine cancer screening.

Signs
- Weakness of the **proximal muscles of the limbs** and neck flexors; **Gowers** sign.
- Purple eyelids = **heliotrope** rash; red scaly papules over knuckles = **Gottron papules**.
- **Calcinosis** much commoner than in adult dermatomyositis.

Dx
- **CK, LDH, AST/ALT, aldolase** + **MRI** (T2/STIR) muscle oedema.
- Biopsy: perifascicular atrophy. SSc antibodies: **anti-Scl-70**, anticentromere.

Rx
- First-line JDM: steroids plus **methotrexate**; IVIG if severe.
- Localised scleroderma: methotrexate ± initial steroids; physiotherapy.
- SSc: CCB for Raynaud, ACE inhibitor for renal crisis.

III-10 Vasculitis. Henoch-Schönlein purpura

Def
- Commonest vasculitis of childhood: **Henoch–Schönlein purpura (IgA vasculitis)**, small vessel, 3–10 years.

Cause
- Follows upper respiratory infection (often streptococcal); autumn and winter.
- **Takayasu** is **large**-vessel, **Kawasaki** **medium**-vessel; granulomatosis with polyangiitis: **c-ANCA (PR3)**.

Patho
- IgA small-vessel vasculitis → **palpable purpura** with a **normal** platelet count.
- Gut → colic, bleeding, **ileo-ileal** intussusception; kidney → haematuria, proteinuria.

Signs
- **Palpable purpura** on buttocks and legs (always).
- Arthritis of ankles/knees (~75%); colicky abdominal pain (~50–75%).
- Nephritis (~20–50%) usually within 4–6 weeks.

Dx
- Clinical: purpura (mandatory) + abdominal pain, IgA histology, arthritis or renal involvement.
- FBC/clotting normal; urine protein–creatinine ratio, **BP**; US if severe pain.

Rx
- Supportive: rest, paracetamol (acetaminophen), fluids; NSAIDs only if kidneys normal.
- Steroids treat severe abdominal pain but **do not prevent** nephritis.
- Monitor urine and blood pressure for **6–12 months**.

III-11 Urinary tract infections, localisation, course, treatment, complications, VUR

Def
- UTI: cystitis (bladder mucosa) vs **pyelonephritis** (renal parenchyma).

Cause
- Most frequent pathogen: **E. coli** (P fimbriae); Klebsiella, Proteus, Enterococcus.
- Risks: **VUR**, posterior urethral valves, constipation, dysfunctional voiding.

Patho
- Bowel flora ascends; primary VUR is caused by a **short intramural ureteric tunnel**.
- Pyelonephritis → fever, high CRP → **renal scarring** → HTN, CKD.

Signs
- Infants: **non-specific** — fever, vomiting, poor feeding, jaundice, sepsis.
- Older: dysuria, frequency (cystitis) vs fever, loin pain (pyelonephritis).

Dx
- Clean/catheter sample: dipstick nitrite + leucocyte esterase, culture.
- Pyelonephritis sediment: leucocytes, **white blood cell casts** and bacteria.
- VUR: **MCU**, **sonocystography**, **dynamic kidney scintigraphy**; **DMSA** scan 4–6 months later shows scarring.

Rx
- Infants under **3 months**: admit, IV antibiotics (cefotaxime/ceftriaxone).
- Pyelonephritis: oral cefalexin or co-amoxiclav 7–10 days; cystitis 3 days.
- VUR: fix bowel/bladder, prophylaxis; surgery for breakthrough infections.

III-12 Nephrolithiasis in childhood

Def
- Urinary tract stones in children; mostly **metabolic**.

Cause
- Commonest metabolic abnormality: **hypercalciuria**; also hypocitraturia, hyperoxaluria, **cystinuria**, distal RTA.
- **Infection** (Proteus), anatomical stasis, drugs (furosemide in preterms, topiramate).

Patho
- Excess solute + little water + low **citrate** (main inhibitor) → supersaturation → crystals.
- Struvite forms in **alkaline** urine via **urease**-producing organisms such as Proteus.

Signs
- Adolescents: **renal colic** flank → groin, nausea, haematuria.
- Young children: **atypical** — vague pain, irritability, UTI, painless haematuria.
- Hexagonal crystals in urine indicate **cystinuria**.

Dx
- First-line imaging: **ultrasound**; KUB for calcium stones; low-dose CT only if unclear.
- Metabolic work-up in every child: blood + urine calcium, oxalate, citrate, cystine; stone analysis.

Rx
- NSAIDs + paracetamol, hydration; obstructed infected kidney → urgent drainage.
- <5 mm usually pass (tamsulosin); else ESWL, ureteroscopy, PCNL for staghorn.
- Prevent: fluids, low salt, **thiazide** (reduces urinary calcium), potassium citrate, tiopronin for cystine.

III-13 Glomerular diseases

Def
- **Glomerulonephritis**: glomerular inflammation → haematuria ± proteinuria; nephritic syndrome.

Cause
- Most frequent type: **IgA nephropathy**; **PSGN** after group A strep.
- Others: lupus, MPGN, HSP nephritis, ANCA, anti-GBM.
- Haematuria with sensorineural deafness suggests **Alport syndrome**.

Patho
- Immune complexes break GBM → red cells + casts, proteinuria.
- ↓ GFR + ongoing Na^+ reabsorption → oedema, hypertension, oliguria.

Signs
- Cola urine, **periorbital oedema, hypertension, oliguria**.
- IgA: macroscopic haematuria **1–2 days** after URTI. PSGN: **1–3 weeks** after pharyngitis, **3–6 weeks** after impetigo.

Dx
- Urine: **red blood cells with red blood cell casts**, most commonly with **proteinuria**.
- IgA: **normal** C3. PSGN: C3 **low**, normalises by **6–8 weeks**; ASO/anti-DNase B.

Rx
- PSGN: supportive — fluid/salt restriction, furosemide, antihypertensives, penicillin.
- IgA: ACE inhibitor; steroids for heavier disease.
- RPGN: pulse methylprednisolone + immunosuppression; plasma exchange if anti-GBM.

III-14 Nephrotic syndrome

Def
- **Heavy proteinuria** (UPCR ≥200 mg/mmol), **albumin <30 g/L**, **oedema**, + hyperlipidaemia.

Cause
- Commonest cause in young children: **minimal change disease**, age 2–6, boys, ~80–90%.
- Atypical if <1 or >12 y, HTN, macroscopic haematuria, low C3, renal impairment.

Patho
- Circulating factor → **podocyte effacement** → selective albumin leak → low oncotic pressure → oedema.
- Thrombosis: urinary loss of **antithrombin III**; lost IgG → infection.

Signs
- **Periorbital oedema** in the morning → legs, scrotum, ascites, effusions.
- Peritonitis classically caused by **Streptococcus pneumoniae**; also renal vein thrombosis, hypovolaemia/AKI.

Dx
- Urine protein–creatinine ratio, serum albumin, lipids.
- No remission after 4 weeks of daily prednisolone = **steroid-resistant**; needs **biopsy and genetic testing**.

Rx
- First episode: **oral prednisolone** daily for 4 weeks, then **alternate days** for 4 weeks.
- Albumin 20% for symptomatic hypovolaemia; pneumococcal vaccine; no added salt.
- Relapsing/dependent → CNI, mycophenolate, rituximab; resistant → CNI + ACE inhibitor.

III-15 Acute and chronic kidney failure. Kidney transplantation.

Def
- KDIGO AKI: creatinine rise ≥**26.5 µmol/L (0.3 mg/dL)** in 48 h or ≥**1.5×** in 7 days.
- **CKD**: abnormal structure/function >3 months; GS <15 mL/min/1.73 m^2.

Cause
- Pre-renal commonest; post-renal: PUV.
- Intrinsic AKI after bloody diarrhoea: **HUS**; commonest CKD cause: **CAKUT**.

Patho
- Low perfusion → Na and water kept → **FENa <1%**; **BUN** rises relatively more.
- No K^+ excretion → **hyperkalaemia** → arrhythmia.
- CKD: ↓ **erythropoietin** → anaemia; PTH ↑, Ca^{2+} ↓, phosphorus ↑.

Signs
- Oliguria, fluid overload, hypertension, uraemia.
- CKD: anaemia, bone disease, HTN, **growth failure**.
- HUS triad: **microangiopathic haemolytic anaemia**, **thrombocytopenia**, **acute kidney injury**.

Dx
- Creatinine, urine sodium, ultrasound; hyperkalaemia ECG: peaked T, wide QRS.

Rx
- Treat cause, stop nephrotoxins, balance fluids (insensible 300–400 mL/m^2/day + UO).
- Hyperkalaemia: IV **calcium gluconate** → salbutamol/insulin-glucose → resin → dialyse.
- CKD: ESA, binders; **pre-emptive living-donor transplantation**; atypical HUS: **eculizumab**.

III-16 Developmental abnormalities of the kidneys and the urinary tract

- **Def**
  - **CAKUT**: congenital kidney/urinary tract anomalies — commonest cause of childhood CKD.
- **Cause**
  - Failed ureteric bud–mesenchyme induction → agenesis, dysplasia; abnormal ascent → **horseshoe** kidney.
  - Duplex kidney: **upper**-pole ureter obstructs, **lower** refluxes; PUJ, PUV obstruct.
- **Patho**
  - Bilateral agenesis → **oligohydramnios** → **pulmonary hypoplasia** (Potter sequence).
  - PUV → back-pressure → renal dysplasia, CKD; short intravesical ureter → **VUR**.
  - VUR → pyelonephritis → **focal scars** → hypertension, CKD.
- **Signs**
  - Antenatal hydronephrosis: commonest cause (normal ureter) **PUJ obstruction**.
  - PUV (boys): commonest lower tract obstruction; poor stream, palpable bladder, UTI.
  - VUR: recurrent pyelonephritis; siblings ~30%. Horseshoe kidney: under **inferior mesenteric artery**.
- **Dx**
  - **US** after first days (early scans underestimate dilatation).
  - VUR diagnosed and graded (I–V) with **MCUG**; **MAG3** drainage; scars on **DMSA**.
- **Rx**
  - PUJ → **pyeloplasty** if function falls.
  - PUV → urgent **bladder catheter** → endoscopic valve ablation.
  - VUR: treat UTIs, prophylaxis if high grade; reimplantation for breakthrough infection.

III-17 Renal tubulopathy

- **Def**
  - Inherited tubular **transporter defects**, mapped to nephron segments.
- **Cause**
  - Proximal: Fanconi (commonest cause in children: **cystinosis**); loop: Bartter; DCT: Gitelman.
  - Collecting duct: distal RTA, type 4 RTA, nephrogenic DI (X-linked **AVPR2**).
- **Patho**
  - Bartter/Gitelman: salt wasting → renin, aldosterone up → **hypokalaemic alkalosis**.
  - Distal RTA: no H+ secretion → normal-gap acidosis; urine pH stays **>5.5**; hypercalciuria, hypocitraturia → **nephrocalcinosis**.
- **Signs**
  - Fanconi: glycosuria with **normal blood glucose** (proximal tubulopathy), phosphaturia → rickets, failure to thrive.
  - Bartter mimics **furosemide** and causes **hypercalciuria**, nephrocalcinosis, polyhydramnios.
  - Gitelman mimics **thiazides** and causes **hypomagnesaemia**, hypocalciuria, cramps, tetany.
- **Dx**
  - Blood gas, K, Mg, urine pH and calcium; cystinosis → corneal crystals.
  - Nephrogenic DI: dilute urine, hypernatraemia, no response to desmopressin.
- **Rx**
  - Cystinosis: **cysteamine** + replace water, bicarbonate, phosphate, K.
  - Bartter: indometacin + K; Gitelman: Mg + K.
  - Distal RTA: alkali (citrate); nephrogenic DI: free water, thiazide.

III-18 Chromosomal abnormalities

- **Def**
  - Numerical or structural chromosome errors: trisomies, microdeletions, sex chromosome aneuploidies.
- **Cause**
  - Down syndrome: ~**95%** non-disjunction (maternal age), ~**4%** translocation (karyotype the parents), ~1% mosaic.
  - Trisomy 18/13; **22q11** deletion; Williams 7q11.23; Turner 45,X; Klinefelter 47,XXY.
- **Patho**
  - Down: extra copies of **APP** gene → early Alzheimer; endocardial cushion defect → **AVSD**.
  - 22q11 deletion: **thymic** hypoplasia → infections; **parathyroid** hypoplasia → hypocalcaemia.
  - Turner short stature: **SHOX** haploinsufficiency.
- **Signs**
  - Down: hypotonia, **duodenal atresia** (trisomy 21), Hirschsprung, leukaemia, hypothyroid.
  - **Edwards**: clenched fists, overlapping fingers, rocker-bottom feet; **Patau**: holoprosencephaly, cleft lip, polydactyly.
  - 22q11: conotruncal heart defects; Turner: webbed neck, coarctation; Klinefelter: tall, small **firm testes**, infertility.
- **Dx**
  - Prenatal: combined first-trimester test, NIPT → CVS/amniocentesis with QF-PCR/karyotype.
  - Translocation Down → karyotype parents; microdeletions → microarray, FISH.
- **Rx**
  - Down: screen heart, thyroid, hearing, eyes, coeliac; multidisciplinary care.
  - Turner: growth hormone + oestrogen; check for Y material (gonadoblastoma).
  - Klinefelter: testosterone at puberty.

III-19 Mendelian inherited diseases (autosomal dominant and recessive), common disorders

- **Def**
  - Single-gene disease: **AD** needs one mutant allele; **AR** needs both.
- **Cause**
  - AD: NF1, **Marfan** (fibrillin-1), **achondroplasia** (FGFR3 gain-of-function), TSC, ADPKD, FH.
  - AR: CF, PKU, SMA (**SMN1** deletion), sickle, CAH, galactosaemia, Wilson.
- **Patho**
  - AD: haploinsufficiency, dominant negative, gain of function; **de novo** mutations common, linked to advanced **paternal** age.
  - AR: most enzyme deficiencies are AR because **50%** activity is sufficient, so carriers are healthy.
- **Signs**
  - AD pedigree: vertical, each child has **50%** risk, male-to-male transmission.
  - AR pedigree: horizontal, two carrier parents → **25%** risk per pregnancy; consanguinity.
  - Healthy sibling of an AR child: **2/3** carrier; meconium ileus → **cystic fibrosis**.
- **Dx**
  - Mutation analysis, gene panels, exome/genome sequencing; carrier screening.
  - Prenatal (CVS, amniocentesis); newborn screening for PKU, CAH, galactosaemia, CF, SMA.
- **Rx**
  - Non-directive genetic counselling with risk calculation.
  - Cascade testing of relatives; disease-specific therapy (e.g. nusinersen for SMA).

III-20 X-linked inherited diseases, common disorders

Def
- **XLR**: affected boys, carrier mothers, no **male-to-male** transmission.
- **XLD**: both sexes; affected father → all daughters affected.

Cause
- A carrier mother has a **50%** risk of each son being affected.
- Haemophilia father: all daughters **obligate carriers**; ~1/3 of Duchenne new mutations.

Patho
- Males hemizygous; lyonisation protects most carriers.
- Duchenne: **out-of-frame** dystrophin deletion; Becker: **in-frame** one.
- Fragile X: **CGG** repeat in FMR1; XLA: **BTK**.

Signs
- **DMD** age 2–5: Gowers sign, calf pseudohypertrophy, falls.
- Haemophilia: haemarthrosis; fragile X: ID, long face; XLA: infections from ~**6 months**.
- WAS: eczema, **thrombocytopenia with small platelets**, infections; XLD: **hypophosphataemic rickets**, **Rett syndrome**.

Dx
- First test in suspected Duchenne: serum **creatine kinase** (>10 000 U/L) → genetic testing.
- Haemophilia: isolated long APTT, normal PT and platelets.

Rx
- DMD: corticosteroids (prednisolone [prednisone] or deflazacort), physio, cardiac/respiratory surveillance.
- Avoid suxamethonium; haemophilia: factor prophylaxis, emicizumab, no IM/NSAIDs.
- **Live vaccines** must be avoided in XLA.

III-21 Multifactorial and polygenic genetic disorders

Def
- Disease appears when combined genetic and environmental liability crosses a **threshold**.

Cause
- Boys: **pyloric stenosis** (about **4–5** times commoner in boys), cleft lip, Hirschsprung, club foot.
- Girls: **DDH**, isolated cleft palate, NTDs; also CHD, diabetes, asthma.

Patho
- Relatives share genes → curve shifts toward threshold → empiric recurrence.
- **Carter** effect: when the less commonly affected sex is affected, relatives have a **higher** recurrence risk.

Signs
- First-degree recurrence of an isolated malformation is **2–5%**; falls sharply in second-degree relatives.
- Risk ↑ with severity, ≥2 affected relatives, affected rarer sex.

Dx
- Other anomalies/dysmorphism/delay → look for syndrome (22q11, trisomy 13) first.
- NTD screening: raised maternal serum **AFP** + anomaly ultrasound.

Rx
- Counsel with empiric risks; NTD recurrence ~3–5% without folate.
- Folic acid **400 µg** (all) or **5 mg** daily (high-risk woman) preconception to 12 weeks.
- US: 4 mg after previous NTD pregnancy.

III-22 Congenital metabolic disorders of amino acid metabolism

Def
- Mostly **autosomal recessive** enzyme defects of amino acid metabolism.

Cause
- Classic **PKU**: deficiency of **phenylalanine hydroxylase**; MSUD: branched-chain ketoacid dehydrogenase.
- Tyrosinaemia 1: fumarylacetoacetate hydrolase; homocystinuria: cystathionine β-synthase.

Patho
- Placenta clears metabolites → well at birth → sick **after protein feeds**.
- PKU: low **tyrosine** (precursor of **melanin**) → fair hair; leucine → cerebral oedema.

Signs
- Sepsis-like: poor feeding, vomiting, lethargy, seizures, coma.
- Untreated PKU: **musty** odour; sweet-smelling urine + neonatal encephalopathy: **maple syrup urine disease**.
- Tyrosinaemia 1: **liver failure** + Fanconi; homocystinuria: marfanoid, lens dislocates **downwards** (Marfan: upwards).

Dx
- PKU: newborn heel-prick phenylalanine.
- Crisis: glucose, gas, lactate, **ammonia**, amino acids, urine organic acids.

Rx
- Acute metabolic crisis: stop protein and give **high-energy IV glucose** (10%).
- Ammonia scavengers/haemofiltration, treat sepsis; PKU: lifelong **low-Phe diet** ± sapropterin.
- Tyrosinaemia → nitisinone; homocystinuria → pyridoxine (B6); MSUD → BCAA-restricted diet.

III-23 Congenital metabolic disorders of glucose and glycogen metabolism

Def
- Inherited defects of glycogen, galactose or fructose metabolism.

Cause
- Von Gierke (GSD I) is deficiency of **glucose-6-phosphatase** (AR).
- Pompe (GSD II), McArdle (GSD V), galactosaemia (GALT), fructose intolerance (aldolase B); MCAD: AR.

Patho
- G6Pase block → raised **lactate**, **uric acid** and **triglycerides**.
- MCAD deficiency causes **hypoketotic** hypoglycaemia during fasting.

Signs
- Low blood sugar in an infant with normal temperature and hepatomegaly: **von Gierke disease**.
- Pompe: **hypertrophic cardiomyopathy** and hypotonia; McArdle: exercise cramps, myoglobinuria, second wind.
- Neonatal jaundice, cataract, **E. coli** sepsis after milk: galactosaemia; hereditary fructose intolerance at **weaning**.

Dx
- GSD I: fasting hypoglycaemia 2–4 h post-feed, lactic acidosis, high urate.
- Galactosaemia: red-cell GALT, urine reducing substances; McArdle: raised CK.

Rx
- GSD I: never fast — frequent/overnight feeds, **uncooked cornstarch**, allopurinol.
- Pompe: enzyme replacement; galactosaemia: stop lactose now, soy formula.
- MCAD: avoid **fasting**; fructose intolerance: exclude fructose, sucrose, sorbitol.

III-24 Lysosomal storage diseases

Def
- Lysosomal enzyme deficiency → substrate stored in the cells that turn over most.

Cause
- Mostly recessive; the two X-linked ones: **Fabry** and **Hunter**.
- Gaucher is deficiency of **β-glucocerebrosidase**; Niemann-Pick, Tay-Sachs (hex A), Hurler (α-L-iduronidase), Pompe.

Patho
- Macrophage → hepatosplenomegaly, cytopenias; neuron → regression, cherry-red spot.
- Connective tissue (GAGs) → coarse face, dysostosis multiplex, valves, corneal clouding.

Signs
- Normal at birth, then **loss of skills** = key alarm.
- Cherry-red spot **without** hepatosplenomegaly: Tay-Sachs; **with** hepatosplenomegaly: Niemann-Pick.
- Gaucher: massive splenomegaly, bone crises; **Hurler**: corneal clouding (Hunter: none); Fabry: burning hands/feet.

Dx
- **Enzyme activity** in leucocytes or dried blood spots.
- Urinary GAGs/oligosaccharides, genetic testing; newborn screening for Pompe, MPS I.

Rx
- ERT does not treat CNS disease: it does not cross the **blood–brain barrier**.
- Gaucher, Fabry, Hunter: ERT; substrate reduction.
- HSCT before ~2 y in Hurler preserves cognition; multidisciplinary supportive care.

III-25 Rhinitis, adenoiditis, tonsillitis

Def
- Upper airway inflammation: rhinitis, **adenoid hypertrophy**, tonsillitis.

Cause
- Rhinitis: viral or allergic (IgE).
- Tonsillitis: mostly viral; **group A strep** commonest bacterial, 5–15 years.

Patho
- Adenoid hypertrophy peaks at **3–7** years: blocked nose, Eustachian tubes → snoring, OSA, otitis media.
- GAS → rheumatic fever; PSGN follows pharyngitis by **1–2 weeks**.

Signs
- Allergic: sneezing, itch; unilateral foul discharge in a toddler = **foreign body**.
- GAS: fever, exudate, tender anterior nodes, **no cough**.
- **Trismus, hot-potato voice and deviated uvula**: peritonsillar abscess (quinsy).

Dx
- GAS: rapid antigen test; negative backed by throat culture (low sensitivity).
- Adenoids: nasendoscopy or lateral neck X-ray; sleep study if OSA uncertain.

Rx
- Allergic rhinitis: most effective is an **intranasal steroid**; viral: no antibiotics.
- GAS: **penicillin V for 10 days** (prevents rheumatic fever, not PSGN); avoid **amoxicillin** if EBV possible.
- Adenectomy: recurrent URTIs, otitis media, blocked nasal breathing, OSA; tonsillectomy: **7**/1 y, 5/y×2, 3/y×3.

III-26 Acute upper respiratory tract infections, acute sinusitis, bronchitis

Def
- Upper airway infections: pharyngitis, sinusitis, croup, epiglottitis, bronchitis — **mostly viral**.

Cause
- Viral: adenovirus, rhinovirus, EBV; croup = parainfluenza.
- Bacterial: GAS; sinusitis and otitis media: **S. pneumoniae**.

Patho
- Cold swells ostiomeatal complex → sinus can't drain → bacterial colonisation.
- Croup: subglottic oedema in cricoid ring → small radius → high resistance.

Signs
- Strep: fever, exudate, tender nodes, no cough.
- Croup: barking cough, stridor; epiglottitis: **drooling and tripod** posture; bronchiolitis: wheeze, not stridor.
- Pertussis in young infants: **apnoea and cyanosis**, often no whoop.

Dx
- Bacterial sinusitis: over **10 days**, double worsening, or severe onset; CT only for complications.
- Strep: rapid antigen test or culture. Epiglottitis: do not **examine the throat**.

Rx
- Antibiotics only: **Streptococcus angina**, acute otitis media, acute bacterial sinusitis; not colds or bronchiolitis.
- Penicillin V for **10 days** to prevent rheumatic fever; croup: single oral **dexamethasone**.
- Epiglottitis: secure the **airway**; oseltamivir within **48 hours**.

III-27 Infections of the ear

Def
- **AOM**: middle-ear infection; **OME**: effusion without infection; **otitis externa**: canal infection.

Cause
- Commonest bacterial cause of AOM: **S. pneumoniae**, NTHi, Moraxella; day-care, smoking, Down.
- Otitis externa: **Pseudomonas aeruginosa**.

Patho
- Children get AOM because their Eustachian tube is **short, horizontal and floppy**.
- Cold → tube blocks → effusion → bulging drum; perforation → otorrhoea.
- OME: fluid dampens drum → **conductive** hearing loss.

Signs
- AOM: otalgia, ear-pulling, fever, bulging drum.
- OME: dull retracted drum, type B tympanogram.
- Typical sign of otitis externa: **pain and/or sensitivity of the tragus**.

Dx
- Otoscopy; tympanometry for OME.
- Protruding ear with postauricular swelling after AOM suggests **mastoiditis** → IV antibiotics, CT.

Rx
- AOM: analgesia first; amoxicillin if under **2 years** with bilateral disease or otorrhoea.
- OME: wait 3 months; **grommets** in chronic serous otitis media or Eustachian tube dysfunction.
- Externa: aural toilet, topical antibiotic ± steroid drops.

III-28 Pneumonia and complications in infants and children

Def
- Lung infection with alveolar exudate; complications are effusion, empyema, abscess.

Cause
- Neonates: GBS, E. coli, Listeria; preschool: viruses commonest overall.
- Typical pneumonia: most commonly **S. pneumoniae**; school age: Mycoplasma; pneumatoceles after influenza → **S. aureus**.

Patho
- Alveolar exudate → low compliance + shunt → tachypnoea, indrawing, grunting, hypoxaemia.
- Parapneumonic effusion stages: exudative, **fibrinopurulent**, organising.

Signs
- WHO tachypnoea threshold for age 1–5 years: **40**/min (≥60 <2 mo, ≥50 2–12 mo).
- Crackles, bronchial breathing, dullness; abdominal pain from diaphragmatic pleura.
- Fever persisting after **48 hours** of antibiotics suggests empyema or abscess.

Dx
- Clinical; CXR not routine if managed at home.
- Admit if SpO_2 <92%, grunting, apnoea, poor feeding; US shows effusion septations.

Rx
- First-line oral antibiotic for childhood CAP: **amoxicillin**; add macrolide if Mycoplasma.
- Severe: IV ampicillin/penicillin G or ceftriaxone (US); co-amoxiclav/cefuroxime (UK).
- Empyema: drain; thick pus dissolved by **fibrinolysis (urokinase/streptokinase)** → VATS if fails.

III-29 Congenital lung malformations

Def
- Congenital anomalies of diaphragm, lung parenchyma and chest wall.

Cause
- CDH is usually a **left** posterolateral (**Bochdalek**) defect: pleuroperitoneal canal fails to close.
- CPAM, sequestration, lobar emphysema; two most frequent chest deformities: **pectus excavatum** and **pectus carinatum**.

Patho
- CDH: gut in chest → pulmonary hypoplasia + pulmonary hypertension drive mortality.
- Lobar emphysema most often affects the **left upper** lobe: deficient cartilage → ball-valve.

Signs
- CDH: cyanosis, **scaphoid** abdomen, heart sounds displaced to the **right**.
- CPAM/sequestration: recurrent pneumonia in the **same lobe**, distress.

Dx
- Often antenatal ultrasound; CPAM = cysts that **communicate with the airway**.
- Sequestration has a **systemic arterial supply, usually from the aorta** (Doppler/CT angiography).
- Pectus: Haller index on CT/MRI, echocardiography, spirometry.

Rx
- In suspected CDH avoid **bag-and-mask ventilation**; intubate, NG tube on suction; NO/ECMO, repair when stable.
- CPAM/sequestration/lobar emphysema: resection if symptomatic.
- Pectus excavatum: **Nuss** procedure; carinatum: compressive brace.

III-30 Cystic fibrosis. Lung transplantation

Def
- **Autosomal recessive CFTR** defect; commonest mutation **F508del** causes protein misfolding.

Cause
- Carrier ~1 in 25 Europeans; class II F508del, class III gating (G551D).

Patho
- No Cl^- secretion + unopposed ENaC → dehydrated mucus → infection → bronchiectasis.
- Blocked pancreatic ducts → exocrine insufficiency; sweat duct cannot **reabsorb** chloride → salty sweat.

Signs
- **Meconium ileus** (X-ray: soap-bubble; contrast enema: microcolon), steatorrhoea, FTT despite good appetite.
- Main chronic lung pathogen: **Pseudomonas aeruginosa** (S. aureus early); nasal polyps, male infertility.
- Salty baby, hyponatraemic dehydration in heat; ADEK deficiency.

Dx
- Newborn screening measures **immunoreactive trypsinogen** (IRT) → CFTR panel.
- Diagnostic sweat chloride: **≥60 mmol/L** (30–59 intermediate); faecal elastase low.

Rx
- Airway clearance, **dornase alfa** (cleaves neutrophil DNA), hypertonic saline, Pseudomonas eradication, azithromycin.
- Enzymes every meal, high-calorie diet, ADEK, salt; CFTR modulators (elexacaftor/tezacaftor/ivacaftor).
- End-stage: **bilateral** lung transplant (avoids seeding from native lung); limited by bronchiolitis obliterans.

III-31 Bronchial asthma

Def
- Asthma = chronic airway inflammation + **bronchial hyperresponsiveness** + reversible obstruction.

Cause
- Atopy (eczema, rhinitis), family history, tobacco smoke, prematurity, obesity.
- Triggers: viruses, exercise, allergens, smoke.

Patho
- Th2/mast cells → IL-4/5/13, histamine, leukotrienes (C4, D4, E4).
- → **bronchospasm** (β2-agonist) + oedema (steroid) + mucus → remodelling.

Signs
- Recurrent wheeze, night cough, exercise symptoms, chest tightness.
- Life-threatening: **silent chest**, SpO_2 <92%, PEF <33%; normal or rising **PaCO2** signals exhaustion.

Dx
- Under 5 years: **clinical** diagnosis (trial of ICS); from 5 years FeNO and spirometry.
- FeNO ≥35 ppb; reversibility: FEV1 increase of at least **12%**; PEF variability.

Rx
- Cornerstone controller: **inhaled corticosteroid** (low dose) → MART/LABA or LTRA → biologics; spacer, action plan.
- Acute: oxygen, **salbutamol**, ipratropium, prednisolone (US: dexamethasone).
- Severe acute asthma not responding to nebulised bronchodilators: add IV **magnesium sulfate** → PICU.

III-32 Allergy in infants and children. Definition of atopy, atopic diseases.

Def
- Atopy is a genetic tendency to produce **IgE** driven by a **Th2** response.

Cause
- Atopic march, in order: **food allergy** - **atopic dermatitis** – **bronchial asthma** - **allergic rhinitis**.
- Loss-of-function mutations in **filaggrin** break the skin barrier; milk, egg, peanut, nuts, fish.

Patho
- Th2 (IL-4/13) → IgE on mast cells → cross-linking → histamine, leukotrienes.
- IL-5 → eosinophils → late-phase chronic inflammation.

Signs
- IgE-mediated (minutes–2 h): urticaria, angioedema, vomiting, wheeze, **anaphylaxis**.
- Non-IgE (hours–days): eczema flare, reflux, proctocolitis, FPIES.

Dx
- Food allergy frequency in childhood: parents report **approx. 7-10%**; in reality **1-2 %**.
- Allergen specific IgE is **not totally reliable**; elimination diet, then allergen re-challenge after symptoms disappear.

Rx
- Avoidance, dietitian, action plan; milk-allergic infant → **extensively hydrolysed formula**, not goat's milk.
- Anaphylaxis: IM adrenaline **10 µg/kg (0.01 mL/kg of 1 mg/mL)**, **anterolateral thigh**; repeat at 5 min.

III-33 Symptoms and treatment of decompensated heart failure

Def
- Heart cannot meet demand → congestion and low output; decompensated = shock/collapse.

Cause
- First week: **duct-dependent** left lesions (HLHS, coarctation), SVT, sepsis.
- Large VSD: heart failure at **4–8 weeks** because pulmonary vascular resistance falls (AVSD, PDA).
- Later: myocarditis, cardiomyopathy, arrhythmia, Kawasaki, anaemia.

Patho
- ↓ Output → sympathetic + RAAS → tachycardia, sweating, salt/water retention.
- ↑ Pulmonary venous pressure → tachypnoea; ↑ systemic venous pressure → **hepatomegaly**.

Signs
- Poor feeding with **sweating**, tachypnoea and failure to thrive.
- Most reliable sign of heart failure in an infant: **hepatomegaly**; gallop, cardiomegaly, weak pulses.

Dx
- **Echocardiography** defines cause; CXR (cardiomegaly), ECG.
- BNP/NT-proBNP, electrolytes, renal function, gas and lactate.

Rx
- Sit up, careful O_2, high-calorie NG feeds (up to 150 kcal/kg/day).
- Furosemide + **spironolactone** (spares potassium, blocks aldosterone), ACE inhibitor; **milrinone** if decompensated.
- First-week duct-dependent lesion → **prostaglandin E1**; then repair defect or transplant.

III-34 Cyanotic congenital heart disease

Def
- Congenital heart lesions causing central cyanosis — the **5 Ts**.

Cause
- R→L shunt, low pulmonary flow: Fallot, pulmonary/tricuspid atresia, critical PS.
- Commonest cyanotic lesion in first days: **transposition of the great arteries**; truncus (22q11), TAPVR.

Patho
- Cyanosis visible at ~3–5 g/dL deoxygenated haemoglobin.
- Fallot: **RVOT obstruction**, VSD, overriding aorta, RV hypertrophy; spell → more R→L shunt.
- Chronic hypoxia → EPO → polycythaemia, clubbing; brain abscess.

Signs
- Fallot: murmur **softens** in a tet spell; boot-shaped heart.
- TGA: profound cyanosis first hours–days, often no murmur; egg on a string.

Dx
- **Hyperoxia test**: PaO2 does not rise with 100% oxygen in cardiac cyanosis.
- Echocardiography; pulse-oximetry screening (right hand + foot) detects critical CHD.

Rx
- Blue newborn not improving on O_2 → **PGE1** (watch apnoea).
- Tet spell: **knee-chest** (raises SVR), O_2, **morphine**, fluids, **propranolol**, phenylephrine; repair 3–6 mo.
- TGA: prostaglandin → **balloon atrial septostomy** → **arterial switch** in first 2–3 weeks.

III-35 Non-cyanotic congenital heart disease

Def
- Congenital heart disease without cyanosis: **left-to-right shunts** or **obstructive lesions**.

Cause
- Commonest defect: **VSD**; also ASD, AVSD (Down), PDA (preterm).
- Obstruction: **coarctation** (Turner), aortic stenosis, pulmonary stenosis (Noonan).

Patho
- Long-standing left-to-right shunt reverses = **Eisenmenger syndrome**.
- A **small** VSD gives a louder murmur: high pressure gradient.

Signs
- 4 signs: **heart murmur**, **heart failure**, **cyanosis**, **peripheral pulse abnormalities**.
- **Fixed** wide splitting of S2 = ASD; PDA: **continuous machinery** murmur, **bounding** pulses.
- Coarctation: upper-limb **hypertension**, weak, delayed **femoral** pulses.

Dx
- Congenital defects are diagnosed with **echocardiography**.
- Holosystolic murmur at left lower sternal border = **VSD**.
- Innocent murmur: soft, systolic, postural, no other signs.

Rx
- Large VSD: surgery in first months; ASD: catheter device at 3–5 years.
- Critical coarctation: **prostaglandin E1**; preterm PDA: ibuprofen, indomethacin or paracetamol.

III-36 Cardiomyopathy

Def
- Heart muscle disease: **dilated** (commonest), **hypertrophic**, **restrictive**.

Cause
- DCM: viral **myocarditis**, **anthracycline (doxorubicin)** chemotherapy, Duchenne.
- HCM: **autosomal dominant** **sarcomere** gene mutation; newborn: **infant of a diabetic mother**.

Patho
- DCM: can't empty → low EF → dilation → systolic failure, arrhythmia, thrombi.
- HCM: can't fill → dynamic **LVOT obstruction** → sudden death.
- RCM: can't stretch → **huge atria**, early pulmonary hypertension.

Signs
- DCM: tachypnoea, poor feeding, sweating, hepatomegaly, **gallop**, cool peripheries.
- HCM: exertional syncope/dyspnoea; murmur **increases** with Valsalva and standing.

Dx
- **Echo**: dilated low-EF LV or hypertrophy; also ECG, CXR, troponin, BNP, cardiac MRI.
- **Screen first-degree relatives**.

Rx
- DCM: **ACE-i, ARB, β-blockers (carvedilol), spironolactone** improve survival; transplant if end-stage.
- HCM: **beta-blockers**; avoid inotropes, vasodilators, excess diuretics; high-risk: **ICD**.
- RCM → early transplant.

III-37 Myocarditis, pericarditis - symptoms and management

Def
- **Myocarditis**: heart muscle inflammation; **pericarditis**: inflamed pericardium ± effusion.

Cause
- Commonest cause of myocarditis in children: **viral infection** (enteroviruses, adenovirus, parvovirus B19).
- Pericarditis: viral, **purulent (Staphylococcus aureus)**, TB, autoimmune, post-pericardiotomy.

Patho
- Viral + immune myocyte loss → ↓ stroke volume → tachycardia, congestion, arrhythmias.
- Tamponade: fluid → ventricles cannot fill → stroke volume falls.

Signs
- Myocarditis: tachycardia out of proportion to fever; **hepatomegaly** = reliable sign of venous congestion.
- Pericarditis: pain relieved by **sitting up and leaning forward**, friction rub.
- Tamponade: **Beck triad**, **pulsus paradoxus** (inspiratory fall in systolic pressure over **10** mmHg).

Dx
- Myocarditis: sinus tachycardia, **low-voltage QRS**; troponin, BNP, echo, cardiac MRI.
- Pericarditis: widespread concave **ST elevation** with **PR depression**; echo shows effusion.

Rx
- Myocarditis: small fluid boluses, furosemide, inotropes, ECMO; no competitive sport for **3–6 months**.
- Pericarditis: NSAID + colchicine; tamponade → urgent pericardiocentesis.
- Purulent: surgical drainage + IV anti-staphylococcal antibiotics.

III-38 Infective endocarditis – risk factors, prophylaxis, management

Def
- Infection of the endocardium, usually on a valve or defect.

Cause
- Risk: CHD with prosthetic material, prosthetic valves, previous IE.
- Neonates with normal hearts: **central venous catheters**.
- **Viridans streptococci**: subacute, after dental bacteraemia; **S. aureus**: acute.

Patho
- Turbulent jet damages endothelium → platelet-fibrin **vegetation** → seeded by bacteraemia.
- Emboli; **immune complexes** → glomerulonephritis, Osler nodes, Roth spots.

Signs
- Prolonged fever, malaise, **new/changing murmur**, splenomegaly (**FROM JANE**).
- Painful finger-pulp nodules are **Osler nodes**; painless palmar macules are **Janeway lesions**.

Dx
- Basis of diagnosis: **haemoculture** and **echocardiography**.
- Before antibiotics take at least **three** blood cultures from separate venepunctures.

Rx
- IV bactericidal antibiotics for **4–6 weeks**: amoxicillin + flucloxacillin [nafcillin] + gentamicin.
- Prosthetic material → vancomycin + gentamicin; surgery for heart failure, emboli.
- Prophylaxis, high-risk: **single oral dose of amoxicillin** 30–60 minutes before dental work.

III-39 Causes of malabsorption. Celiac disease.

Def
- **Maldigestion** (no enzymes) vs **mucosal malabsorption**.
- **Coeliac**: permanent immune enteropathy triggered by gluten.

Cause
- Maldigestion: **cystic fibrosis** (commonest cause of pancreatic insufficiency in children).
- Mucosal: coeliac, cow's milk allergy, Giardia, Crohn, lactase deficiency.
- Coeliac needs HLA-DQ2/DQ8; at risk: T1DM, Down, Turner, IgA deficiency.

Patho
- Gliadin → immune villous atrophy, **crypt hyperplasia**, intraepithelial lymphocytosis → malabsorption; fat loss drags ADEK.

Signs
- Classical 6 m–2 y: diarrhoea, distension, **wasted buttocks**, FTT.
- Extraintestinal: **hepatitis, osteoporosis, arthritis, isolated iron deficiency, dermatitis herpetiformis**.

Dx
- First-line: **anti-tTG IgA plus total IgA**, on a **gluten-containing diet**.
- IgA at least **0.2 g/l** to read tTG and EMA reliably.
- Avoid biopsy: positive symptoms, tTG AND EMA, positive HLA-DQ2/DQ8.

Rx
- **Strict lifelong gluten-free diet**; antibodies monitor adherence.
- CMPA → extensively hydrolysed formula; amino-acid formula if severe.

III-40 Diseases of oral cavity. Diseases of the oesophagus.

- **Def**
  - Mouth: mostly infections; oesophagus: reflux, EoE, achalasia, **TEF**, foreign bodies.
- **Cause**
  - **HSV-1** gingivostomatitis, **coxsackie A** herpangina, **Candida** thrush (persistent: **immunodeficiency**).
  - TEF: **VACTERL** association; commonest type is atresia with distal fistula (85%).
- **Patho**
  - Atopy → eosinophilic oesophagitis → rings, strictures → food impaction.
  - Button battery burns through the oesophageal wall in **2 hours**.
- **Signs**
  - Gingivostomatitis: swollen bleeding gums, anterior ulcers; herpangina: soft palate only.
  - Physiological reflux resolves by **12–18 months**.
  - TEF: **coughing, choking, cyanosis**; achalasia: dysphagia, regurgitation.
- **Dx**
  - EoE biopsy: at least **15 eosinophils per high-power field**.
  - Achalasia: **bird's-beak** on barium, absent peristalsis on **manometry**.
  - TEF: NG tube coiling on X-ray; pure atresia: gasless abdomen.
- **Rx**
  - Gingivostomatitis: fluids, analgesia, aciclovir within 72 h.
  - Severe reflux: **fundoplication**; achalasia: **Heller myotomy**.
  - TEF: **surgery**, look for VACTERL; swallowed coin: **emergency endoscopy (within 2 hours)**.

III-41 Chronic liver failure in children

- **Def**
  - Chronic loss of hepatic clearance, synthesis and flow in children.
- **Cause**
  - Commonest cause of chronic liver failure in children: **biliary atresia**.
  - Alpha-1 antitrypsin, Alagille; **Wilson** (AR, ATP7B), autoimmune hepatitis, hepatitis B/C.
- **Patho**
  - Can't clear (jaundice, itch), **can't build** (INR, albumin), can't flow (portal hypertension).
  - Wilson: copper can't reach bile → liver, basal ganglia, cornea, haemolysis.
- **Signs**
  - Jaundice, pale stools, pruritus, bruising, ascites, splenomegaly, varices, ADEK deficiency.
  - Wilson symptoms start **after 4–5 years of age**.
- **Dx**
  - INR not correcting with IV vitamin K = **synthetic liver failure**.
  - Wilson: serum caeruloplasmin **low**, 24-hour urinary copper **high**, KF rings, ATP7B genetics.
  - Acute liver failure with **Coombs-negative haemolysis** suggests Wilson disease.
- **Rx**
  - Cholestatic children: **MCT** formula + ADEK; ursodeoxycholic acid, cholestyramine for itch.
  - Spironolactone, lactulose, band ligation; rifaximin lowers ammoniagenic bacteria.
  - Wilson: penicillamine/trientine, then **zinc**; end-stage: **liver transplantation**.

III-42 Biliary atresia. Liver transplantation in children.

- **Def**
  - Progressive **fibro-obliteration of extrahepatic bile ducts** in infancy.
- **Cause**
  - Probably perinatal viral/immune trigger; isolated, or with polysplenia/situs anomalies, cystic.
- **Patho**
  - Blocked ducts → conjugated bilirubin refluxes → jaundice, dark urine; no stercobilin → acholic stool.
  - Retained bile → fibrosis → cirrhosis, portal hypertension; no bile → vitamin K bleeding.
- **Signs**
  - Typical stool feature: **clay-coloured (acholic)** stool, dark urine.
  - Term infant jaundiced beyond **2 weeks**; thriving, so missed as breast-milk jaundice.
- **Dx**
  - Jaundice beyond **2 weeks** needs a **conjugated (split) bilirubin**; raised GGT, INR.
  - Ultrasound: **triangular cord sign**, small or absent gallbladder; definitive: intraoperative cholangiogram.
  - Diagnose **earlier, but no later than 6 weeks** of infancy.
- **Rx**
  - **Kasai portoenterostomy** ASAP; ursodiol, prophylactic antibiotics, ADEK, MCT formula.
  - Fever and returning jaundice = **ascending cholangitis** → IV antibiotics.
  - Most common cause of **liver transplantation** in children; tacrolimus.

III-43 Inflammatory bowel disease. Crohn disease and ulcerative colitis in childhood

- **Def**
  - **Crohn**: **transmural** with **skip lesions**; **UC**: continuous from the **rectum**.
- **Cause**
  - Chronic immune-mediated gut inflammation (IBD) of childhood.
- **Patho**
  - Crohn: transmural, **non-caseating granulomas** → strictures, fistulae, perianal disease.
  - UC: mucosal only, **crypt abscesses** → bloody diarrhoea; linked to PSC.
- **Signs**
  - Crohn: weight loss; often presents first with **growth failure and delayed puberty**.
  - Perianal fissures: **not in the midline, deep, not painful**.
  - UC: bloody diarrhoea, urgency, tenesmus; high GGT → think **sclerosing cholangitis**.
- **Dx**
  - Bloods, stool culture; **faecal calprotectin** gatekeeps endoscopy.
  - Ileocolonoscopy + upper endoscopy with biopsies + MR enterography.
- **Rx**
  - Crohn: first choice **exclusive enteral nutrition** 6–8 wk, then thiopurine/methotrexate, anti-TNF.
  - UC: 5-ASA (mesalamine) → steroids → thiopurine/anti-TNF; colectomy curative.
  - **PUCAI ≥65**: IV steroids; days 3/5 → infliximab or ciclosporin.

III-44 Pancreatitis in childhood

Def
- Acute pancreatic inflammation with autodigestion, may progress to SIRS.

Cause
- **Biliary**, trauma (handlebar, NAI), drugs: **valproate**, azathioprine, **L-asparaginase**.
- Infection (mumps), HUS, HSP, metabolic; recurrent → genetic testing: **PRSS1**, **SPINK1**, CFTR.

Patho
- Premature intra-acinar activation of **trypsinogen** → autodigestion → cytokines → SIRS, capillary leak.
- **Fat saponification** binds calcium → hypocalcaemia.

Signs
- Epigastric pain radiating to back, eased by leaning forward; vomiting, fever.
- Tachycardia, ileus; rarely Grey Turner or Cullen sign.

Dx
- INSPPIRE needs **2 of 3**: typical pain, amylase/lipase ≥**3×** ULN, imaging.
- Ultrasound first; check calcium, glucose, triglycerides, LFTs.

Rx
- Fluids ~1.5–2× maintenance, analgesia incl. opioids; **enteral feeding** within **48–72 hours**.
- No routine antibiotics — only infected necrosis or cholangitis.
- Treat cause: stop drug, cholecystectomy same admission; pseudocyst forms after more than **4 weeks**.

III-45 Cleft lip and cleft palate

Def
- Congenital cleft from failed fusion of facial processes; ~1 in 700 births.

Cause
- Maternal smoking, alcohol, **phenytoin/valproate**, folate deficiency.
- Isolated cleft palate is commoner in **girls** and more often syndromic, e.g. **22q11 deletion**.

Patho
- Cleft lip: failed fusion of the **medial nasal** process with the **maxillary** process.
- Palatal shelves fail → no suction seal, VPI.
- **Tensor veli palatini** dysfunction → Eustachian tube can't open → otitis media with effusion.

Signs
- Feeding difficulty, nasal milk reflux; glue ear.
- Hypernasal speech (VPI); **Pierre Robin**: micrognathia, glossoptosis, U-shaped cleft palate.

Dx
- Often seen at 20-week anomaly scan (lip more than palate).
- Any cleft palate → hearing test; syndromic features → 22q11 testing.

Rx
- Cleft team; squeezable bottles, upright feeding, weight monitoring.
- Lip repaired at about **3 months**; soft palate closed at **1 year**, before first words.
- Grommets, speech therapy, alveolar bone graft 8–11 years.

III-46 Ileus, intussusception

Def
- **Ileus**: bowel obstruction, mechanical (lumen blocked) or paralytic (peristalsis stops).

Cause
- Mechanical: adhesions, incarcerated hernia, intussusception, volvulus, atresia.
- Paralytic: post-surgery, sepsis, peritonitis, **hypokalaemia**, opioids, **vincristine**.
- Intussusception: usual lead point **Peyer's patches**; over 3 y: Meckel's, polyp, lymphoma.

Patho
- Mechanical → tinkling bowel sounds; paralytic → silent, distended abdomen.
- Colicky pain turning constant suggests **strangulation**: ischaemia, fever, rising lactate.
- Ileocolic intussusception → mucosal ischaemia → redcurrant-jelly stool (**late**).

Signs
- Intussusception **3 months – 3 years**: paroxysmal colic, pallor, legs drawn up, lethargy.
- Sausage mass (RUQ); vomiting, later bilious.

Dx
- Ultrasound: **target (doughnut) sign**.
- Commonest cause of strangulation ileus in children: **incarcerated hernia**.
- Suspected SBO: unstable → X-ray; stable → CT with IV contrast.

Rx
- Obstruction: drip and suck, analgesia, surgical review; complete SBO: laparotomy.
- Intussusception: resuscitate → **air (pneumatic) enema reduction**; ~10% recur.
- Surgery: strangulation, peritonitis, perforation, shock, failed reduction.

III-47 Diseases of the colon. Obstipation. Hirschsprung disease.

Def
- About **95%** of childhood constipation is functional; Hirschsprung = congenital aganglionosis.

Cause
- Functional: weaning, toilet training, school, anal fissure (tight sphincter → mucosal tear).
- Hirschsprung: failed neural crest migration; Down syndrome association.

Patho
- Withholding → harder stool → pain → dilated rectum → **overflow soiling** (liquid stool around impacted faeces).
- Aganglionic rectum can't relax → functional obstruction.

Signs
- Functional: withholding postures, packed rectum, soiling; **infantile dyschezia**: infant strains ~10 minutes.
- Hirschsprung: meconium delayed beyond **48 hours**, distension, bilious vomiting.
- Rectum **empty** with explosive release on withdrawal.

Dx
- **Rectal suction biopsy**: absent ganglion cells, raised acetylcholinesterase.
- Contrast enema: transition zone; manometry: absent rectoanal inhibitory reflex.
- Anal fissure: physical exam (DRE, anoscopy).

Rx
- Functional: disimpaction with macrogol, then maintenance for **months**, not days; toileting.
- Hirschsprung: rectal washouts → pull-through operation.
- Feared complication: **enterocolitis** — fever, distension, explosive foul diarrhoea.

III-48 Acute appendicitis

Def
- Acute inflammation of the appendix from **luminal obstruction**.

Cause
- Obstruction of the lumen by a **faecolith** (older) or **lymphoid hyperplasia** (younger, after viral illness).

Patho
- Obstruction → pressure, venous failure, bacterial invasion → gangrene → perforation after **36–48 hours**.
- Visceral T10 → periumbilical pain → parietal peritoneum → localised RIF pain.
- Children under 5 perforate more often: short **omentum**.

Signs
- Pain first epigastric/periumbilical, later **right lower quadrant**; vomiting comes **after** the pain.
- McBurney tenderness, pain on hopping/cough; **Rovsing's** sign: pressing LIF hurts RIF.

Dx
- Paediatric Appendicitis or Alvarado score: low discharge, high surgery, middle imaging.
- First-line imaging is **ultrasound**: non-compressible tube >6 mm; CT only if inconclusive.

Rx
- Analgesia given freely **does not mask** the diagnosis; IV fluids, antibiotics.
- Laparoscopic appendicectomy.
- Appendix mass → antibiotics first ± drainage, interval appendicectomy.

III-49 Testis and ovarian torsion. Cryptorchidism.

Def
- **Torsion**: testis/ovary twists on its pedicle; **cryptorchidism**: testis not in scrotum.

Cause
- Testis: **bell-clapper** deformity, usually **bilateral**; ovary: cyst or teratoma as lead weight.
- Cryptorchidism: preterm (up to 30%) vs term 2–4%.

Patho
- Torsion: veins occlude first → arterial failure → infarction.
- Undescended testis → germ cells degenerate → **infertility** and **testicular germ cell tumour**.

Signs
- Testis: sudden pain, vomiting, high-riding horizontal lie, **cremasteric reflex** absent.
- Appendix testis torsion: **blue dot** sign, reflex preserved.
- Ovary: sudden unilateral lower abdominal pain, vomiting, may mimic appendicitis.

Dx
- Testicular torsion is clinical → theatre, not scan.
- Ovary: US shows enlarged oedematous ovary; preserved Doppler does not exclude.
- Bilateral impalpable testes → karyotype, exclude CAH.

Rx
- Testis: explore within **6 hours** (~90% salvage), detort, fix both sides.
- Ovary: laparoscopic detorsion, **untwisted and preserved** even if dusky.
- Cryptorchidism: refer by 3–6 months, **orchidopexy** at **6–18 months**.

III-50 Neural tube defects

Def
- Failed neural tube closure (week 4): cranial **day 25**, caudal **day 28**.
- Anencephaly cranially; spina bifida (occulta, meningocele, **myelomeningocele**) caudally.

Cause
- Folate deficiency (**TMP-SMX** in 1st trimester), previous NTD, valproate/carbamazepine, maternal diabetes, obesity.

Patho
- Exposed placode damaged → deficits **below the level**.
- Most myelomeningoceles: tethered cord → **Chiari II** → **hydrocephalus**.

Signs
- Flaccid leg weakness, sensory loss, neurogenic bladder/bowel, club feet, scoliosis.
- Hydrocephalus; stridor/apnoea from brainstem compression; latex allergy.

Dx
- Raised maternal serum **alpha-fetoprotein**; ultrasound **lemon** (frontal bones) and **banana** (cerebellum) signs.
- At birth: level, head circumference, fontanelle, hips, feet, bladder.

Rx
- Prevent: folic acid **400 µg** daily routinely, **5 mg** if high risk.
- Birth: moist sterile dressing, prone, no latex; close within **48 hours**.
- VP shunt/ETV; clean intermittent catheterisation, oxybutynin, physio.

III-51 Perinatal asphyxia, Hypoxic-ischemic encephalopathy (HIE)

Def
- Perinatal oxygen deprivation causing encephalopathy (**HIE**), staged by Sarnat.

Cause
- Abruption, cord prolapse, uterine rupture, shoulder dystocia, maternal hypotension.
- Supports: sentinel event, Apgar ≤5 at 10 min, pH <7.0, base deficit ≥12–16.

Patho
- Hypoxia → ATP fails → primary energy failure → oedema, excitotoxicity.
- Latent phase 1–6 h (window) → **secondary energy failure** 6–72 h → neuronal death.
- **Diving reflex** spares brain, heart, adrenals; kidneys, gut, liver suffer.

Signs
- Mild: hyperalert, jittery; moderate: lethargy, hypotonia, **seizures**.
- Severe: coma, flaccid, apnoea; multiorgan ATN, PPHN, DIC, hypoglycaemia.

Dx
- Cord gas, aEEG (normalising by 24–48 h good).
- MRI days 4–10: **basal ganglia and thalamus** injury = poor prognosis.
- Apgar: Appearance, Pulse, Grimace, Activity, Respiration (10-point scale).

Rx
- Cooling: **moderate or severe** HIE, babies ≥36 weeks.
- **33.5 °C** for **72 hours**, started within **6 hours**; rewarm 0.5 °C per hour.
- Normocapnia, avoid hyperthermia, restrict fluids; seizures → **phenobarbital**.

III-S2 Hydrocephalus. Infantile cerebral palsy

Def
- **Hydrocephalus**: CSF accumulation with ventricular dilatation.
- **CP**: essential criterion **non-progressive, residual brain injury** of developing brain.

Cause
- Hydrocephalus: aqueduct stenosis, Chiari II, tumour, post-IVH/meningitis, congenital toxoplasmosis.
- CP: antenatal (most), **prematurity (PVL, IVH)** strongest, HIE, kernicterus.

Patho
- Obstruction or failed absorption → CSF builds → skull expands if sutures open.
- Preterm IVH arises in the **germinal matrix**.
- CP lesion static, picture evolves: hypotonia → spasticity, contractures.

Signs
- Infant: head circumference crossing centiles, bulging fontanelle, **sunsetting** eyes.
- CP: **spastic** commonest; **dyskinetic** (basal ganglia), ataxic (cerebellum).
- Hand preference before **12 months** suggests hemiplegia; loss of skills is not CP.

Dx
- Hydrocephalus/IVH: **cranial ultrasound** (infant), MRI; bulging fontanelle + vomiting → brain imaging.
- CP: standardised movement assessment + MRI; classify function by GMFCS.

Rx
- **Ventriculoperitoneal shunt** or **endoscopic third ventriculostomy**; headache/vomiting/drowsiness = blockage.
- CP: multidisciplinary physio/OT/SLT; **baclofen**, botulinum toxin, hip surveillance.

III-S3 Meningitis, brain abscess

Def
- Bacterial meningeal infection; brain abscess = walled-off pus in brain.

Cause
- Neonates: **Streptococcus agalactiae (GBS)** and **E. coli**; Listeria.
- Later: meningococcus, pneumococcus; abscess: ear/sinus spread, cyanotic CHD (right-to-left shunt bypasses lung filter).

Patho
- Cytokines open BBB → **protein elevated, glucose decreased**, neutrophils.
- Oedema, impaired CSF absorption → raised ICP.

Signs
- Neck stiffness, Kernig, Brudzinski; infants: fever, irritability, bulging fontanelle.
- Non-blanching rash → meningococcal sepsis.
- Abscess: headache, focal signs, seizures; fever in only half.

Dx
- LP: neutrophils >1000/μL, protein >1 g/L, CSF:blood glucose <0.4.
- Raised ICP, shock, coagulopathy: delay **LP**, never delay **antibiotics**.
- Abscess: **ring-enhancing** lesion; **lumbar puncture** contraindicated.

Rx
- Antibiotics without delay: under 3 months cefotaxime + amoxicillin (covers **Listeria**); over 3 months ceftriaxone.
- **Dexamethasone** over **3 months** reduces **hearing loss**; test hearing within 4 weeks; contact prophylaxis.
- Abscess: drain + IV antibiotics 4–8 weeks.

III-S4 Encephalitis. Facial palsy.

Def
- **Encephalitis** = brain parenchyma inflammation; **facial palsy** = central (UMN) or peripheral (LMN).

Cause
- Encephalitis: **HSV-1** (key treatable cause), enterovirus, VZV; neonates HSV-2; anti-NMDAR, ADEM.
- Facial palsy: Bell (exclusion), otitis, Lyme, **Ramsay Hunt** (VZV), hypertension, leukaemia.

Patho
- HSV encephalitis typically affects the **temporal** lobe.
- Upper face has **bilateral** cortical supply → forehead spared in central lesions.

Signs
- Encephalitis: fever + altered behaviour/consciousness, seizures, focal signs.
- Peripheral: whole half-face, can't close eye, taste loss; central: forehead sparing.
- Bilateral palsy: Lyme, **Guillain-Barre**, leukaemia; Ramsay Hunt: ear vesicles.

Dx
- LP: lymphocytic CSF, normal glucose; **HSV PCR**; MRI, EEG.
- Every facial palsy: examine the **ear**, measure **blood pressure**; atypical course → image brain.

Rx
- **IV aciclovir** on suspicion, 14–21 days (21 neonates); crystal-induced AKI risk.
- ADEM methylprednisolone; anti-NMDAR in girls → search for **ovarian teratoma**.
- Palsy: protect the eye, treat cause; prednisolone within 72 h.

III-S5 Neurocutaneous syndromes

Def
- **Neurocutaneous syndromes**: skin + nervous system lesions (shared ectodermal origin).

Cause
- **NF1**: neurofibromin, chromosome **17**; **NF2**: merlin, chromosome 22 (AD).
- **TSC1/TSC2** (AD); Sturge–Weber: somatic **GNAQ** mutation, not inherited.

Patho
- NF1: neurofibromin loss → **Ras** unchecked → neural crest tumours.
- TSC: **mTOR** overactivity → hamartomas; SWS: leptomeningeal angioma → cortical ischaemia.

Signs
- NF1: ≥**6** café-au-lait macules, freckling, neurofibromas, optic glioma, **Lisch nodules**.
- NF2: bilateral **vestibular schwannomas**; TSC: ash-leaf macules, angiofibromas, rhabdomyomas, **infantile spasms**.
- SWS: **V1** port-wine stain, seizures, hemiparesis, **glaucoma**.

Dx
- NF1: two or more diagnostic features; annual eye checks.
- TSC: Wood lamp for ash-leaf; infantile spasms show hypsarrhythmia (West syndrome).
- SWS imaging: tram-track calcification.

Rx
- NF1 plexiform neurofibroma: selumetinib (MEK inhibitor).
- TSC spasms: first-line **vigabatrin**; SEGA: **everolimus**.
- SWS: seizure control, low-dose aspirin, glaucoma follow-up, pulsed-dye laser.